MAT3253 Complex Variables
Lecture Notes

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Contents

0.1 Notation v

1 Arithmetic and Geometry of Complex Numbers **1**

1.1 Introduction: from natural numbers to complex numbers 1

1.2 Models of complex field 3

1.2.1 Complex numbers as points in the plane 4

1.2.2 Complex numbers as matrices 5

1.2.3 Complex numbers as polynomials 6

1.3 Complex conjugate, and modulus 8

1.4 Polar form and DeMoivre formula 13

1.5 Applications in plane geometry 20

1.5.1 A criterion of collinear points 20

1.5.2 Points lying on an equilateral triangle 21

1.5.3 A proof of Napoleon's theorem 21

1.6 Appendix: Solving cubic equation using discrete Fourier transform 24

1.7 Appendix: Complex numbers in Python 26

2 Linear Fractional Transformation and Riemann sphere **37**

2.1 Complex inversion 37

2.2 Spherical representation of complex numbers 42

2.3 Linear fractional transformations 44

2.4 3-Transitivity of linear fractional transformation 53

2.5 Cross ratio 55

2.6 Python Program 58

2.7 Appendix: Details of stereographic projection 58

2.8 Appendix: Complex projective line . . . 62

3 Complex Exponential Function and Euler's formula 68

3.1 Convergent sequences . . . 68
3.2 Complex series . . . 71
3.3 Convergence of power series . . . 75
3.4 Complex exponential function . . . 78
3.5 Euler's formula . . . 81
3.6 Application to alternating current . . . 83

4 Complex Trigonometric Function and Log Function 88

4.1 Trigonometric functions . . . 88
4.2 Log function . . . 92
4.3 Power function . . . 94
4.4 Inverse of complex trigonometric functions . . . 95
4.5 Visualization of complex function using phase portrait . . . 96
4.6 Numerical calculations . . . 97
4.7 Summary . . . 100

5 Topology of the Complex Plane and Continuity 103

5.1 Point-set topology in the (extended) complex plane . . . 103
5.2 Limit and continuity of complex function . . . 106
5.3 Paths and parametric curves . . . 109
5.4 Connected set . . . 112
5.5 Uniform convergence and continuity of power series . . . 120

6 Complex Differentiability 126

6.1 Differentiable real functions from multi-variable calculus . . . 126
6.2 Complex differentiability and Cauchy-Riemann condition . . . 129
6.3 Complex differential form and Wirtinger derivatives . . . 136
6.3.1 Vector field and differential form . . . 137
6.3.2 Wirtinger derivatives . . . 146
6.4 Appendix: Fréchet Derivatives in Real and Complex Variables . . . 149

6.5 Appendix: A necessary and sufficient condition for complex differentiability at a given point . . . 152

7 Holomorphic Functions **157**

7.1 Holomorphic functions and entire functions . . . 157
7.2 Term-wise differentiation of power series . . . 159
7.3 Differentiation rules and differentiation at the point at infinity . . . 163
7.4 Chain rule for parametric curve . . . 166
7.5 Conformal property . . . 167
7.6 Harmonic functions . . . 171
7.7 Application: Smith Chart as a conformal Möbius transformation . . . 175
7.8 Additional proof for existence of harmonic conjugate . . . 177

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0.1 Notation

List of symbols and notation

i	Imaginary unit
$\mathbb{N}$	Set of integers larger than or equal to 1
$\mathbb{N}_{\geq 0}$	Set of integers larger than or equal to 0
$\mathbb{Z}$	Set of integers
$\mathbb{Q}$	Set of rational numbers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$\bar{\mathbb{C}}$	Extended complex plane
sup, inf	Supremum and infinum
$\limsup$	Limit superior
$\liminf$	Limit inferior
$\bar{z}$, z^*	Conjugate of complex number z
$\mathrm{Re}(z)$	Real part of z
$\mathrm{Im}(z)$	Imaginary part of z
$\lvert z\rvert$, $\mathrm{abs}(z)$	The modulus/absolute value/magnitude of z
$\arg(z)$	The argument/angle of z
$\mathrm{Arg}(z)$	Principal argument of z
$[z_0, z_1, z_2, z_3]$	Cross ratio
∞	Point at infinity
$\exp(z)$	Complex exponential function
$\log(z)$	Complex log function
$\mathrm{Log}(z)$	Principal branch of complex log function
$f^{-1}(U)$	Inverse image of a set U through the function f
$\overline{S}$	Closure of a set S in $\mathbb{C}$
$\overline{S}^X$	Relative closure of S in X
$\mathcal{T}_X$	Relative topology of a set X in $\mathbb{C}$
$\mathbf{1}_S$	Indicator function of a set S
$D(z,r)$	Open disk with radius r centered at z
$\overline{D}(z,r)$	Closed disk with radius r centered at z
$(df)_{\mathbf{p}}$	Differential of f at a point $\mathbf{p}$

$f(z)dz$	Complex differential form
$f'(z)$, df/dz	Complex derivative of function f
$f^{(n)}$	the n-th order derivative of function $f(z)$
$\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$	Partial differential operators with respect to x and y
$\frac{\partial}{\partial z}$, $\frac{\partial}{\partial \bar{z}}$	Partial differential operators with respect to z and $\bar{z}$
$-C$	Reverse of contour C
$C_1 + C_2$	Concatenation of contours C_1 and C_2
$\int_C f\,dz$	Complex integral of f along a curve C
$\oint_C f\,dz$	Complex integral of f along a closed curve C
$\mathrm{Res}(f; z_0)$	Residue of function f at the point z_0
$\mathcal{O}(D)$	The ring of holomorphic functions defined on a domain D
$\mathcal{M}(D)$	The field of meromorphic functions defined on a domain D
$n(C; z_0)$	Winding number of a closed curve C relative to the point z_0

Chapter 1

Arithmetic and Geometry of Complex Numbers

1.1 Introduction: from natural numbers to complex numbers

We begin with a famous remark of the German mathematician Leopold Kronecker:

> God created the natural numbers, the rest is the work of man.

The natural numbers $\mathbb{N}$ allow us to perform addition and multiplication, but they are not closed under subtraction or division. For instance, the equations $x + 5 = 3$ and $2x = 3$ have no solutions in $\mathbb{N}$. To overcome this limitation, we enlarge $\mathbb{N}$ first to the integers $\mathbb{Z}$, and then to the rational numbers $\mathbb{Q}$. Let us briefly recall how $\mathbb{Q}$ can be constructed from $\mathbb{Z}$.

A rational number is usually written as a/b, where a and b are integers with $b \neq 0$. Formally, we represent such a number by an ordered pair (a, b) of integers. Two pairs (a, b) and (c, d) represent the same rational number if and only if $ad = bc$. Thus a rational number is, strictly speaking, an equivalence class of such pairs.

Addition and multiplication are defined by

$$(a, b) + (c, d) \triangleq (ad + bc,\, bd), \qquad (a, b) \cdot (c, d) \triangleq (ac,\, bd).$$

For example, the rational numbers $1/2$ and $1/3$ are represented by $(1, 2)$ and $(1, 3)$, and their sum is

$$(1, 2) + (1, 3) = (1 \cdot 3 + 2 \cdot 1,\, 2 \cdot 3) = (5, 6).$$

Pairs of the form $(k, 1)$ behave exactly like the integer k, so $\mathbb{Z}$ naturally embeds into this system. Moreover, every nonzero rational number has a multiplicative inverse: if $a \neq 0 \neq b$, then

$$(a, b) \cdot (b, a) = (ab, ab),$$

which is equivalent to $(1, 1)$.

Although $\mathbb{Q}$ is closed under the usual algebraic operations, it is not complete in an analytic sense. In particular, it is not closed under taking supremum and infimum. Recall that the *supremum* of a set A is the least upper bound of A, and the *infimum* is the greatest lower bound. A classical example shows the failure of completeness: the set

$$\{x \in \mathbb{Q} : x^2 < 2\}$$

has no smallest upper bound in $\mathbb{Q}$.

To remedy this, we complete $\mathbb{Q}$ with respect to the Euclidean metric. This can be done using Dedekind cuts or Cauchy sequences, and the resulting number system is the field of real numbers $\mathbb{R}$. By construction, $\mathbb{R}$ has the property that every nonempty subset bounded above has a supremum, every Cauchy sequence converges, and $\mathbb{Q}$ sits densely inside $\mathbb{R}$.

However, even $\mathbb{R}$ is not the end of the story. The real numbers are not algebraically closed: a polynomial

$$c_0 + c_1 x + c_2 x^2 + \cdots + c_d x^n$$

with real coefficients need not have a real root. The simplest example is $x^2 + 1$, which has no solution in $\mathbb{R}$. The remarkable fact is that adjoining a single new element i, defined formally by $i^2 = -1$, is enough to remedy this deficiency. The resulting number system,

$$\mathbb{C} = \{\, a + bi : a, b \in \mathbb{R} \,\},$$

is called the *complex numbers*. It contains $\mathbb{R}$ as a subfield (via the identification $a \leftrightarrow a + 0i$), and it turns out to be *algebraically closed*: every nonconstant polynomial with complex coefficients has a root in $\mathbb{C}$.

This property is known as the *Fundamental Theorem of Algebra*. It is not obvious from the definition of $\mathbb{C}$, and we will give a proof using complex analysis. Nevertheless, the theorem justifies the introduction of complex numbers as the natural setting for algebraic equations.

1.2 Models of complex field

The rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$ share a common algebraic structure. In abstract algebra, this structure is called a *field*.

Definition 1.2.1. A number system $(F, +, \cdot)$ is called a *field* if it satisfies the following axioms:

1. (closure) $a + b \in F$ for all $a, b \in F$.
2. (associativity) $(a + b) + c = a + (b + c)$ for all $a, b, c \in F$.
3. (commutativity) $a + b = b + a$ for all $a, b \in F$.
4. (existence of zero) There exists $0 \in F$ such that $0 + a = a + 0 = a$ for all $a \in F$.
5. (additive inverse) For every $a \in F$, there exists $a' \in F$ such that $a + a' = 0$.
6. (closure) $a \cdot b \in F$ for all $a, b \in F$.
7. (associativity) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in F$.
8. (commutativity) $a \cdot b = b \cdot a$ for all $a, b \in F$.
9. (existence of one) There exists $1 \in F$ with $1 \neq 0$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in F$.
10. (multiplicative inverse) For every $a \in F \setminus \{0\}$, there exists $a'' \in F$ such that $a \cdot a'' = 1$.
11. (distributivity) $a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in F$.

A subset K of a field F is called a *subfield* if the elements of K satisfy all the field axioms. In this case, F is called an *extension* of K.

We require a "number system" to be a field so that addition, subtraction, multiplication, and division behave according to these axioms.

Definition 1.2.2. A *complex field* is a field that extends $\mathbb{R}$ and contains a special element i satisfying $i^2 = -1$, and is generated by $\mathbb{R}$ and i. Such a field is denoted by $\mathbb{C}$.

The special number i that satisfies $i^2 = -1$ is called the *imaginary unit*.

Remark. The phrase "generated by $\mathbb{R}$ and i" means that there is no proper subfield containing all real numbers together with the element i. For example, consider the field of rational functions $f(T)/g(T)$, where $f(T)$ and $g(T)$ are polynomials in T with complex coefficients. This field contains $\mathbb{R}$ and i, but it is far too large to be considered the complex field: mixing real numbers with i never produces the transcendental variable T.

1.2.1 Complex numbers as points in the plane

One way to construct a complex field is to regard each ordered pair (x, y) of real numbers as a number. The set of all such pairs forms a plane called the *complex plane* or *Argand diagram*. The point (x, y) may also be viewed as a vector, and addition is defined by vector addition (Fig. 1.1).

Define

$$F_1 \triangleq \{(a, b) : a, b \in \mathbb{R}\}.$$

Two pairs (a, b) and (c, d) are *equal* if $a = c$ and $b = d$.

Addition and multiplication in F_1 are defined by

$$(a, b) + (c, d) \triangleq (a + c,\ b + d), \qquad (a, b) \cdot (c, d) \triangleq (ac - bd,\ ad + bc).$$

The real numbers embed into F_1 via $x \mapsto (x, 0)$. Under this identification,

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0), \qquad (x_1, 0) \cdot (x_2, 0) = (x_1 x_2, 0),$$

so the horizontal axis of the complex plane corresponds to the real line.

The element $(1, 0)$ is the multiplicative identity. The element $(0, 1)$ satisfies

$$(0, 1) \cdot (0, 1) = (-1, 0),$$

so it plays the role of the imaginary unit i.

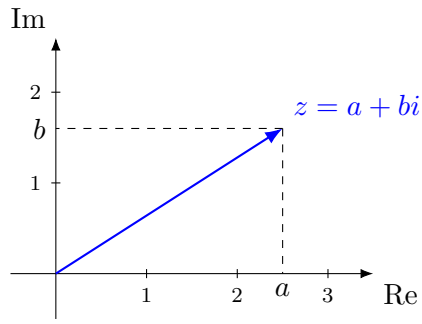


Figure 1.1: Argand diagram

Every element (x, y) can be written as

$$(x, y) = x(1, 0) + y(0, 1) = x + iy.$$

This shows that $1 = (1, 0)$ and $i = (0, 1)$ form a basis of this number system over $\mathbb{R}$.

Multiplication by i has the geometric meaning of rotation by 90 degrees (Fig. 1.2).

1.2.2 Complex numbers as matrices

In the second model, a complex number is represented by a 2×2 real matrix:

$$F_2 \triangleq \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} : a, b \in \mathbb{R} \right\}.$$

Addition and multiplication are the usual matrix operations. The arithmetic mirrors that of the first model:

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{bmatrix}.$$

A real number x corresponds to the diagonal matrix

$$\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}.$$

The matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

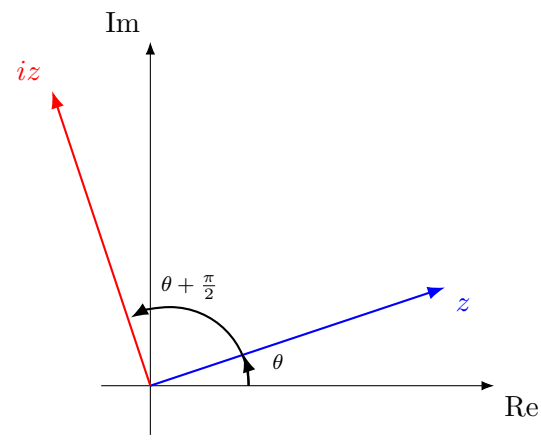


Figure 1.2: Multiply $z = a + bi$ by i gives $iz = -b + ai$.

satisfies

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix},$$

so it plays the role of the imaginary unit. Every matrix in F_2 decomposes as

$$\begin{bmatrix} x & -y \\ y & x \end{bmatrix} = x \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + y \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The first part $\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}$ is a diagonal matrix, while the second part $\begin{bmatrix} 0 & -y \\ y & 0 \end{bmatrix}$ is a skew symmetric matrix.

1.2.3 Complex numbers as polynomials

Let $\mathbb{R}[T]$ denote the set of all polynomials in the form $a_0 + a_1 T + \cdots + a_d T^d$, where with coefficients a_0, $a_1 \ldots, a_d$ in $\mathbb{R}$. Define an equivalence relation on $\mathbb{R}[T]$ by declaring $f(T) \sim g(T)$ if $f(T) - g(T)$ is divisible by $T^2 + 1$. Each equivalence class has a unique representative of the form $a + bT$. For example, the imaginary unit is represented by the equivalence class

$$\{T + p(T)(T^2 + 1) : p(T) \in R[T]\},$$

a real number a is represented by

$$\{a + p(T)(T^2 + 1) : p(T) \in R[T]\}.$$

In each equivalence class, we select a representative in the form $a + bT$, for some real numbers a and b. Addition and subtraction is performed modulo $T^2 + 1$. The quotient ring is denoted $\mathbb{R}[T]/(T^2 + 1)$. This quotient ring has the structure of a field, and is a model of the complex field.

All three constructions are isomorphic. The correspondences are:

$$(a, b) \longleftrightarrow \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \longleftrightarrow a + bT.$$

The real number 1 can be represented by

$$(1, 0), \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad 1.$$

The vector $(1, 0)$ corresponds ot the unit vector in the x-axis, the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity matrix, and "1" is the polynomial consisting of the constant term 1.

The imaginary unit corresponds to

$$(0, 1), \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad T.$$

Usually, we adopt the notation

$$a + bi$$

for a complex number. The elements 1 and i form a basis over $\mathbb{R}$, so $\mathbb{C}$ is a 2-dimensional real vector space. Multiplication is given by

$$(a + bi)(c + di) = ac - bd + i(ad + bc).$$

Real numbers embed into $\mathbb{C}$ via $r \mapsto r + 0i$, so $\mathbb{C}$ is indeed an extension of $\mathbb{R}$.

Theorem 1.2.3. *Let*

$$\mathbb{C} \triangleq \{a + bi : a, b \in \mathbb{R}\},$$

where i is a symbol satisfying $i^2 = -1$. Then $\mathbb{C}$ is a field and, simultaneously, a real vector space of dimension 2.

Remark. In the vector representation of complex numbers, the element $(0, 1)$ is chosen to represent the imaginary unit i. However, this choice is not unique. We could instead take $(0, -1)$ to be the imaginary unit. If we do so, all definitions and operations can be reformulated consistently, and the resulting number system is isomorphic to the usual complex numbers. In other words, choosing i or $-i$ is an arbitrary convention in the construction.

Definition 1.2.4. For a complex number $z = a+bi$ in $\mathbb{C}$, the *real part* and *imaginary part* of z are defined by

$$\mathrm{Re}(z) \triangleq a, \qquad \mathrm{Im}(z) \triangleq b.$$

A complex number z is said to be *purely imaginary* if $\mathrm{Re}(z) = 0$, and *real* if $\mathrm{Im}(z) = 0$.

1.3 Complex conjugate, and modulus

Definition 1.3.1. The *complex conjugate* of a complex number $z = a + bi$ is defined as

$$\bar{z} \triangleq z^* \triangleq a - bi.$$

Geometrically, the real and imaginary parts of z are simply the x- and y-coordinates of the point representing z in the complex plane. The complex conjugate $\bar{z}$ is the reflection of z across the real axis (See Fig. 1.3).

We list some elementary properties of complex conjugation in the next proposition. The proof is straightforward.

Proposition 1.3.2. *For complex numbers $z \in \mathbb{C}$:*

(a). $(z^*)^* = z$.

(b). $z^* = z$ *if and only if* z *is real.*

(c). $z^* = -z$ *if and only if* z *is purely imaginary.*

Proof. Part (i) says that the conjugate of a conjugate is the same as itself (LEAN proof for part (i)). Parts (ii) and (iii) give the relation between complex conjugate and the real and imaginary part of a complex number. (LEAN proof for part (ii) and (iii)) □

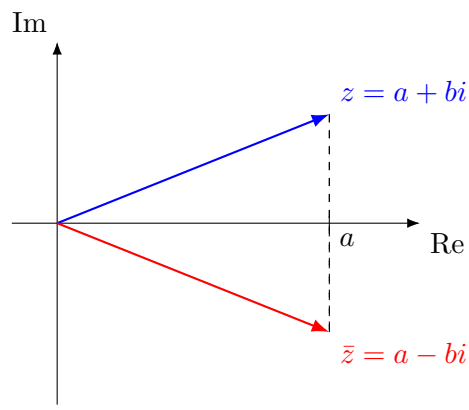


Figure 1.3: Complex conjugation

Theorem 1.3.3.

(a). $(z_1 + z_2)^* = z_1^* + z_2^*$, *for* $z_1, z_2 \in \mathbb{C}$,

(b). $(z_1 z_2)^* = z_1^* z_2^*$, *for* $z_1, z_2 \in \mathbb{C}$,

(c). $(1/z)^* = 1/(z^*)$ *for* $z \neq 0$.

This theorem says that the conjugate of a sum is equal to the sum of the conjugates, and the conjugate of a product is equal to the product of the conjugates.

Proof. Suppose $z_1 = a + bi$ and $z_2 = c + di$. For the conjugate of sum, we calculate

$$\begin{aligned} z_1^* + z_2^* &= (a + bi)^* + (c + di)^* \\ &= a - bi + (c - di) \\ &= (a + c) - i(b + d) \\ &= (z_1 + z_2)^*. \end{aligned}$$

Likewise, we prove the second part by

$$\begin{aligned} z_1^* z_2^* &= (a - bi)(c - di) \\ &= ac - bd - i(ac + bd) \\ &= ((ac - bd) + i(ac + bd))^* \\ &= (z_1 z_2)^*. \end{aligned}$$

For the last part, we can apply part (ii) and check that for $z \neq 0$,

$$(1/z)^* z^* = ((1/z)z)^* = 1^* = 1.$$

Since the product of $(1/z)^*$ and z^* is equal to 1, the inverse of z^* is equal to $(1/z)^*$. (LEAN proof) □

By Theorem 1.3.3, we see that complex conjugation is an *automorphism* of the complex field. It fixes all the real numbers and maps i to $-i$, preserving the algebraic structure. (cf. the remark before Definition 1.2.4)

We now illustrate how complex conjugation is used to perform division. Suppose we want to find the complex number z satisfying $(c + di)z = a + bi$, assuming $c + di \neq 0$. The solution is the quotient $(a + bi)/(c + di)$. Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} = \frac{ac + bd}{c^2 + d^2} + i\,\frac{bc - ad}{c^2 + d^2}. \tag{1.3.1}$$

Complex division can also be performed using the 2×2 matrix representation. Division corresponds to matrix inversion:

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} c & -d \\ d & c \end{bmatrix}^{-1} = \frac{1}{c^2 + d^2} \begin{bmatrix} ac + bd & ad - bc \\ bc - ad & ac + bd \end{bmatrix}.$$

Remark. The quantity $c^2 + d^2$ is the determinant of the matrix $\begin{bmatrix} c & -d \\ d & c \end{bmatrix}$. In the matrix model, complex conjugation corresponds to taking the transpose.

Example 1.3.1. Solve the equation

$$3z + iz + 2 = 1 - i.$$

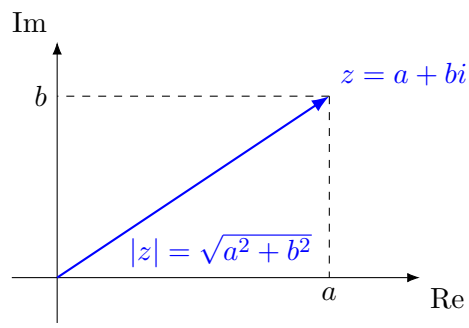


Figure 1.4: Geometric meaning of the modulus of a complex number.

First isolate z:

$$\begin{aligned} 3z + iz + 2 &= 1 - i, \\ 3z + iz &= -1 - i, \\ z &= \frac{-1 - i}{3 + i}. \end{aligned}$$

Now compute the inverse of $3 + i$:

$$\begin{aligned} z &= \frac{-1 - i}{3 + i} \cdot \frac{3 - i}{3 - i} \\ &= \frac{(-1 - i)(3 - i)}{10} \\ &= -0.4 - 0.2i. \end{aligned}$$

The product of z and its conjugate is real.

Theorem 1.3.4. *If $z = a + bi$ for $a, b \in \mathbb{R}$, then*

(a). $zz^* = a^2 + b^2$.

(b).

$$a = \frac{z + z^*}{2}, \qquad b = \frac{z - z^*}{2i}. \tag{1.3.2}$$

Proof. We prove part (i) by calculation,

$$zz^* = (a + bi)(a - bi) = a^2 - (bi)^2 = a^2 - b^2(i^2) = a^2 + b^2.$$

The proof of Part (ii) is straightforward. (LEAN proof) □

Definition 1.3.5. The *modulus* of $z = a + bi$ is

$$|z| = \sqrt{a^2 + b^2}.$$

It is also called the *absolute value*, *magnitude*, or *radius*.

The geometric meaning of the modulus is depicted in Fig. 1.4.

Proposition 1.3.6.

(a). $|z|^2 = zz^*$ *for all* $z \in \mathbb{C}$.

(b). For all $z_1, z_2 \in \mathbb{C}$,

$$|z_1 z_2| = |z_1|\,|z_2|.$$

Proof. Part (i) is a re-statement of part (i) of Theorem 1.3.4. Using part (i) and commutativity of multiplication, we can obtain part (ii) by

$$|z_1 z_2|^2 = (z_1 z_2)(z_1 z_2)^* = z_1 z_2 z_1^* z_2^* = (z_1 z_1^*)(z_2 z_2^*) = |z_1|^2 |z_2|^2.$$

(LEAN proof) □

Substituting $z_1 = a + bi$ and $z_2 = c + di$ into part (ii) yields the classical identity

$$(ac - bd)^2 + (ad + bc)^2 = (a^2 + b^2)(c^2 + d^2). \tag{1.3.3}$$

The Euclidean distance between two complex numbers can be expressed in terms of the complex absolute value:

$$d(z_1, z_2) \triangleq |z_1 - z_2|.$$

It is easy to check that d is nonnegative, symmetric, and vanishes only when $z_1 = z_2$. The remaining property is the triangle inequality.

Proposition 1.3.7 (Triangle inequality)**.** *For any* $z_1, z_2 \in \mathbb{C}$,

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

Although the triangle inequality is geometrically clear, we give an algebraic proof to show its compatibility with complex arithmetic.

Proof. Compute the square of the left-hand side:

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(z_1^* + z_2^*) \\ &= |z_1|^2 + 2\operatorname{Re}(z_1 z_2^*) + |z_2|^2 \\ &\leq |z_1|^2 + 2|\operatorname{Re}(z_1 z_2^*)| + |z_2|^2. \end{aligned} \tag{1.3.4}$$

Since $|\operatorname{Re}(w)| \leq |w|$ for any complex number w,

$$|\operatorname{Re}(z_1 z_2^*)| \leq |z_1 z_2^*| = |z_1|\,|z_2|. \tag{1.3.5}$$

Substituting (1.3.5) into (1.3.4) gives

$$|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2.$$

The proof is complete after taken square of both sides. (LEAN proof) □

1.4 Polar form and DeMoivre formula

A nonzero complex number, like a vector in $\mathbb{R}^2$, can be written in *polar form*. Given any nonzero complex number $x + iy$, we may write

$$x + iy = r(\cos\theta + i\sin\theta),$$

where $r > 0$ and θ is a real number. The number r is the modulus of $x + iy$, representing the length of the corresponding vector, and θ is the angle from the positive real axis to that vector (Fig. 1.5). In contrast to polar form, we will call $x + iy$ the Cartesian form of a complex number.

Using the matrix representation of complex numbers (Section 1.2.2), the 2×2 matrix corresponding to $a + bi$ can be written as

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = r \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

The matrix

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

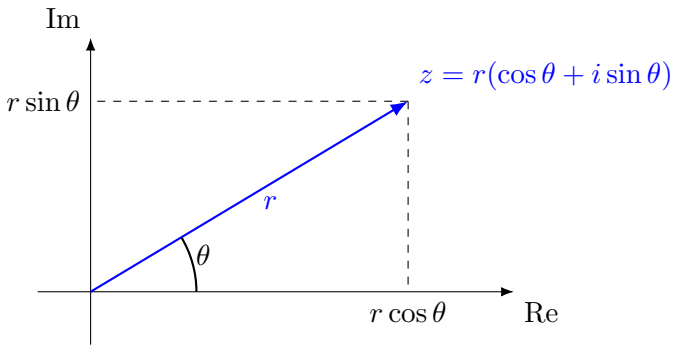


Figure 1.5: Polar form diagram

is the standard rotation matrix, representing counterclockwise rotation by θ.

Proposition 1.4.1. *Let*

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1), \qquad z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

be two complex numbers in polar form. Then

$$z_1 z_2 = r_1 r_2\big[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big].$$

Proof. Using the definition of complex multiplication and the angle addition formulas,

$$\begin{aligned} z_1 z_2 &= r_1 r_2(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \\ &= r_1 r_2\big[\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i(\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2)\big] \\ &= r_1 r_2\big[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big]. \end{aligned}$$

□

Thus, multiplication by a complex number

$$a + bi = r(\cos\theta + i\sin\theta)$$

has a clear geometric interpretation: (i) rotate by θ counterclockwise, and (ii) scale by a factor of r. In particular, multiplication by i corresponds to a $90°$ counterclockwise rotation.

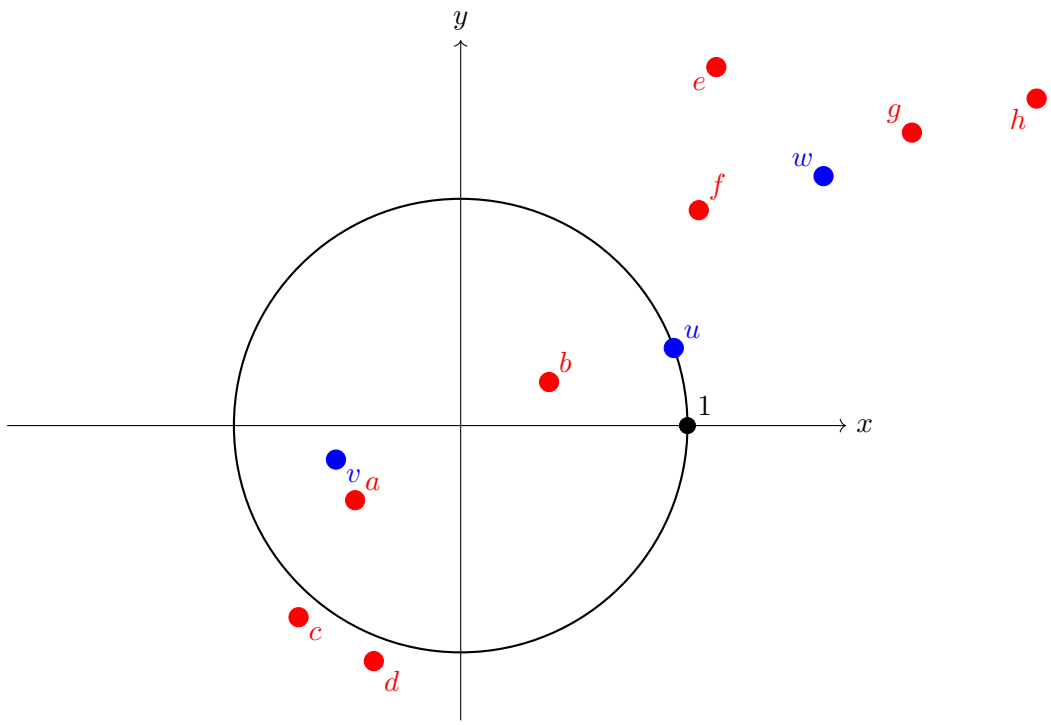


Figure 1.6: Example 1.4.1. The points a to h are obtained by adding or multiplying some combinations of u, v, and w.

Example 1.4.1. This example is adapted from [StewartTall]. In Fig. 1.6, the circle has radius 1 and is centered at the origin. The points u, v, and w have modulus 1, less than 1, and greater than 1, respectively.

The sums and products of various combinations of u, v, and w are shown as points a through h. Locate the points

$$u+v, \quad u+w, \quad v+w, \quad u+v+w,$$

$$uv, \quad uw, \quad vw, \quad uvw.$$

Definition 1.4.2. The *argument* of a nonzero complex number z is the angle from the positive real axis to the line segment joining 0 to z. We denote it by $\arg(z)$. The argument is also called the *phase* or *angle* of z.

The argument of a complex number is defined only up to integer multiples of 2π, since rotating by 2π radians returns to the same point. The argument of 0 is not defined.

Example 1.4.2. The number -1 can be written in polar form as

$$-1 = 1 \cdot (\cos(\pi + 2\pi k) + i \sin(\pi + 2\pi k)),$$

where $k \in \mathbb{Z}$. Thus

$$\arg(-1) = \pi + 2\pi k, \qquad k \in \mathbb{Z}.$$

The function $\arg(z)$ is a *multi-valued* function, meaning that there are more than one possible values for $\arg(z)$. To be more rigorous, we may interpret the argument function as a *set-valued function*. For instance, we may define the $\arg(-1)$ as the set $\{\pi + 2\pi k : k \in \mathbb{Z}\}$. However, for simplicity, we will just write $\pi + 2\pi k$, and understand that the variable k can assume any integral value.

To remove the ambiguity in the argument, one often restricts θ to a specific interval. Common choices are (i) $(-\pi, \pi]$, or (ii) $[0, 2\pi)$. The unique value of the argument lying in the chosen interval is called the *principal argument*. This choice is purely conventional and serves to simplify notation.

Definition 1.4.3. If we choose the interval $(-\pi, \pi]$ as the range, then for any nonzero complex number z the *principal argument* θ_0 is the unique angle in $(-\pi, \pi]$ satisfying

$$\cos\theta_0 = \frac{\mathrm{Re}(z)}{|z|}, \qquad \sin\theta_0 = \frac{\mathrm{Im}(z)}{|z|}.$$

We write $\mathrm{Arg}(z)$ for the principal argument.

The choice of the interval $(-\pi, \pi]$ as the range of $\mathrm{Arg}(z)$ agrees with convention of the argument function in Python.

Example 1.4.3. If we take $(\pi, \pi]$ as the range, the principal argument of $-1 - i$ is $-3\pi/4$.

The DeMoivre formula describes how the argument of a complex number behaves when the number is raised to an integer power.

Theorem 1.4.4 (DeMoivre formula)**.** *For any $n \in \mathbb{Z}$ and $\theta \in \mathbb{R}$,*

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta). \tag{1.4.1}$$

Proof. The formula is trivial when $n = 0$ and $n = 1$. Assume it holds for some positive integer n. Then

$$\begin{aligned}
(\cos\theta + i\sin\theta)^{n+1} &= (\cos\theta + i\sin\theta)^n(\cos\theta + i\sin\theta) \\
&= (\cos n\theta + i\sin n\theta)(\cos\theta + i\sin\theta) \\
&= (\cos n\theta\cos\theta - \sin n\theta\sin\theta) \\
&\qquad + i(\cos n\theta\sin\theta + \sin n\theta\cos\theta) \\
&= \cos((n+1)\theta) + i\sin((n+1)\theta).
\end{aligned}$$

Thus the formula holds for all nonnegative integers.

For negative n, write $n = -m$ with $m > 0$. Then

$$\begin{aligned}
(\cos\theta + i\sin\theta)^n &= \frac{1}{(\cos\theta + i\sin\theta)^m} \\
&= \frac{1}{\cos m\theta + i\sin m\theta} \cdot \frac{\cos m\theta - i\sin m\theta}{\cos m\theta - i\sin m\theta} \\
&= \cos m\theta - i\sin m\theta \\
&= \cos(n\theta) + i\sin(n\theta).
\end{aligned}$$

This proves the formula for all integers n. (LEAN proof) □

Using the matrix representation of complex numbers, DeMoivre's formula becomes

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^n = \begin{bmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{bmatrix}.$$

Geometrically, this states that rotating by an angle n times is equivalent to a single rotation by n times the angle.

Remark. Because the expression "$\cos\theta + i\sin\theta$" occurs very frequently, some people use the short-hand notation

$$cis(\theta) \triangleq \cos\theta + i\sin\theta.$$

Once we have proved Euler's identity, we will write it as $e^{i\theta}$.

Example 1.4.4. Compute $(-1+i\sqrt{3})^8$.

The number $-1+i\sqrt{3}$ has polar form

$$2\Big(-\tfrac{1}{2} + i\tfrac{\sqrt{3}}{2}\Big) = 2(\cos(2\pi/3) + i\sin(2\pi/3)).$$

Thus

$$(-1+i\sqrt{3})^8 = 2^8(\cos(8\cdot 2\pi/3) + i\sin(8\cdot 2\pi/3)) = 256(\cos(4\pi/3) + i\sin(4\pi/3)).$$

In Cartesian form, this is

$$128(-1-i\sqrt{3}).$$

Example 1.4.5. Compute $(-1+i)^{20}$.

We write

$$-1+i = \sqrt{2}\,(\cos(3\pi/4) + i\sin(3\pi/4)).$$

Then

$$\begin{aligned}(-1+i)^{20} &= (\sqrt{2})^{20}\big(\cos(20\cdot 3\pi/4) + i\sin(20\cdot 3\pi/4)\big)\\ &= 2^{10}(\cos\pi + i\sin\pi)\\ &= -1024.\end{aligned}$$

Example 1.4.6. Express $\sin(5\theta)$ as a polynomial in $\sin\theta$.

Using DeMoivre's formula,

$$\begin{aligned}\cos(5\theta) + i\sin(5\theta) &= (\cos\theta + i\sin\theta)^5\\ &= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta\\ &\quad - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta.\end{aligned}$$

Taking imaginary parts,

$$\begin{aligned}\sin(5\theta) &= 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta \\ &= 5(1-\sin^2\theta)^2\sin\theta - 10(1-\sin^2\theta)\sin^3\theta + \sin^5\theta \\ &= 5\sin\theta - 20\sin^3\theta + 16\sin^5\theta.\end{aligned}$$

We can use DeMoivre's formula to extract the n-th roots of a complex number. Let $w \neq 0$ be written in polar form

$$w = |w|(\cos(\operatorname{Arg} w) + i\sin(\operatorname{Arg} w)).$$

We seek all z such that $z^n = w$. Write $z = r(\cos\theta + i\sin\theta)$. Then

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta)) = w.$$

Thus

$$r = |w|^{1/n}, \qquad \theta = \frac{\operatorname{Arg} w + 2k\pi}{n}, \quad k = 0, 1, \ldots, n-1.$$

Therefore the n-th roots of w are

$$|w|^{1/n}\cos\left(\frac{\operatorname{Arg} w + 2k\pi}{n}\right) + i\,|w|^{1/n}\sin\left(\frac{\operatorname{Arg} w + 2k\pi}{n}\right), \tag{1.4.2}$$

for $k = 0, 1, \ldots, n-1$. The n-th root of a nonzero complex number w are the vertices of a regular n-gon in the complex plane.

The case of the n-th roots of unity is of special importance.

Definition 1.4.5. Given a positive integer n, a complex number ζ is called an *n-th root of unity* if $\zeta^n = 1$.

An example of seventh roots of unity is shown in Fig. 1.7.

We can express the complex n-th roots of any nonzero complex number w using n-th roots of unity. Suppose z_0 is one of the n-th roots of w. If z_1 is another n-th root of w, then the ratio z_0/z_1 satisfies

$$\left(\frac{z_0}{z_1}\right)^n = \frac{z_0^n}{z_1^n} = \frac{w}{w} = 1.$$

Hence, z_0/z_1 is a root of unity. We can write z_1 as

$$z_1 = z_0\zeta$$

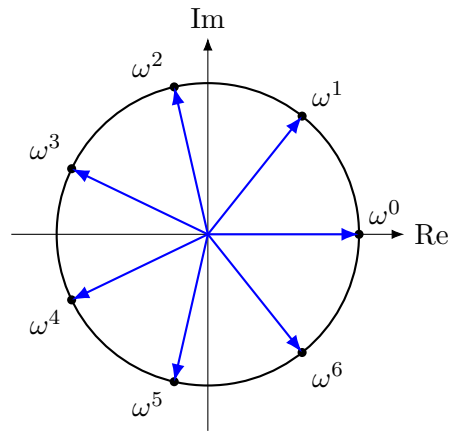


Figure 1.7: The powers of $\omega = (\cos(2\pi/7) + i\sin(2\pi/7))$ are the seventh roots of unity.

for some n-th root of unity ζ. Conversely if ζ is an n-th root of unity, $z_0\zeta$ is also an n-th root of unity. Therefore, once we have obtain one particular n-th root z_0 of w, we can obtain the complete set of the n-th roots of w:

$$z_0,\ z_0\zeta,\ z_0\zeta^2,\ \ldots,\ z_0\zeta^{n-1}.$$

Example 1.4.7. Let n be a positive integer. To find all the n-th root of a positive real number a, we first note that $\sqrt[n]{a}$ is one of the solutions to $z^n = a$. The n distinct roots are characterized by $\sqrt[n]{a}\zeta^k$, where $\zeta = \cos(2\pi/n) + \sin(2\pi/n)$, and $k = 0, 1, 2, \ldots, n-1$.

1.5 Applications in plane geometry

1.5.1 A criterion of collinear points

We start with the simple but useful observation that

$$\arg(z_1/z_2) = \arg(z_1) - \arg(z_2) \bmod 2\pi.$$

This device allows us to compute the angle between two line segments in terms of the argument function.

We can test whether three points z_1, z_2 and z_3 lies on the same line in terms of complex operations, without going to the real domain. The idea is: if we divide $z_1 - z_2$ by $z_3 - z_2$,

the argument of the quotient is the angle between the line segment between z_1 and z_2, and the line segment between z_3 and z_2. They are collinear if and only if the angle is 0 or π,

$$\arg\left(\frac{z_1 - z_2}{z_3 - z_2}\right) = 0 \text{ or } \pi.$$

Equivalently, this condition can be written as

$$\operatorname{Im}\left(\frac{z_1 - z_2}{z_3 - z_2}\right) = 0. \tag{1.5.1}$$

1.5.2 Points lying on an equilateral triangle

We can derive a condition to test whether three complex points form an equilateral triangle. If z_1, z_2 and z_3 are the vertices of an equilateral triangle, then the line segments $z_1 - z_2$ and $z_3 - z_2$ must have the same length, and the angle between them is equal to $\pm\pi/3$. Hence,

$$\frac{z_1 - z_2}{z_3 - z_2} = \cos(\pm\pi/3) + i\sin(\pm\pi/3) = \frac{1 \pm \sqrt{3}i}{2}.$$

This condition is equivalent to

$$\begin{aligned} \frac{z_1 - z_2}{z_3 - z_2} - \frac{1}{2} &= \frac{\pm\sqrt{3}i}{2} \\ \Leftrightarrow \quad (2z_1 - z_2 - z_3) &= \pm\sqrt{3}i(z_3 - z_2) \\ \Leftrightarrow \quad (2z_1 - z_2 - z_3)^2 &= -3(z_3 - z_2)^2 \\ \Leftrightarrow \quad z_1^2 + z_2^2 + z_3^2 &= z_1z_2 + z_1z_3 + z_2z_3. \end{aligned}$$

This gives a concise criterion to determine whether z_1, z_2, and z_3 form an equilateral triangle.

1.5.3 A proof of Napoleon's theorem

Draw an arbitrary polygon with 4 sides. Label the vertices by A, B, C and D. On each side, we draw a square with this side as the base. Label the midpoint of the four squares by P, Q, R and S. Show that the line PR is perpendicular to the line QS. (See Fig. 1.8)

We can use complex numbers to solve this puzzle. More precisely, we use the fact that multiplication by i means rotation by 90 degrees counter-clockwise. Indeed, the matrix representation of i is $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, and the multiplication by this matrix from the left is the same as rotation by 90 degrees.

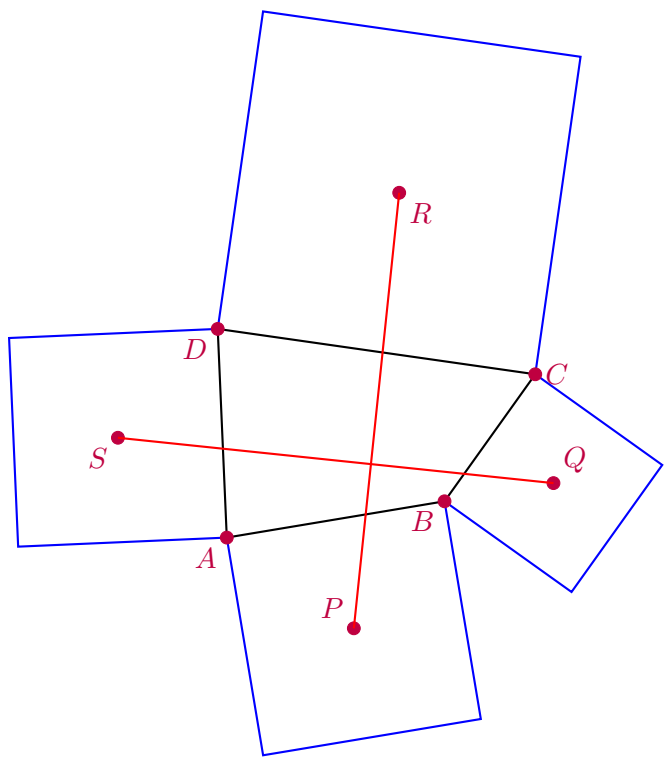


Figure 1.8: Napoleon's theorem for quadrilateral: the lines PR and QS are perpendicular.

We first represent each points in the question by complex numbers. Suppose the point A is represented by complex number z_A, point B by z_B, etc. The complex number z_P is related to z_A and z_B through the following equation

$$z_P = \frac{z_A + z_B}{2} + i\frac{z_A - z_B}{2}.$$

The first term $(z_A - z_B)/2$ is the midpoint of the line segment AB. The second term is the vector from the midpoint of AB to the point P.

By the same argument, we can express z_Q, z_R and z_S in terms of the four vertices of the quadrilateral,

$$\begin{aligned} z_Q &= \frac{z_B + z_C}{2} + i\frac{z_B - z_C}{2}, \\ z_R &= \frac{z_C + z_D}{2} + i\frac{z_C - z_D}{2}, \\ z_S &= \frac{z_D + z_A}{2} + i\frac{z_D - z_A}{2}. \end{aligned}$$

After some simplification, we get

$$\begin{aligned} z_P &= \frac{1}{2}\Big[(1+i)z_A + (1-i)z_B\Big], \\ z_Q &= \frac{1}{2}\Big[(1+i)z_B + (1-i)z_C\Big], \\ z_R &= \frac{1}{2}\Big[(1+i)z_C + (1-i)z_D\Big], \\ z_S &= \frac{1}{2}\Big[(1+i)z_D + (1-i)z_A\Big]. \end{aligned}$$

The two line segments PR and QS are represented by $z_P - z_R$ and $z_Q - z_S$, respectively,

$$\begin{aligned} z_P - z_R &= \frac{1}{2}\Big[(1+i)(z_A - z_C) + (1-i)(z_B - z_D)\Big] \\ z_Q - z_S &= \frac{1}{2}\Big[(1+i)(z_B - z_D) + (1-i)(z_C - z_A)\Big]. \end{aligned}$$

We can now verify that

$$\begin{aligned} i(z_P - z_R) &= \frac{1}{2}\Big[i(1+i)(z_A - z_C) + i(1-i)(z_B - z_D)\Big] \\ &= \frac{1}{2}\Big[(i-1)(z_A - z_C) + (i+1)(z_B - z_D)\Big] \\ &= \frac{1}{2}\Big[(1-i)(z_C - z_A) + (1+i)(z_B - z_D)\Big] \\ &= z_Q - z_S. \end{aligned}$$

This proves not only that PR is perpendicular to QS, it also proves that they have the same length.

1.6 Appendix: Solving cubic equation using discrete Fourier transform

We outline a method of finding the roots of a polynomial of degree 3

$$ax^3 + bx^2 + cx + d$$

using discrete Fourier transform (DFT). In classical literature, this is called the method of *Lagrange resolvent*.

The first ingredient in this method is the DFT of order 3. Let $\omega = (1/2) + i\sqrt{3}/2$ be a cube root of unity. We will use the following property

$$1 + \omega + \omega^2 = 1$$

about cube root of unity. Let W be the 3×3 matrix

$$W \triangleq \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$$

The discrete Fourier transform of order 3 is the linear transformation that takes a column vector $\mathbf{v} = \begin{bmatrix} z_1 & z_2 & z_3 \end{bmatrix}^T$ as input and return $W\mathbf{v}$. We can show that

$$W^H W = 3\mathbf{I}_{3\times 3},$$

where W^H denotes the conjugate transpose of matrix W (Exercise 14)). The inverse transform is thus given by

$$W^{-1} = \frac{1}{3} W^H.$$

The second idea is to simplify the cubic polynomial by reducing the coefficients of x^2 to 0. We first assume without loss of generality that the leading coefficient is 1 and consider the a cubic equation in the form

$$x^3 + bx^2 + cx + d = 0.$$

By letting $u = x + b/3$, we can transform this equation to

$$u^3 + \big(c - \frac{b^2}{3}\big)u + \big(d - \frac{cb}{3} + \frac{2b^3}{27}\big) = 0. \tag{1.6.1}$$

Let α, β and γ be the roots of (2.7.1).

We re-define c and d by

$$\begin{aligned} c &\leftarrow c - b^2/3 \\ d &\leftarrow d - cb/3 + 2b^3/27, \end{aligned}$$

so that

$$(u - \alpha)(u - \beta)(u - \gamma) = u^3 + cu + d.$$

By taking the inverse DFT, we can find y_0, y_1 and y_2 such that

$$\begin{aligned} \alpha &= y_0 + y_1 + y_2 \\ \beta &= y_0 + y_1\omega + y_2\omega^2 \\ \gamma &= y_0 + y_1\omega^2 + y_2\omega. \end{aligned}$$

Because the sum of roots $\alpha + \beta + \gamma$ is zero, it is guaranteed that $y_0 = 0$. Hence, we can write

$$\alpha = y_1 + y_2 \tag{1.6.2}$$

$$\beta = y_1\omega + y_2\omega^2 \tag{1.6.3}$$

$$\gamma = y_1\omega^2 + y_2\omega \tag{1.6.4}$$

and our task is to solve for y_1 and y_2.

We derive the relationship between y_1, y_2, c and d by

$$\begin{aligned} u^3 + cu + d &= (u - \alpha)(u - \beta)(u - \gamma) \\ &= (u - y_1 - y_2)(u - y_1\omega - y_2\omega^2)(u - y_1\omega^2 - y_2\omega) \\ &= u^3 + \big[(y_1 + y_2)(y_1\omega + y_2\omega^2) + (y_1 + y_2)(y_1\omega^2 + y_2\omega) + \\ &\quad (y_1\omega + y_2\omega^2)(y_1\omega^2 + y_2\omega)\big]u + (y_1 + y_2)(y_1\omega + y_2\omega^2)(y_1\omega^2 + y_2\omega). \end{aligned}$$

By comparing the coefficients on both sides, after some simplification, we obtain

$$\begin{aligned} c &= -3y_1y_2 \\ d &= -(y_1^3 + y_2^3). \end{aligned}$$

We can solve for y_1^3 and y_2^3 by solving the following quadratic equation

$$(y - y_1^3)(y - y_2^3) = y^2 + dy - \frac{c^3}{27}.$$

The solutions are

$$\frac{1}{2}\big(-d \pm \sqrt{d^2 + 4c^3/27}\big).$$

We can let y_1 to be a cube root of one of the above solutions,

$$y_1 := \Big(\frac{1}{2}\big(-d \pm \sqrt{d^2 + 4c^3/27}\big)\Big)^{1/3},$$

and let y_2 be the complex number such that $c = -3y_1y_2$. (If y_1 is zero, then we cannot calculate y_2 by $-c/(3y_1)$ and we need to switch the role of y_1 and y_2, or assign another value to y_1.)

Once we know y_1 and y_2, we can compute α, β and γ by equations (7.8.3) to (1.6.4). This can be done in one step by calling the DFT function. Then, the final solution is obtained by subtracting $b/3$.

1.7 Appendix: Complex numbers in Python

To emphasize the computation aspect of complex numbers, we describe how to calculate complex numbers in Python. Complex number is a built-in type in Python. The imaginary unit is represented by the symbol j.

For further references see Sample Python code

```
# Basic arithemtic of complex numbers
z1 = 2 + 3j
z2 = -1 + 0.2j

print(f"The sum of {z1} and {z2} is {z1+z2}.")
print(f"The difference of {z1} and {z2} is {z1-z2}.")
print(f"The product of {z1} and {z2} is {z1*z2:.3}.")
print(f"The quotient of {z1} and {z2} is {z1/z2:.3}.")
```

The program output is:

```
The sum of (2+3j) and (-1+0.2j) is (1+3.2j).
The difference of (2+3j) and (-1+0.2j) is (3+2.8j).
```

```
The product of (2+3j) and (-1+0.2j) is (-2.6-2.6j).
The quotient of (2+3j) and (-1+0.2j) is (-1.35-3.27j).
```

Commands for the real part, imaginary part and conjugate is part of the data structure.

```
# Real part, imaginary part, and conjugate

z = 3.2-5.3j
print(f"Real Part of {z} = {z.real}")
print(f"Imaginary Part of {z} = {z.imag}")
print(f"Conjugate of {z} = {z.conjugate()}")
```

If we run this program, we get

```
Real Part of (3.2-5.3j) = 3.2
Imaginary Part of (3.2-5.3j) = -5.3
Conjugate of (3.2-5.3j) = (3.2+5.3j)
```

The argument and modulus are called absolute value and phase in Python.

```
# Absolute value and phase
from cmath import phase
from numpy import abs

z = 12+34j
print (f"z = {z}")
print (f"Argument of z = {phase(z):.4} (rad)")
print (f"Magnitude of z = {abs(z):.4}")
```

The command phase and abs are imported from library cmath and numpy, respectively. The returned value of phase() is in radian.

```
z = (12+34j)
Argument of z = 1.232 (rad)
Magnitude of z = 36.06
```

We can check Proposition 1.4.1 by the Python program below.

```
from cmath import phase
from numpy import abs, sin, cos
```

```
from math import isclose

z1 = 12+34j
z2 = 3-6j
r1 = abs(z1)   # absolute value of z1
r2 = abs(z2)   # absolute value of z2
theta1 = phase(z1)   # argument of theta1
theta2 = phase(z2)   # argument of theta2
print (f"z1 = {z1}, z2 = {z2}")
print (f"z1*z2 = {z1*z2}")   # compute the product directly

w = r1*r2*(cos(theta1+theta2) + sin(theta1+theta2)*1j)
assert isclose(abs(w- z1*z2), 0, abs_tol=1e-8)
```

The variables z1 and z2 can be any complex numbers. The result of z1*z2 and the value of w are the same, up to numerical error.

```
z1 = (12+34j), z2 = (3-6j)
z1*z2 = (240+30j)
```

In Python, we can compute the the n-th root of a complex number by the command z**(1/n). Raising to a rational power m/n can be computed by z**(m/n), for integers m and n. Python will return one of the n possible answers. Compare the following calculations in Python.

```
# Computation of roots of complex numbers in Python
z = -1+1j
w1 = (z**3)**(1/4) # raise to the power 3 and then take 4th root
w2 = z**(3/4)  # raise to the power 3/4
print(f"w1 = {w1:.5}")
print(f"w2 = {w2:.5}")

print(f"\nThey are both fourth roots of {z*z*z}.")
print(f"w1^4={w1**4:.5}")
print(f"w2^4={w2**4:.5}")
```

The number w1 is obtained by first raising z to the third power and then taking the fourth root. The number w2 is computed by directly raising z to the power $3/4$. Mathematically, they are supposed to be the same, but the actually results are not the same. It is because

there are four possible answers when we take the fourth root. The fourth power of both of them are equal to $(-1+i)^3 = 2+2i$.

```
w1 = (1.2719+0.253j)
w2 = (-0.253+1.2719j)

They are both fourth roots of (2+2j).
w1^4=(2+2j)
w2^4=(2+2j)
```

Using complex numbers, we can implement the condition of collinearity in (1.5.1) by Python.

```
# Determine whether three points lie on a straight line
# Input three tuples:  pt1= (x1,y1), pt2= (x2,y2), pt3= (x3,y3)
# return True if the three points pt1, pt2, pt3 are collienar
def is_collinear(pt1, pt2, pt3):
  z1 = complex(pt1[0],pt1[1])   # convert to complex numbers
  z2 = complex(pt2[0],pt2[1])
  z3 = complex(pt3[0],pt3[1])
  return abs(((z1-z2)/(z3-z2)).imag) <= 1e-8
  # test whether the imaginary part equals 0, up to numerical error
```

Samples are shown below. We checked that the points $(1,2)$, $(2,3)$, and $(3,4)$ are collinear, but $(1,2)$, $(2,3)$, and $(3,3)$ are not collinear.

```
In [1]: is_collinear((1,2),(2,3),(3,4))
Out[1]: True

In [2]: is_collinear((1,2),(2,3),(3,3))
Out[2]: False
```

Summary

- A complex number $a+bi$ can be represented by 2×2 matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$. When the determinant a^2+b^2 is equal to 1, this is a rotation matrix.
- Geometric meaning of the operations of complex numbers:

- Addition of complex numbers is vector addition.
- Multiplying by a complex number is the same as rotation followed by an expansion.
- Complex conjugate is the mirror image with respect to the real axis.
- The modulus of a complex number is the distance to the origin.

- $\mathrm{Re}(z) = (z + z^*)/2$, $\mathrm{Im}(z) = (z - z^*)/(2i)$, for all $z \in \mathbb{C}$.
- Complex conjugate respects addition, multiplication, and taking inverse: $(z_1 + z_2)^* = z_1^* + z_2^*$, $(z_1 z_2)^* = z_1^* z_2^*$, $(1/z)^* = 1/(z^*)$.
- $|z|^2 = zz^*$ for all $z \in \mathbb{C}$.
- $|z_1 z_2| = |z_1|\,|z_2|$, for any $z_1, z_2 \in \mathbb{C}$.
- Triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$ for all $z_1, z_2 \in \mathbb{C}$.
- DeMoivre formula: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$. for $n \in \mathbb{Z}$ and $\theta \in \mathbb{R}$.
- A complex number z is a called an n-th root of unity if $z^n = 1$.

Exercises

1. Write the following complex numbers in Cartesian form

   (a) $(2 - i)(3 + 2i)$, (b) $\dfrac{(4+i)^2}{i^3}$, (c) $\dfrac{1 + 2i + 3i^2}{(3+4i)^2}$,

   (d) $(1 + i)^{1024}$, (e) $\dfrac{1}{i + \frac{1}{i + \frac{1}{i+1}}}$, (f) $\left(1 + \dfrac{i}{5}\right)^5$.

2. Solve the following linear equations.

   (a) $(2 + i)z = (-1 + i/2)$, (b) $3 - iz = 2z - i$.

3. Explain what is wrong in the proof of $1 = -1$:

$$1 = \sqrt{1} = \sqrt{(-1)(-1)} = \sqrt{-1}\sqrt{-1} = i \cdot i = -1$$

4. By writing a complex number as a pair (x, y) and following the computation rules in Section 1.2.1, find all solutions to the following quadratic equation,

$$z^2 + 4iz - 3 = 0.$$

5. Let $\mathrm{Arg}(z)$ denote the principal argument of nonzero complex number z. Determine whether the following equations hold for all nonzero complex numbers z_1 and z_2. If yes, provide a proof, otherwise, give a counter-example.

   (a) $\mathrm{Arg}(z_1) + \mathrm{Arg}(z_2) = \mathrm{Arg}(z_1 z_2)$.

   (b) $\mathrm{Arg}(z_1) - \mathrm{Arg}(z_2) = \mathrm{Arg}(z_1/z_2)$.

6. Determine the value of z that satisfies the system of equations:

$$\begin{cases} 2z + 4\bar{z} = -2, \\ z - 2\bar{z} = 1. \end{cases}$$

7. Solve the following equations.

   (a) $z^2 = -i$.

   (b) $z^2 = 1 + \sqrt{3}i$.

   (c) $z^8 = 2 - 2i$.

8. Use DeMoivre's formula to derive

   (a) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.

   (b) $\sin 3\theta = -4\sin^3\theta + 3\sin\theta$.

   (c) $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$.

   (d) $\sin 4\theta = 4\cos\theta(\sin\theta - 2\sin^3\theta)$.

9. Suppose z lie on the unit circle. Show that for any complex number w that is not equal to z, we have

$$\left|\frac{w - z}{w\bar{z} - 1}\right| = 1.$$

10. Prove the reverse triangle inequality for complex numbers:

$$\big||z_1| - |z_2|\big| \leq |z_1 - z_2|,$$

for $z_1, z_2 \in \mathbb{C}$.

11. Show that the following identity holds for any nonzero complex number z and positive integer n.

$$\begin{aligned}&(z - z^{-1})(z^{2n} + z^{2n-2} + \cdots + z^4 + z^2 + 1 + z^{-2} + z^{-4} + \cdots + z^{-2n+2} + z^{-2n})\\ &= z^{2n+1} - z^{-2n-1}.\end{aligned}$$

Substitute z by $\cos\theta + i\sin\theta$ and show that

$$\frac{\sin(2n+1)\theta}{\sin\theta} = 1 + 2\cos 2\theta + 2\cos 4\theta + \cdots + 2\cos 2n\theta.$$

12. (a) Find all the square roots of $3i$.
    (b) Find all the fourth roots of $-1 + i$.
    (c) Find all the sixth roots of unity.

13. (Complex matrices) Many theorems in linear algebra can be formulated over the complex field. A major difference with the real vectors and matrices is that the transpose operator must be replaced by the Hermitian operator, and the dot product of two column vectors $\mathbf{v}$ and $\mathbf{u}$ with complex components is defined by $\mathbf{v}^H \cdot \mathbf{u}$.

Definition. The *conjugate* of a vector or matrix with complex entries is computed by taking the complex conjugate of each entry in the vector or matrix. The *Hermitian* of a matrix is defined as the conjugate of the transpose of the matrix. The Hermitian of a matrix M is denoted by M^H.

For example, the Hermitian of $\begin{bmatrix} i & 2-i \\ -3 & 4+3i \end{bmatrix}$ is $\begin{bmatrix} -i & -3 \\ 2+i & 4-3i \end{bmatrix}$.

Definition. An $n \times n$ complex matrix A is called

(a) *Hermitian* if $A = A^H$,
(b) *skew-Hermitian* if $A = -A^H$,
(c) *unitary* if $A \cdot A^H = I$,

where I is the $n \times n$ identity matrix.

Show that

(a) the eigenvalues of Hermitian matrix is real,

(b) the eigenvalues of skew-Hermitian matrix is purely imaginary (the eigenvalue is in the form αi for some $\alpha \in \mathbb{R}$),

(c) the eigenvalues of unitary matrix has modulus 1.

14. (Discrete Fourier transform) Fix a positive integer n and let ω be an n-th root of unity defined by

$$\omega = \cos(2\pi/n) - i\sin(2\pi/n).$$

The complex number ω has modulus 1 and argument $-2\pi/n$. Define an $n \times n$ matrix

$$W_n \triangleq \begin{bmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \cdots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \omega^{3(n-1)} & \cdots & \omega^{(n-1)^2} \end{bmatrix}.$$

The (i,j)-entry of W_n is $\omega^{(i-1)(j-1)}$, for $i, j = 1, 2, \ldots, n$.

Given a column vector $\mathbf{v}$ of length n (whose entries can be real or complex), the discrete Fourier transform (DFT) $\hat{\mathbf{v}}$ of $\mathbf{v}$ is computed by a matrix multiplication

$$\hat{\mathbf{v}} \triangleq W_n \mathbf{v}.$$

(a) Write down the the matrix W_n for $n = 3$. Find the DFT of $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

(b) Let W_n^H denote the conjugate transpose of W_n. Show that

$$\frac{1}{n} W_n^H W_n = \frac{1}{n} I.$$

Hint: for any integers a and b between 0 and $n-1$, show that

$$\frac{1}{n} \sum_{k=0}^{n-1} \omega^{ak} (\omega^{bk})^* = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b. \end{cases}$$

15. (Quaternions as 2×2 complex matrices) In this exercise we construct the quaternions as complex matrices. Let $M_2(\mathbb{C})$ denote the set of 2×2 matrices over the complex numbers. Consider the set

$$\mathcal{A} = \left\{ \begin{pmatrix} z & w \\ -\bar{w} & \bar{z} \end{pmatrix} : z, w \in \mathbb{C} \right\} \subseteq M_2(\mathbb{C}).$$

(a) Show that $\mathcal{A}$ is closed under matrix addition and multiplication, and contains the identity matrix. Conclude that $\mathcal{A}$ is a subalgebra of $M_2(\mathbb{C})$.

(b) Define the matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{i} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \mathbf{j} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \mathbf{k} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Verify that all four matrices lie in $\mathcal{A}$.

(c) Compute the products

$$\mathbf{i}^2, \qquad \mathbf{j}^2, \qquad \mathbf{k}^2, \qquad \mathbf{ij}, \qquad \mathbf{ji},$$

and show that

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -I, \qquad \mathbf{ij} = \mathbf{k}, \qquad \mathbf{ji} = -\mathbf{k}.$$

(d) Prove that every element of $\mathcal{A}$ can be written uniquely in the form

$$aI + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}, \qquad a, b, c, d \in \mathbb{R}.$$

(e) Conclude that $\mathcal{A}$ is a 4-dimensional real algebra with basis I, $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ satisfying the multiplication rules

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -I, \quad \mathbf{ij} = \mathbf{k}, \quad \mathbf{jk} = \mathbf{i}, \quad \mathbf{ki} = \mathbf{j}, \quad \mathbf{ji} = -\mathbf{k}, \quad \mathbf{kj} = -\mathbf{i}, \quad \mathbf{ik} = -\mathbf{j}.$$

This algebra is (by definition) isomorphic to the quaternion algebra.

16. (Pirate and treasure chest) A pirate arrived at an island and he found three trees on the island: an alma fig tree (A), a breadfruit tree (B), and a coconut palm (C). The pirate buried a treasure chest according to the following rule:

(a) Starting at tree A, he walked to tree B, counting the number of steps. He then turned $90°$ to the left and walked the same number of steps from tree A. He placed a stake in the ground at that point.

(b) Starting again at tree A, he walked to tree C, counting the number of steps. He then turned $90°$ to the right and walked the same number of steps from tree A. He placed another stake at that point.

(c) The treasure is buried at the midpoint of the two stakes.

The pirate then removed the two stakes and left the island.

After some years, the pirate returned to the island and wants to dig up the treasure. However, tree A was gone. Could the pirate still find the exact location of the treasure?

(Optional: Suppose we modify the rule as follows. Fix a positive real number r. In steps (a) (resp. step b), after turning left (resp. right), we walk for a distance of r times the distance from tree A to tree B (resp. C). Can we locate the treasure without knowing the location of tree A?)

17. (Medians of a triangle) Use complex numbers to show that the three medians of a triangle intersects at the same point.

18. (Two-lighthouse problem) Two lighthouses stand at complex positions a and b. A ship at point z sees the angle between the two lighthouses as exactly $60°$.

    (a) Show that the condition can be written as

    $$\arg\left(\frac{z-a}{z-b}\right) = \pm\frac{\pi}{3}.$$

    (b) Deduce that the locus of such points z consists of two circular arcs.

    (c) Identify the centers and radii of these circles.

19. (Napoleon's triangle via complex numbers) Given a triangle ABC, construct equilateral triangles externally on each side. Let their outer vertices be A', B', and C'.

    (a) Using complex numbers, express A', B', and C' in terms of A, B, C.

    (b) Show that triangle $A'B'C'$ is equilateral.

20. (Area of triangle)

    (a) Consider a triangle in the complex plane with vertices 0, z_1 and z_2. Prove that the area of the triangle is equal to $\frac{1}{2}|\operatorname{Im}(z_2\bar{z}_1)|$

    (b) More generally, let z_1, z_2 and z_3 be the vertices of a triangle in the complex plane. Suppose that the paths $z_1 \to z_2 \to z_3 \to z_1$ is in the counter-clockwise direction. Prove that the area of the triangle can be computed by the determinant

$$\frac{i}{4}\begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ \bar{z}_1 & \bar{z}_2 & \bar{z}_3 \end{vmatrix}.$$

21. Implement the algorithm in Section 1.6 to calculate the roots of a cubic equation with complex coefficients.

Chapter 2

Linear Fractional Transformation and Riemann sphere

2.1 Complex inversion

In Euclidean geometry, *circle inversion* (a.k.a. geometric inversion or circular inversion) is the transformation

$$(x, y) \mapsto (\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2}).$$

The origin is excluded from the domain of this mapping because x^2+y^2 has no inverse when $x = y = 0$. Let the inversion be taken in the circle with center O and radius R. For a point $P \neq O$, its inverse of P is defined as the point P' such that

$$|OP| \cdot |OP'| = R^2.$$

In particular, every point on the circle is mapped to itself under the circle inversion. In terms of polar coordinates, the point $z = r(\cos\theta + i\sin\theta)$ is mapped to the point $(1/r)(\cos\theta + i\sin\theta)$. Although this transformation is nonlinear in coordinates, the point P' can be constructed using only straightedge and compass.

The circle inversion is closely related to complex division. Geometrically, the complex inversion function $f(z) = 1/z$ can be interpreted as a circle inversion function followed by a reflection along the x-axis. We can take the complex conjugate of $(1/r)(\cos\theta + i\sin\theta)$ to obtain

$$(1/r)(\cos\theta - i\sin\theta) = \frac{1}{r(\cos\theta + i\sin\theta)} = \frac{1}{z}.$$

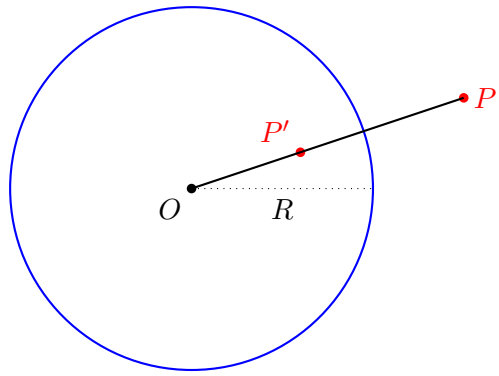


Figure 2.1: Circle inversion. The point P outside is mapped to the point P' inside the circle.

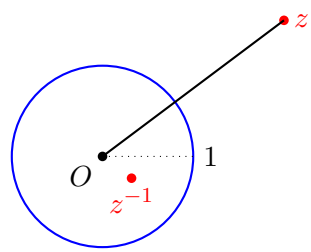


Figure 2.2: Complex inversion. The point z is mapped to the point z^{-1}.

Hence, the function $z \mapsto 1/z$ and the circle inversion with radius $R = 1$ differ only by a mirror reflection. We will see that the complex inversion map preserves orientation, but the circle inversion reverses orientation.

The inversion function has a very special property: it transforms circles and straight lines to circles and straight lines. Consider a quadratic equation in two variables,

$$A(x^2 + y^2) + Bx + Cy + D = 0, \tag{2.1.1}$$

where A, B, C and D are real numbers such that $A \neq 0$. By re-writing it as

$$x^2 + y^2 + \frac{B}{A}x + \frac{C}{A}y = -\frac{D}{A}$$
$$(x + \frac{B}{2A})^2 + (y + \frac{C}{2A})^2 = \frac{B^2 + C^2 - 4DA}{4A^2},$$

it is the equation of a circle if $(B^2 + C^2 - 4DA) > 0$.

When A is zero but B or C is nonzero, (2.1.1) is an equation of a straight line. When A, B, and C are all zero but D is nonzero, the solution set is empty. Finally, when all coefficients are zero, the the solution set is the whole x-y plane. Hence, in general, the equation in (2.1.1) defines a circle, a straight line, an empty set, or the plane.

Theorem 2.1.1. *The inversion function $f(z) = 1/z$ transforms circles and straight lines to circles and straight lines.*

Proof. Suppose the coefficients in (2.1.1) are not all zero, so that the solution set is a circle, a straight line or empty set. Represent x and y by complex variable $z = x + iy$, and re-write (2.1.1) as

$$Azz^* + B\frac{z + z^*}{2} + C\frac{z - z^*}{2i} + D = 0,$$

and simplify it to

$$azz^* + bz + b^*z^* + c = 0, \tag{2.1.2}$$

where $a = A$, $b = (B - iC)/2$, and $c = D$. The coefficients a and c are real numbers, and the coefficient b is a complex number. If we map z to $w = 1/z$, the equation becomes

$$a\frac{1}{ww^*} + b\frac{1}{w} + b^*\frac{1}{w^*} + c = 0,$$

which is the same as

$$a + bw^* + b^*w + cww^* = 0.$$

This has the same form as in (2.1.2), and hence it represents a circle or a straight line. □

Using complex numbers, We can specify a circle in the complex plane using the modulus. Given a point z_0 in the complex plane and a positive real number, the circle with radius r and center z_0 has equation

$$|z - z_0| = r.$$

Using the fact that $|z|^2 = zz^*$, we can write the equation in an alternate form:

$$(z - z_0)(z - z_0)^* = r^2.$$

If we write $z = x + iy$ and $z_0 = x_0 + iy_0$, the above equation becomes the usual equation of circle:

$$(x - x_0)^2 + (y - y_0)^2 = r^2.$$

A straight line in the x-y plane can be represented by equation

$$ax + by = c, \tag{2.1.3}$$

for some real numbers a, b and c. Using the relation $x = \mathrm{Re}(z) = (z + z^*)/2$ and $y = \mathrm{Im}(z) = (z - z^*)/(2i)$ in (1.3.2), we can write this equation as

$$a\frac{z + z^*}{2} + b\frac{z - z^*}{2i} = c.$$

We can simplify this equation to

$$\begin{aligned} ai(z + z^*) + b(z - z^*) &= 2ic \\ (b + ai)z - (b - ai)z^* &= 2ic. \end{aligned} \tag{2.1.4}$$

If we define $\alpha = b + ai$, then we can further simplify it to

$$\mathrm{Im}(\alpha z) = c. \tag{2.1.5}$$

Alternately, we can write the equation of a straight line using Re instead of Im. We let $\beta = a - ib$ in (2.1.4), we obtain

$$\mathrm{Re}(\beta z) = c. \tag{2.1.6}$$

A straight line can thus be (2.1.3) can be represented by (2.1.5) or (2.1.6).

Example 2.1.1. By identifying a point (x, y) with complex variable z, we can represent the straight line $x = y$ by

$$\mathrm{Re}(\frac{1 + i}{2}z) = 0.$$

In words, this equation says that if we rotate the points on the straight line $x = y$ by $45°$, we obtain the vertical line passing through the origin. Because the denominator 2 is a real constant, we can simplify the equation to

$$\mathrm{Re}((1 + i)z) = 0.$$

Example 2.1.2. Find the image of the circle $|z-2i| = 1$ under the transformation $f(z) = 1/z$.

Let $w = 1/z$.

$$\begin{aligned} \left|\frac{1}{w} - 2i\right| &= 1 \\ \Leftrightarrow \quad |1 - 2wi|^2 &= |w|^2 \\ \Leftrightarrow \quad 3|w|^2 - 2wi + 2w^*i &= -1. \end{aligned}$$

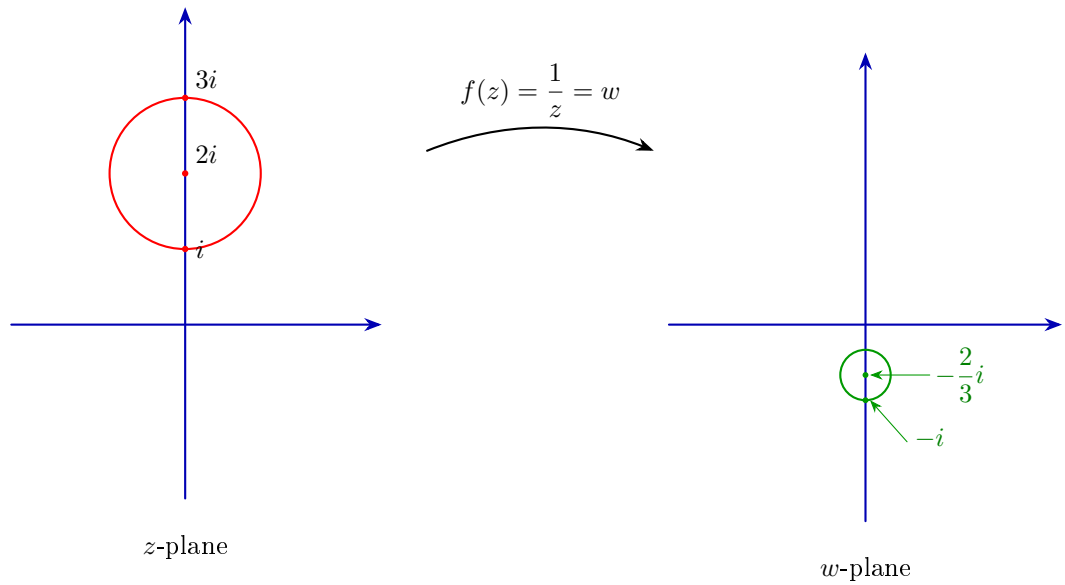


Figure 2.3: The image of a circle under the inversion transformation.

By completing square, the last equation is equivalent to

$$\left|w + \frac{2i}{3}\right|^2 = \frac{1}{9},$$

which is the equation of a circle with radius $1/3$ and center $-2i/3$. (See Fig. 2.3)

Example 2.1.3. Equation $|z - z_0| = |z_0|$ describes a circle that passes through the origin. Find the image of this circle under the transformation $f(z) = 1/z$.

Define a variable $w = 1/z$.

$$\begin{aligned} \left|\frac{1}{w} - z_0\right| &= |z_0| \\ \Leftrightarrow \quad |1 - wz_0|^2 &= |wz_0|^2 \\ \Leftrightarrow \quad 1 - wz_0 - w^* {z_0}^* &= 0 \\ \Leftrightarrow \operatorname{Re}(wz_0) &= 1/2. \end{aligned}$$

This equation defines a straight line in the w-plane. Indeed, if we write $w = u + iv$ and $z_0 = x_0 + iy_0$, then

$$\operatorname{Re}(wz_0) = 1/2 \quad \Longleftrightarrow \quad ux_0 - vy_0 = 0.5.$$

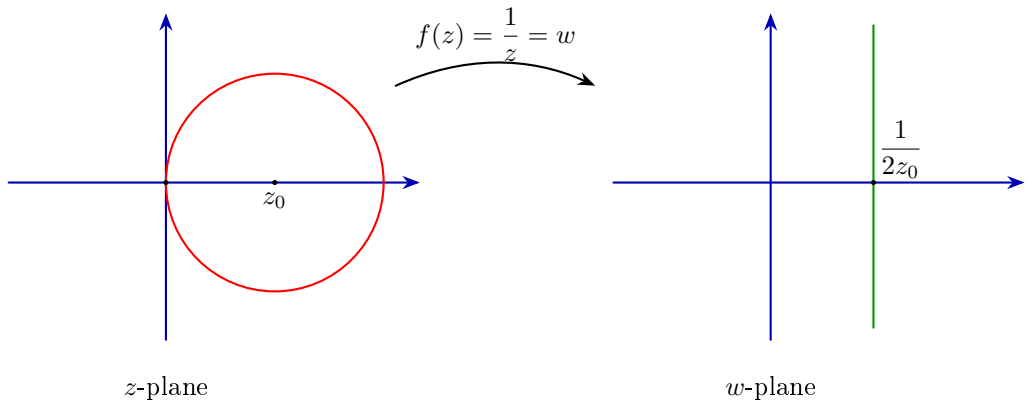


Figure 2.4: Mapping a circle to a straight line.

Hence, the image of the circle $|z - z_0| = |z_0|$ is a straight line with equation $ux_0 - vy_0 = 0.5$.

2.2 Spherical representation of complex numbers

The complex inversion $z \mapsto z^{-1}$ function is not defined when $z = 0$. However, it is tempting to claim that "1/0 is infinity". We extend the domain of the complex inverse function by the following procedure.

We identify a complex number $x + iy$ with the point $(x, y, 0)$ in $\mathbb{R}^3$. Consider the sphere

$$S = \{(x, y, z) \in \mathbb{R}^3 : \alpha^2 + \beta^2 + (\gamma - r)^2 = r^2\} \tag{2.2.1}$$

which has radius r and is tangent to the x-y plane at the origin. The *stereographic projection* $P(x, y)$ maps a complex number $x + iy$ to the point $P(x, y)$ on this sphere obtained by drawing the line through $(x, y, 0)$ and the north pole $(0, 0, 2r)$. The point $P(x, y)$ is defined to be the intersection of this line with the sphere. The mapping is one-to-one; a proof is given in Section 2.7.

Definition 2.2.1. The north pole of the sphere is called the *point at infinity*, denoted by the symbol ∞. The sphere itself is referred to as the *Riemann sphere*.

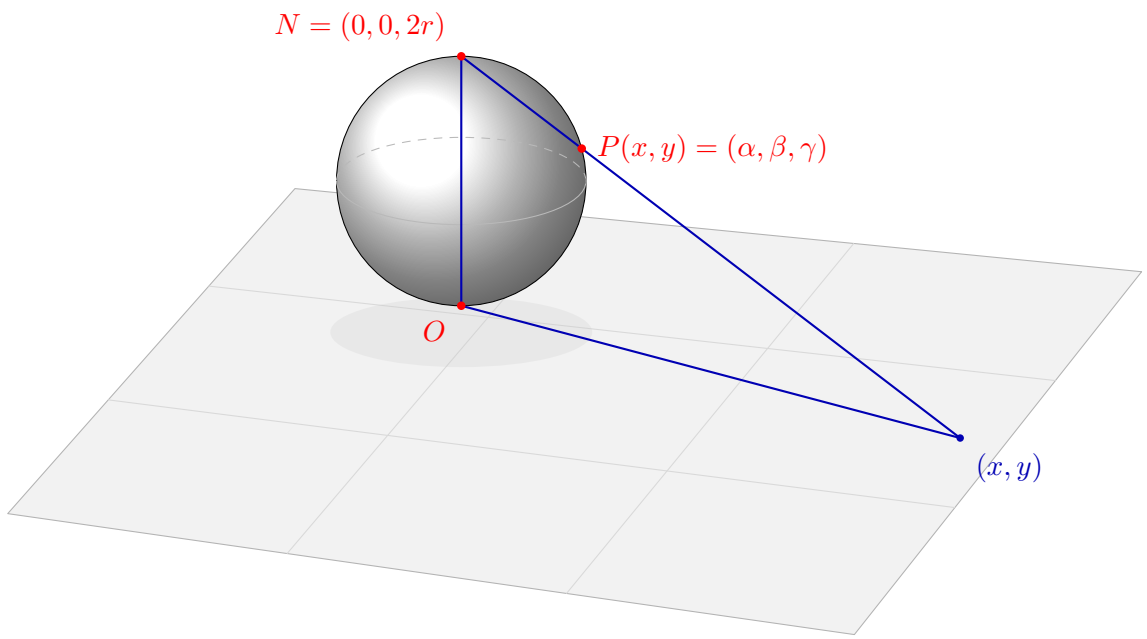


Figure 2.5: Riemann sphere and stereographic projection

Motivated by the stereographic projection, we enlarge the complex number system by adjoining a single extra point, denoted by ∞, which does not belong to $\mathbb{C}$. After adding this point, the resulting set becomes bijection with the Riemann sphere under stereographic projection.

Definition 2.2.2. The *extended complex number system* is the set $\mathbb{C} \cup \{\infty\}$, also called the *extended complex plane*. We will denote it by $\bar{\mathbb{C}}$. Other common notation include includes $\hat{\mathbb{C}}$ and $\mathbb{C}_\infty$. A number z in the extended number plane is said to be *finite* if $z \neq \infty$.

There is no easy way to perform addition and multiplication on the Riemann sphere. The main purpose of introducing the Riemann sphere is to bring in the point of infinity, which can be treated as an ordinary point like the other complex numbers, and making the resulting Riemann sphere a compact set. Hence, the Riemann sphere is often called the *one-point compactification* of the complex plane.

Remark. In contrast to the extended complex number system, the *extended real number system* is obtained by adding two special points to the real number system, i.e., $\mathbb{R} \cup \{\infty, -\infty\}$. In the real case, we have both $+\infty$ and $-\infty$ due to the total ordering of the real number line. However, since there is no ordering in the complex case, we only have one point at infinity.

2.3 Linear fractional transformations

We first treat the special case of affine transformations, which can be classified into three types.

A *translation* is a function $f : \mathbb{C} \to \mathbb{C}$ in the form

$$f_w(z) = z + w$$

for some complex constant w. Geometrically it is just moving horizontally by $\mathrm{Re}(w)$ and vertically by $\mathrm{Im}(w)$.

A *scaling function* is simply expanding by a constant real factor. It can be written as

$$g_a(z) = az$$

for some real constant a. This is also called a *dilation*.

A *rotation* by angle θ can be realized by multiplication by $(\cos\theta + i\sin\theta)$,

$$h_\theta(z) = (\cos\theta + i\sin\theta)z.$$

All three types of mappings preserve the shape of a geometric object in the domain. For example, if we draw a triangle on the complex plane and apply a translation or a dilation to it, we will get a triangle that is similar to the original one. In particular, straight line is mapped to a straight line. If we have two intersecting lines, the angle between them is invariant under translation, scaling and rotation.

We define the class of *affine transformations*/*affine functions* as the set of functions that can be obtained by composing translation, scaling, dilation and rotation. A general form of an affine function is

$$z \mapsto az + b,$$

where b is a complex constant and a is a nonzero complex number. For example, when $a = 1$, we obtain a translation. When $b = 0$, the function $f(z) = az$ is a scaling followed by a rotation. This class of affine functions form a set that is closed under composition.

Moreover, the inverse function of an affine function is also affine. In algebraic terms, the affine functions form a group under composition.

In general, a *group* is a set of elements G on which we can define a binary operation $\circ$, satisfying the following axioms:

1. (closedness) For all $g, h \in G$, $g \circ h \in G$.
2. (associative) For all $f, g, h \in G$, $(f \circ g) \circ h = f \circ (g \circ h)$.
3. (identity) There exists an element $e \in G$ such that $e \circ g = g \circ e = g$ for all $g \in G$.
4. (inverse) For all $g \in G$, there exists an element g' such that $g \circ g' = g' \circ g = e$.

It is straightforward to check that the composition of two affine transformation is also an affine transformation. Given two affine functions $f(z) = az + b$ and $g(z) = a'z + b$, the composition $g(f(z))$ is equal to

$$g(az + b) = a'(az + b) + b' = a'az + a'b + b'.$$

Since a and a' are both nonzero, the coefficient $a'a$ is also nonzero.

It is obvious that a translation, dilation and rotation preserve shape. From Theorem 2.1.1, we know that the inversion function also transforms circle/straight line to circle/straight line. This motivate the study the class of function obtained by mixing and composing affine functions and the inversion function $z \mapsto 1/z$. The resulting class of function is called *linear fractional transformation*. By construction, this class of complex functions has the following property.

Theorem 2.3.1. *A linear fractional transformation $f(z) = (az + b)/(cz + d)$ transforms circles and straight lines to circles and straight lines.*

However, in contrast to affine transformations, the composition of functions is not well defined, as the inverse function is not defined at $z = 0$. We cannot take $\mathbb{C}$ as the domain for a general linear fractional transformation. To remedy this problem, we may define $1/0$ as

the point at infinity in the extended complex number system, and treat a linear fractional transformation as a function from $\bar{\mathbb{C}}$ to $\bar{\mathbb{C}}$.

We adopt the following convention for the computation with the point at infinity.

- For any extended complex number z, which may be a finite complex number or infinity, we define $z + \infty = \infty + z = \infty$.
- For any $b \in \bar{\mathbb{C}} \setminus \{0\}$, we define $b \cdot \infty = \infty \cdot b = \infty$
- For any $c \in \mathbb{C} \setminus \{0\}$, we define $c/\infty = 0$ and $c/0 = \infty$.
- $\infty - \infty$ and $0 \cdot \infty$ are undefined, and they will not occur in any calculations.

We now know how to deal with the reciprocal of 0, and how to do arithmetic with ∞. A *linear fractional transformation* (or *Möbius transformation*, or *bilinear transformation*) has the following from:

$$f(z) = \frac{az + b}{cz + d} \tag{2.3.1}$$

where a, b, c and d are complex numbers with $ad - bc \neq 0$. The number $ad - bc$ is called the *determinant*.

If $b = 0$, $c = 0$ and $d = 1$, a linear fractional transformation reduces to a rotation followed by a dilation, i.e., $f(z) = az$.

If $a = 1$, $c = 0$ and $d = 1$, a linear fractional transformation is a translation $f(z) = z + b$.

If $a = 0$, $b = 1$, $c = 1$ and $d = 0$, a linear fractional transformation becomes the inversion function $f(z) = 1/z$.

We can verify that any function in the form in (2.3.1) can be expressed as a composition of translation, scaling, rotation, and inversion. If $c = 0$ and $d \neq 0$, then $(az + b)/d$ is just an affine transformation, and hence can be computed by first multiplying by a/d and then adding the constant b/d. When $c \neq 0$, by applying a partial fraction expansion, we can see that a linear fractional transformation can be written as:

$$\frac{az + b}{cz + d} = \frac{a}{c} + \frac{bc - ad}{c} \frac{1}{cz + d}. \tag{2.3.2}$$

The condition $ad - bc \neq 0$ in the definition of linear fractional transformation ensures that it is a non-constant function. In fact, we may scale all four parameters a, b, c and d by a nonzero complex constant such that $ad - bc = 1$.

This decomposition suggests that a proper definition of $f(\infty)$ is a/c. When z has very large modulus, the modulus of the fraction $1/(cz+d)$ becomes negligible. The value of $(az+b)/(cz+d)$ tends to a/c when z tends to the point at infinity.

The function induced by a linear fractional transformation in (2.3.1) is given as follows.

Definition 2.3.2. Given complex numbers a, b, c and d with $ad-bc \neq 0$, we define a function $f(z) = (az+b)/(cz+d)$ on the extended complex plane.
When $c = 0$,

$$f(z) \triangleq \begin{cases} (a/d)z + (b/d) & \text{if } z \neq \infty, \\ \infty & \text{if } z = \infty. \end{cases}$$

When $c \neq 0$,

$$f(z) \triangleq \begin{cases} \frac{az+b}{cz+d} & \text{if } z \neq -d/c,\ z \neq \infty \\ \infty & \text{if } z = -d/c \\ a/c & \text{if } z = \infty. \end{cases}$$

We record several remarks about this computational formula. When $c = 0$, the condition $ad - bc \neq 0$ forces $a \neq 0$ and $d \neq 0$. In this case, the LFT reduces to an affine map $az + b$, which sends the point at infinity to itself.

When $c \neq 0$. There are two exceptional input values: $z = -d/c$ and $z = \infty$. At $z = -d/c$, the denominator $cz + d$ vanishes, so we define $f(-d/c) = \infty$. For any finite complex number $z \neq -d/c$, the value $f(z)$ cannot equal a/c. Indeed, if

$$\frac{az+b}{cz+d} = \frac{a}{c},$$

then

$$c(az+b) = a(cz+d) \iff caz + cb = acz + ad \iff ad - bc = 0,$$

contradicting the assumption that $ad - bc \neq 0$.

This observation justifies defining $f(\infty) = a/c$. For all other finite values $z \neq -d/c$, we set $f(z) = (az+b)/(cz+d)$. With these conventions, f becomes a bijection on the extended complex plane.

Theorem 2.3.3. *Let $f(z) = (az+b)/(cz+d)$ be a linear fractional transformation. Then exactly one of the following holds:*

1. *If $f(\infty) = \infty$, then $c = 0$ and $f(z)$ is an affine transformation.*
2. *If $f(\infty)$ is finite, then $c \neq 0$, and $f(z)$ can be expressed as a composition of a translation, a scaling, and an inversion.*

Proof. By the rule of evaluating an LFT in (2.3.2), we have $f(\infty) = \infty$ if and only if the coefficient $c = 0$. When $c = 0$, $f(z)$ reduces to an affine transformation. When $c \neq 0$, the decomposition in (2.3.2) shows that $f(z)$ can be computed by first multiplying z by c, translate by d, then multiply by the $\frac{bc-ad}{c}$, and finally translate by a/c. □

Next, we study the group properties of linear fractional transformations.

Theorem 2.3.4. *The composition of two linear fractional transformations is a linear fractional transformation.*

Proof. (sketch) In the followings, $f(z)$ is a linear fractional transformation $f(z) = (az + b)/(cz + d)$. It suffices to consider three cases.
(i) The function $f(z) + \alpha$ is a linear fractional transformation for any complex constant α.
(ii) The function $\beta f(z)$ is a linear fractional transformation for any nonzero complex constant β.
(iii) The function $1/f(z)$ is a linear fractional transformation. □

We next consider the inverse of a linear fractional transformation. We can use the fact that a linear fractional transformation can be written as the composition of four types of functions — translation, rotation, dilation, and inversion. Since all of them are invertible function, we obtain

Theorem 2.3.5. *The inverse of a linear fractional function exists, and is a linear fractional transformation.*

We give a more direct proof by computing the inverse of a linear fractional transformation, and showing that the inverse is a linear fractional transformation.

Proof. Let $f(z)$ be a linear fractional transformation defined by $f(z) = (az+b)/(cz+d)$. We consider two cases.

Suppose that $c = 0$. In this case, we must have $a \neq 0$ and $d \neq 0$ (because we assume $ad - bc \neq 0$). The function $f(z)$ can be written in the form $az + b$, by writing a/d as a and b/d as b. The inverse of $f(z)$ is $g(w) = (w-b)/a$. We also have $f(\infty) = \infty$ in this case. The inverse of $f(z)$ is

$$g(w) = \begin{cases} (w-b)/a & \text{if } w \neq \infty, \\ \infty & \text{if } w = \infty. \end{cases}$$

We can check that $g \circ f$ and $f \circ g$ are both equal to the identity function.

Next suppose $c \neq 0$. Using (2.3.2), we write the equation $f(z) = w$ as

$$\frac{a}{c} + \frac{bc-ad}{c}\frac{1}{cz+d} = w,$$

and try to express z in terms of w. We first simplify it to

$$\frac{cw-a}{bc-ad} = \frac{1}{cz+d}.$$

If $w \neq a/c$, we can solve for z in $\mathbb{C}$

$$z = \frac{1}{c}\Big(\frac{bc-ad}{cw-a} - d\Big) = \frac{b-dw}{cw-a}.$$

If $w = a/c$, we have $f(\infty) = w = a/c$. The inverse function of $f(z)$ is

$$g(w) = \begin{cases} \frac{b-dw}{cw-a} & \text{if } \infty \neq w \neq a/c \\ \infty & \text{if } w = a/c \\ -d/c & \text{if } w = \infty. \end{cases}$$

Since $g(f(z))$ and $f(g(w))$ are identity functions from Riemann sphere to itself, we conclude that f is a bijection. $\square$

In general, if $f(z) = \frac{az+b}{cz+d}$ is a linear fractional transformation, the inverse exists and is given by

$$f^{-1}(z) = \frac{dz-b}{-cz+a}.$$

Example 2.3.1. Consider the function $f(z) = (z+1)/(z-i)$. When $z = i$, we define $f(i) = \infty$ and when $z = \infty$, we define $f(\infty) = 1$. To compute the inverse function of $f(z)$, we solve for z in the equation $w = f(z)$,

$$\begin{aligned} w &= \frac{z+1}{z-i} \\ wz - iw &= z + 1 \\ (w-1)z &= 1 + iw \\ z &= \frac{1+iw}{w-1}. \end{aligned}$$

The last line requires $w \neq 1$. The inverse of $f(z)$ is thus

$$g(w) = \begin{cases} \frac{1+iw}{w-1} & \text{if } w \neq 1,\ w \neq \infty, \\ \infty & \text{if } w = 1, \\ i & \text{if } w = \infty. \end{cases}$$

Consider two LFTs $f(z)$ and $g(z)$ represented by $\frac{az+b}{cz+d}$ and $\frac{\alpha z+\beta}{\gamma z+\delta}$, respectively. The composition $g \circ f$ is the linear fractional transformation given by

$$\begin{aligned} g(f(z)) &= g\Big(\frac{az+b}{cz+d}\Big) \\ &= \frac{(\alpha a + \beta c)z + (\alpha b + \beta d)}{(\gamma a + \delta c)z + (\gamma b + \delta d)}. \end{aligned}$$

We note the similarity to matrix multiplication

$$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \alpha a + \beta c & \alpha b + \beta d \\ \gamma a + \delta c & \gamma b + \delta d \end{bmatrix}.$$

We may represent a linear fractional transformation $\frac{az+b}{cz+d}$ by 2×2 complex matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, and "define" the composition of two LFT's by matrix multiplication. Since the determinant $ad - bc$ is nonzero by assumption, inverse of the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ exists. Moreover, matrix multiplication is associative, and identity exists. Hence, the set of nonsingular 2×2 complex matrices form a group under matrix multiplication.

Linear fractional transformations can be formulated naturally as a group action on the extended complex plane. In general, an *action* of a group G with identity element e on a set X is a function

$$\alpha : G \times X \to X$$

satisfying the axioms:

1. $\alpha(e, x) = x$ for all $x \in X$.
2. $\alpha(g \circ h, x) = \alpha(g, \alpha(h, x))$ for all $g, h \in G$ and $x \in X$.

Using this terminology, we can state the main theorem of this section.

Theorem 2.3.6. *Let G be the group of nonsingular 2×2 complex matrix, with composition $\circ$ given by matrix multiplication. For a matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and a point z in the extended complex number system, define α by the linear fractional transformation in Definition 2.3.2. Then α defines a group action of G on the extended complex number plane.*

A direct verification of this theorem requires checking several cases. A formal LEAN proof is available on this github page. A more conceptual approach is to view the extended complex numbers as the complex projective line; we outline this perspective in Section 2.8.

Since multiplying all entries of a matrix by a nonzero complex scalar does not change the associated linear fractional transformation, we obtain an equivalence relation on the set of 2×2 nonsingular matrices: two matrices are equivalent if one is a nonzero scalar multiple of the other. Two linear fractional transformations represent the same function precisely if their corresponding matrices are equivalent.

Remark. The quotient fo the group of 2×2 matrices by this equivalence relation is the projective general linear group $PGL(2, \mathbb{C})$. The group $PGL(2, \mathbb{C})$ acts faithfully on the Riemann sphere. A discrete subgroup of $PGL(2, \mathbb{C})$ is called a *Kleinian group*. The next example illustrates a finite subgroup of linear fractional transformations.

Example 2.3.2. Consider the linear fractional transformations $f_1(z) = 1/z$ and $f_2(z) = 1 - z$.

Both of them induce a permutation on the three points $0, 1, \infty$ on the Riemann sphere.

$$\begin{aligned} f_1(0) &= \infty, & f_1(1) &= 1, & f_1(\infty) &= 0, \\ f_2(0) &= 1, & f_2(1) &= 0, & f_2(\infty) &= \infty. \end{aligned}$$

In this example we want to exhaust all function that can be obtained by composing f_1 and f_2 recursively. For example, using f_1 and f_2, we can obtain functions in the form $f_1(f_2(z))$, $f_2(f_1(z))$, $f_1(f_1(f_2(z)))$, $f_2(f_1(f_2(z)))$, etc. How many functions can we get?

First, we see that we can get the identity function from

$$f_1(f_1(z)) = f_2(f_2(z)) = z.$$

We can discover two new functions from

$$\begin{aligned} f_1(f_2(z)) &= \frac{1}{1-z}, \\ f_2(f_1(z)) &= 1 - \frac{1}{z} = \frac{z-1}{z}. \end{aligned}$$

If we apply the inversion function to $f_2(f_1(z))$, we get

$$f_1(f_2(f_1(z))) = \frac{z}{z-1}.$$

We claim that these six functions

$$f_0(z) = z, \quad f_1(z) = 1/z, \quad f_2(z) = 1 - z, \quad f_3(z) = \frac{1}{1-z}, \quad f_4(z) = \frac{z-1}{z}, \quad f_5(z) = \frac{z}{z-1}$$

are all we can get. We can organize the relationship among them in a group composition table.

$\circ$	$f_0(z) = z$	$f_1(z) = \frac{1}{z}$	$f_2(z) = 1-z$	$f_3(z) = \frac{1}{1-z}$	$f_4(z) = \frac{z-1}{z}$	$f_5(z) = \frac{z}{z-1}$
$f_0(z) = z$	z	$\frac{1}{z}$	$1-z$	$\frac{1}{1-z}$	$\frac{z-1}{z}$	$\frac{z}{z-1}$
$f_1(z) = \frac{1}{z}$	$\frac{1}{z}$	z	$\frac{1}{1-z}$	$1-z$	$\frac{z}{z-1}$	$\frac{z-1}{z}$
$f_2(z) = 1-z$	$1-z$	$\frac{z-1}{z}$	z	$\frac{z}{z-1}$	$\frac{1}{z}$	$\frac{1}{1-z}$
$f_3(z) = \frac{1}{1-z}$	$\frac{1}{1-z}$	$\frac{z}{z-1}$	$\frac{1}{z}$	$\frac{z-1}{z}$	z	$1-z$
$f_4(z) = \frac{z-1}{z}$	$\frac{z-1}{z}$	$1-z$	$\frac{z}{z-1}$	z	$\frac{1}{1-z}$	$\frac{1}{z}$
$f_5(z) = \frac{z}{z-1}$	$\frac{z}{z-1}$	$\frac{1}{1-z}$	$\frac{z-1}{z}$	$\frac{1}{z}$	$1-z$	z

In every column and every row, each of the six functions appears once and only once.

A notable feature in this example is that the group generated by $f_1(z)$ and $f_2(z)$ is finite. In fact, each LFT in this example is determined by its action on three special points 0, 1 and ∞. The functions correspond to the six permutations of 0, 1 and ∞ on the Riemann sphere. We will see in the next section that an LFT is determined by its action on three distinct points.

2.4 3-Transitivity of linear fractional transformation

In this section we consider the question: Given two circles (or straight lines) on the complex plane, can we fine a linear fractional transformation that maps the first circle to the second circle. The answer is yes, and we can establish a stronger result. It is known that a circle is uniquely determined by three points. We prove in the next theorem that we can find a linear fractional transformation that maps any three distinct points to any three other distinct points. Moreover, the LFT is unique up to multiplying all coefficients a, b, c and d by a nonzero constant.

We first establish a special property of linear fractional transformation. We will use the notation of "fixed point." Given a function f, we say that a number $z_0 \in \bar{\mathbb{C}}$ is a *fixed point* of f if $f(z_0) = z_0$.

Theorem 2.4.1. *If linear fractional transformation f satisfies $f(0) = 0$, $f(1) = 1$ and $f(\infty) = \infty$, then $f(z) = z$ for all $z \in \bar{\mathbb{C}}$.*

Proof. Since $f(\infty) = \infty$, we know that $f(z)$ is an affine transformation by Theorem 2.3.3. We can find complex numbers a and b such that $f(z) = az + b$ for all $z \in \mathbb{C}$. Substituting $f(0)$ by 0 and $f(1)$ by 1, the only possibility is $a = 1$ and $b = 0$. This proves that $f(z) = z$ for all $z \in \bar{\mathbb{C}}$. □

Theorem 2.4.2. *Given any three distinct points z_1, z_2 and z_3 all in the extended complex plane $\bar{\mathbb{C}}$, we can find a linear fractional transformation that maps z_1 to 0, z_2 to 1, and z_3 to ∞. Moreover, this linear fractional transformation is unique.*

Proof. We first assume that z_1, z_2 and z_3 are finite complex numbers. The function

$$f(z) = \frac{z - z_1}{z - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1} \tag{2.4.1}$$

is a linear fractional transformation that satisfies the requirements, mapping z_1 to 0, z_2 to 1, and z_3 to ∞.

There are three boundary cases to consider. When $z_1 = \infty$ and z_2 and z_3 are finite, we can map ∞ to 0, z_2 to 1, and z_3 to ∞ by

$$f(z) = \frac{z_2 - z_3}{z - z_3}. \tag{2.4.2}$$

When $z_2 = \infty$ and z_1 and z_3 are finite, we can map z_1 to 0, ∞ to 1, and z_3 to ∞ by

$$f(z) = \frac{z - z_1}{z - z_3}. \tag{2.4.3}$$

Finally, when $z_3 = \infty$ and z_1 and z_2 are finite, we can map z_1 to 0, z_2 to 1, and ∞ to ∞ by

$$f(z) = \frac{z - z_1}{z_2 - z_1}. \tag{2.4.4}$$

To prove uniqueness, suppose g is another linear fractional transformation that satisfies the requirements in the theorem. Then $f \circ g^{-1}$ is a linear fractional transformation that fixed 0, 1 and ∞. By Theorem 2.4.1, $f \circ g^{-1}$ is the identity function. By the group property of linear fractional transformation, we have $g = f$. □

Example 2.4.1. Find a linear fractional transformation that maps 0 to 0, 1 to 1, and z_0 to ∞, where z_0 is a complex number different from 0 and 1.

Because we want to take z_0 to ∞, the linear fractional transformation has the form

$$f(z) = \frac{az + b}{z - z_0}$$

for some choice of complex constants a and b. Because we want $f(0) = 0$, the only choice of constant b is $b = 0$. From the requirement $f(1) = 1$, we derive that a is equal to $1 - z_0$. The answer is

$$f(z) = (1 - z_0)\frac{z}{z - z_0}.$$

Consider the case that z_0 is equal to i. The three points 0, 1 and i in the domain of the transformation lie on the circle with equation $|z - (1 + i)/2|^2 = 1/2$. The three points 0, 1, ∞ in the range lie on the line $\mathrm{Im}(z) = 0$ (the real axis plus the point at infinity). In this case, the fractional linear transformation

$$z \mapsto \frac{(1 - i)z}{z - i}$$

maps the circle with and radius $1/\sqrt{2}$ and center at $(1 + i)/2$ to the real axis.

Theorem 2.4.3. *Given any three distinct points z_1, z_2 and z_3 and any three distinct points w_1, w_2 and w_3, all in the extended complex plane $\bar{\mathbb{C}}$, there is a unique linear fractional transformation that maps z_1 to w_1, z_2 to w_2, and z_3 to w_3.*

Proof. By Theorem 2.4.2, there exists a unique LFT $f_1(z)$ that maps z_1, z_2 and z_3 to 0, 1, and ∞, and another unique LFT $f_2(z)$ that maps w_1, w_2 and w_3 to 0, 1, and ∞. The function $f_2^{-1} \circ f_1$ is the required LFT.

To prove uniqueness, suppose $g(z)$ is an LFT that satisfies the conditions in the theorem. The composed function $f_2 \circ g \circ f_1^{-1}$ is an LFT that fixed 0, and 1 and ∞, and hence is the identity function. This implies that $g(z)$ must be equal to $f_2^{-1} \circ f_1$ by Theorem 2.4.1. $\square$

2.5 Cross ratio

Given four distinct numbers z_0, z_1, z_2 and z_3 in $\bar{\mathbb{C}}$, we define the cross ratio as $f(z_0)$, where $f(z)$ is the unique function that maps z_1 to 0, z_2 to 1, and z_3 to ∞.

Definition 2.5.1. Given four complex numbers z_0, z_1, z_2 and z_3, we define the *cross ratio* as

$$[z_0, z_1, z_2, z_3] \triangleq \begin{cases} \frac{z_0-z_1}{z_0-z_3} \cdot \frac{z_2-z_3}{z_2-z_1} & \text{if } z_0 \neq \infty, \\ \frac{z_2-z_3}{z_2-z_1} & \text{if } z_0 = \infty, \end{cases}$$

when z_1, z_2 and z_3 are all finite. If one of z_1, z_2 and z_3 is ∞, the cross ratio is computed by (2.4.2), (2.4.3) or (2.4.4).

When z_0, z_1, z_2 and z_3 are not distinct, then the cross-ratio is not defined, or defined to be some junk value.

An important property of the cross ratio is that it is invariant under linear fractional transformation.

Theorem 2.5.2. *Let $f(z)$ be a linear fractional transformation mapping z_k to z'_k, for $k = 0, 1, 2, 3$, then*

$$[z_0, z_1, z_2, z_3] = [z'_0, z'_1, z'_2, z'_3].$$

Proof. (sketch) Since a linear fractional transformation can be obtained by composing an affine transformation and a complex inversion (see (2.3.2)), we can study the effect of applying a translation, a scaling and an inversion to the cross ratio.

Let z_0, z_1, z_2 and z_3 be four complex numbers. If we add a complex number ζ to all of them, the cross ratio of the resulting points is

$$\begin{aligned}[z_0+\zeta, z_1+\zeta, z_2+\zeta, z_3+\zeta] &= \frac{z_0+\zeta-(z_1+\zeta)}{z_0+\zeta-(z_3+\zeta)} \cdot \frac{z_2+\zeta-(z_3+\zeta)}{z_2+\zeta-(z_1+\zeta)} \\ &= \frac{z_0-z_1}{z_0-z_3} \cdot \frac{z_2-z_3}{z_2-z_1} \\ &= [z_0, z_1, z_2, z_3].\end{aligned}$$

By similar calculation, we can easily verify that, for any nonzero complex number α,

$$[\alpha z_0, \alpha z_1, \alpha z_2, \alpha z_3] = [z_0, z_1, z_2, z_3].$$

When z_0, z_1, z_2, and z_3 are all nonzero complex number in $\mathbb{C}$, we have

$$\begin{aligned}[\frac{1}{z_0}, \frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}] &= \frac{\frac{1}{z_0}-\frac{1}{z_1}}{\frac{1}{z_0}-\frac{1}{z_3}} \cdot \frac{\frac{1}{z_2}-\frac{1}{z_3}}{\frac{1}{z_2}-\frac{1}{z_1}} \\ &= \frac{z_1-z_0}{z_3-z_0} \cdot \frac{z_3-z_2}{z_1-z_2} \\ &= [z_0, z_1, z_2, z_3].\end{aligned}$$

There are few more cases to check when z_0, z_1, z_2 and z_3 can ∞ as the value. They are left as exercise. □

Suppose $f(z)$ is an LFT that map z_k to w_k for $k = 1, 2, 3$. From Theorem 2.4.2 we know that such a function exists and is unique. We can compute $f(z)$ using of cross ratio. For any complex number z not equal to z_1, z_2 and z_k, suppose $f(z) = w$. Since the cross ratio of $[z, z_1, z_2, z_3]$ is equal to $[w, w_1, w_2, w_3]$, we obtain

$$\frac{z-z_1}{z-z_3} \cdot \frac{z_2-z_3}{z_2-z_1} = \frac{w-w_1}{w-w_3} \cdot \frac{w_2-w_3}{w_2-w_1}. \tag{2.5.1}$$

We can solve for w as a function of z. This gives an explicit form for the function $f(z)$.

Example 2.5.1. We construct a linear fractional transformation that maps three points $z_1 = 1$, $z_2 = 2$, and $z_3 = 3$ on the real axis to three points $w_1 = i$, $w_2 = 3i$, and $w_3 = 5i$ on

the imaginary axis. We apply the equation in (2.5.1),

$$\frac{z-1}{z-3}\cdot\frac{2-3}{2-1}=\frac{w-i}{w-5i}\cdot\frac{3i-5i}{3i-i}.$$

After some algebraic calculations, we can simplify it to

$$w=2iz-i.$$

The desired linear fractional transformation is

$$f(z)=2iz-i=i(2z-1).$$

Using the fact that linear fractional transformation maps circle/straight line to circle/straight line, we have a concise condition for four points lying on a circle. We say that a set of points are *concyclic* if they are on the same circle.

Recall that $[z,z_1,z_2,z_3]$ is a linear fractional transformation that maps z_1 to 0, z_2 to 1, and z_3 to the point of infinity. Since 0, 1 and the point at infinity lies on a straight line, namely the real axis, any point z_0 that is concyclic with z_1, z_2 and z_3 must also be mapped to the real axis by the same fractional linear transformation. Hence,

$$[z_0,z_1,z_2,z_3] \text{ is real iff } z_0,\ z_1,\ z_2 \text{ and } z_3 \text{ are concyclic.} \tag{2.5.2}$$

This gives a convenient test to determine whether four points are located on a circle.

Example 2.5.2. Derive the equation of the circle that passes through $(0,0)$, $(1,0)$ and $(0,2)$ in the x-y plane.

We take $z_1=0$, $z_2=1$ and $z_3=2i$. Let $z=x+iy$ be a point that lie on the circle that contains z_1, z_2 and z_3. Applying the criterion in (2.5.2), we obtain the condition

$$\operatorname{Im}\left(\frac{x+iy}{x+iy-2i}\cdot\frac{1-2i}{1}\right)=0,$$

which can then be simplified to

$$x^2+y^2-x-2y=0,$$

or equivalently

$$(x-1/2)^2+(y-1)^2=\frac{5}{4}.$$

This is the equation of a circle with center at $(1/2,1)$ and radius $\sqrt{5}/2$.

2.6 Python Program

We implement a test of concyclicity in Python. This program takes for points with real coordinates as inputs, and returns 1 if the four points lie on a circle. Even though this geometrical question itself has nothing to do with complex numbers, we can use complex numbers and complex arithmetic to get the answer easily. The calculation is done in one line.

```
# Determine whether four distinct points lie on a circle

def test_concyclic(pt0, pt1, pt2, pt3):
"""
  Input four tuples:  pt1= (x1,y1), pt2= (x2,y2), pt3= (x3,y3),
     pt4=(x4,y4)
  return True if the four points pt1, pt2, pt3, pt4 are concyclic,
     False if not concyclic
"""
  z0 = complex(pt0[0],pt0[1])   # convert to complex numbers
  z1 = complex(pt1[0],pt1[1])
  z2 = complex(pt2[0],pt2[1])
  z3 = complex(pt3[0],pt3[1])
  return abs(((z0-z1)*(z2-z3)/((z0-z3)*(z2-z1))).imag)< 1e-8
```

For example, we can use this Python function to check that four points $(0,0)$, $(1,0)$, $(1,2)$, and $(0,2)$ lie on the circle in the previous example.

```
In [1]: test_concyclic((0,0), (1,2), (0,2), (1,0))
Out[1]: True
```

2.7 Appendix: Details of stereographic projection

Stereographic projection can be explicitly computed using the formula below. Recall that the sphere S is centered at $(0,0,r)$ with radius r.

Theorem 2.7.1. *The functions $P : \mathbb{R}^2 \to S \setminus \{(0,0,2r)\}$ described in the previous paragraph is given by*

$$P(x,y) = \Big(\frac{4r^2x}{4r^2+x^2+y^2}, \frac{4r^2y}{4r^2+x^2+y^2}, \frac{2r(x^2+y^2)}{4r^2+x^2+y^2}\Big).$$

This gives a bijection between the complex plane $\mathbb{R}^2$ and the punctured sphere $S \setminus \{(0,0,2r)\}$.

We may alternately write the projection function P with $z = x + iy$ as input,

$$P(x+iy) = \Big(\frac{4r^2 \operatorname{Re} z}{4r^2+|z|^2}, \frac{4r^2 \operatorname{Im} z}{4r^2+|z|^2}, \frac{2r|z|^2}{4r^2+|z|^2}\Big).$$

Proof. Let N denote the north pole $(0,0,2r)$ of S. The triangle with vertices N, $(0,0,0)$ and $(x,y,0)$ and the triangle with vertices N, $(0,0,\gamma)$, $P(x,y)$ are similar. This gives

$$\frac{x}{\alpha} = \frac{y}{\beta} = \frac{2r}{2r-\gamma}. \tag{2.7.1}$$

Substitute $\alpha = \frac{2r-\gamma}{2r}x$ and $\beta = \frac{2r-\gamma}{2r}y$ into the equation of sphere S (Equation 2.2.1), we get

$$\begin{aligned}(\frac{2r-\gamma}{2r})^2(x^2+y^2) + (\gamma - r)^2 &= r^2 \\ (2r-\gamma)(x^2+y^2) &= 4r^2\gamma.\end{aligned}$$

In the last step, we cancel the factor $2r-\gamma$ from both sides. This is possible because $\gamma < 2r$. We can express γ in terms of x and y

$$\gamma = \frac{2r(x^2+y^2)}{4r^2+x^2+y^2}.$$

Hence,

$$2r - \gamma = \frac{8r^3}{4r^2+x^2+y^2}.$$

Substituting the expression of $2r-\gamma$ back to (2.7.1) gives

$$\alpha = \frac{4r^2x}{4r^2+x^2+y^2} \text{ and } \beta = \frac{4r^2y}{4r^2+x^2+y^2}.$$

This proves that $P(x,y)$ can be written as in the proposition.

We now verify that P is a bijection by checking that P has an inverse function

$$Q(\alpha,\beta,\gamma) \triangleq \big(\frac{2r\alpha}{2r-\gamma}, \frac{2r\beta}{2r-\gamma}\big). \tag{2.7.2}$$

For any point (α,β,γ) on $S\setminus\{N\}$, we can compute

$$\begin{aligned} P(Q(\alpha,\beta,\gamma)) &= P\Big(\frac{2r\alpha}{2r-\gamma}, \frac{2r\beta}{2r-\gamma}\Big) \\ &= \Big(\alpha\frac{2r(2r-\gamma)}{(2r-\gamma)^2+\alpha^2+\beta^2}, \beta\frac{2r(2r-\gamma)}{(2r-\gamma)^2+\alpha^2+\beta^2}, \frac{2r(\alpha^2+\beta^2)}{(2r-\gamma)^2+\alpha^2+\beta^2}\Big). \end{aligned} \tag{2.7.3}$$

From the equation of S, we have

$$\alpha^2+\beta^2 = 2r\gamma-\gamma^2. \tag{2.7.4}$$

The demoninators in (2.7.3) can be simplified to

$$(2r-\gamma)^2+\alpha^2+\beta^2 = 4r^2-4r\gamma+\gamma^2+\alpha^2+\beta^2 = 4r^2-2r\gamma.$$

The point in (2.7.3) can then be simplified to (α,β,γ). This proves that the composite function $P\circ Q$ is the identity function on the punctured sphere.

In the other direction, we want to show that

$$Q(P(x,y)) = Q\Big(\frac{4r^2x}{4r^2+x^2+y^2}, \frac{4r^2y}{4r^2+x^2+y^2}, \frac{2r(x^2+y^2)}{4r^2+x^2+y^2}\Big)$$

is equal to the point (x,y). To this end, we compute

$$\frac{2r\alpha}{2r-\gamma} = \frac{2r\frac{4r^2x}{4r^2+x^2+y^2}}{2r-\frac{2r(x^2+y^2)}{4r^2+x^2+y^2}} = x$$

and

$$\frac{2r\beta}{2r-\gamma} = \frac{2r\frac{4r^2y}{4r^2+x^2+y^2}}{2r-\frac{2r(x^2+y^2)}{4r^2+x^2+y^2}} = y.$$

This proves that the composite function $Q\circ P$ is the identity function on the complex plane.

Because $Q\circ P$ and $P\circ Q$ are identity functions, we conclude that P is a bijection. $\square$

The stereographic projection is not merely a bijection, it has the feature of preserving circle and straight line. Suppose that a plane with equation

$$a\alpha + b\beta + c\gamma = d \tag{2.7.5}$$

for some constants a, b, c and d intersects with the sphere S. We know that the cross-section is circular in shape. We derive the equation that describes the stereographic projection of the cross-section as follows.

Consider a plane Π that contains the north pole of S. This plane intersects the sphere S at a circle as long as it is not a horizontal plane, i.e., when a and b are not both zero. Using the fact that the vector (a, b, c) is a normal vector of Π, we can write the equation of Π as

$$\begin{aligned} [(\alpha, \beta, \gamma) - (0, 0, 2r)] \cdot (a, b, c) &= 0 \\ a\alpha + b\beta + \gamma c &= 2rc. \end{aligned}$$

We note that when the plane Π passes through the north pole, the constant d in (2.7.5) is equal to $d = 2rc$. The intersection of Π and the horizontal plane is the straight line described by the equation $a\alpha + b\beta = 2rc$.

Next, suppose $a\alpha + b\beta + c\gamma = d$ is the equation of a plane Π that cut through the sphere S but does not pass through the north pole N. Let (x, y) be a point on the horizontal plane such that the point $(\alpha, \beta, \gamma) = P(x, y)$ lies on the plane Π. We obtain the equation

$$\begin{aligned} a\frac{4r^2x}{4r^2 + x^2 + y^2} + b\frac{4r^2y}{4r^2 + x^2 + y^2} + c\frac{2r(x^2 + y^2)}{4r^2 + x^2 + y^2} &= d \\ 4r^2ax + 4r^2by + 2rc(x^2 + y^2) &= d(4r^2 + x^2 + y^2) \\ (d - 2rc)(x^2 + y^2) - 4r^2ax - 4r^2by + 4r^2d &= 0. \end{aligned}$$

This equation defines a circle provided that $d \neq 2rc$, which happens if and only if the plane Π does not contain the north pole. By completing the squares we see that the center of the circle is located at

$$(x, y) = \frac{2r^2}{d - 2rc}(a, b)$$

and the radius is

$$\frac{2r}{|d - 2rc|}\sqrt{r^2(a^2 + b^2) - d(d - 2rc)}. \tag{2.7.6}$$

The number $r^2(a^2+b^2)-d(d-2rc)$ should be positive, because the radius is supposed to be positive. We can show this algebraically. We are assuming that the perpendicular distance between the plane and the center of S is less than r, otherwise the plane Π and the sphere S has no intersection, or intersect at a single point. We express this condition using the dot product

$$\left|(\alpha,\beta,\gamma-r)\cdot\frac{(a,b,c)}{\sqrt{a^2+b^2+c^2}}\right|<r.$$

This condition can be simplified to

$$(\alpha a+\beta b+\gamma c-rc)^2<r^2(a^2+b^2+c^2)$$

which can be further simplified to

$$d^2-2drc<r^2(a^2+b^2).$$

This is equivalent to the condition that $r^2(a^2+b^2)-d(d-2rc)$ is positive.

2.8 Appendix: Complex projective line

There is alternate description of the extended complex plane using projective geometry. We use the notation $\mathbb{C}^2$ to represent the complex vector space consisting of column vector $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$, with z_1 and z_2 not both zero. Define an equivalence relation by declaring $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \sim \begin{bmatrix} z_1' \\ z_2' \end{bmatrix}$ if there exists a complex number λ such that $z_1'=\lambda z_1$ and $z_2'=\lambda z_2$. Two pairs are thus equivalent if they lie on the same complex line that passes through the origin in $\mathbb{C}^2$.

The *complex projective line* is defined as the set that consists of all equivalence classes defined in the previous paragraph. Common notation for the complex projective line are $\mathbb{CP}_1$ or $\mathbb{C}P^1$. In an equivalence class, the exact values of z_1 and z_2 are not relevant, but only the fraction z_1/z_2 is relevant. When z_2 is nonzero, the fraction z_1/z_2 is a complex number, and we can pick the vector $\begin{bmatrix} z \\ 1 \end{bmatrix}$, with $z=z_1/z_2$, as the representative of this equivalence class. If $z_2=0$, we can select the vector $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ as the representative.

We have a bijection between the extended complex plane and the complex projective line. A finite complex number z corresponds to the equivalence class that contains $\begin{bmatrix} z \\ 1 \end{bmatrix}$. The point at infinity is the equivalence class that contains $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. We can thus regard the extended complex number system as a set consisting of elements in the form $\begin{bmatrix} z \\ 1 \end{bmatrix}$, for $z \in \mathbb{C}$, and a special element $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

We can perform a linear fractional transformation by multiplying $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$ from the left by a 2×2 matrix,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

One can show that it does not depend on the choice of $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$ in an equivalence class. If $z_2 \neq 0$, the resulting vector is equivalent to

$$\begin{bmatrix} \frac{az+b}{cz+d} \\ 1 \end{bmatrix}$$

where $z = z_1/z_2$.

If we multiply with the point at infinity $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, we obtain

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \sim \begin{bmatrix} a \\ c \end{bmatrix}$$

If we multiply with the projective point that corresponds to the complex number $-d/c$,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} -d/c \\ 1 \end{bmatrix} \sim \begin{bmatrix} a(-d/c) + b \\ 0 \end{bmatrix},$$

which is equivalent to the point at infinity (using the assumption that $ad - bc \neq 0$.) Therefore, the definition using matrix multiplication is isomorphic to Definition 2.3.2.

Summary

- A linear fractional transformation, also known as Möbius transformation, is a function in the form $f(z) = \frac{az+b}{cz+d}$, where a, b, c and d are complex numbers satisfying $ad-bc \neq 0$. This class of functions include translation, rotation, expansion, and inversion as special cases.

- The stereographic projection maps the points on the complex plane to the Riemann sphere injectively. The extended complex number system $\bar{\mathbb{C}}$ is obtained by adjoining a point at infinity to $\mathbb{C}$.

- Straight lines and circles on the complex plane are mapped to circles on the Riemann sphere.

- We can extend the domain and range of the inversion function $f(z) = 1/z$ by defining the point at infinity as the inverse of 0.

- Properties of linear fractional transformation:

  1. Composition of two LFT's is an LFT. The inverse function of LFT is an LFT.
  2. The class of linear fractional transformation can map any three distinct points in the extended complex plane to three other distinct points.

- The cross ratio of four complex numbers z_1, z_2, z_3 and z_4 is defined as

$$[z_0, z_1, z_2, z_3] \triangleq \frac{z_0 - z_1}{z_0 - z_3} \cdot \frac{z_2 - z_3}{z_2 - z_1}$$

  when $z_0, \ldots, z_3$ are all finite complex numbers. When one of $z_0, \ldots, z_3$ is ∞, the value can be computed by applying the convention $\infty/\infty = 1$.

- The value of cross ratio is invariant under linear fractional transformation.

Exercises

1. Let $A, B \in \mathbb{C}$ and let $k > 0$ be a constant. Consider the locus of points P satisfying

$$\frac{|P - A|}{|P - B|} = k.$$

(a) Show that this is the equation of a circle (or a line when $k = 1$) by rewriting the condition in complex form as $|z - a| = k|z - b|$.

(b) Find the center and radius of the circle explicitly.

2. Find the image of the half plane $x > c$, where c is a positive real number, under the inversion function $f(z) = 1/z$.

3. Find the image of the following geometric shapes under the map

$$f(z) = \frac{z - i}{z + i}.$$

(a) the closed disk with radius 1 and center at i, i.e., the set of points that satisfy $|z - i| \leq 1$.

(b) the closed unit disk $|z| \leq 1$.

4. Show that the linear fractional transformation

$$w = \frac{z - z_0}{z_0^* z - 1}$$

where $|z_0| < 1$, maps the unit circle $|z| = 1$ onto the unit circle $|w| = 1$.

5. Find the linear fractional transformation $w = \frac{az+b}{cz+d}$ that maps:

(a) $z_1 = 1$ to $w_1 = i, z_2 = i$ to $w_2 = -1, z_3 = -1$ to $w_3 = -i$;

(b) $z_1 = \infty$ to $w_1 = 1, z_2 = 0$ to $w_2 = i, z_3 = 1$ to $w_3 = \infty$.

6. Prove that the solutions to $z^n = (z + 1)^n$ lie on the vertical line $\mathrm{Re}(z) = -1/2$ in the complex plane.

7. Describe geometrically the transformation in parts (a) and (b) by decomposing them into elementary transformations (translation, rotation, dilation, inversion).

(a) $w = (1 + i)z + 2$ $\qquad$ (b) $w = \dfrac{z + 1}{z - 1}$

8. Write the linear fractional transformation

$$f(z) = \frac{z - 2}{z + 2}$$

as a composition of translation, scaling, rotation, and inversion.

9. Show that a linear fractional transformation has either 1 or 2 fixed points in the extended complex plane. Suppose $f(z) = (az + b)/(cz + d)$. Derive a condition on coefficients a, b, c and d under which $f(z)$ has two distinct fixed points.

10. Let C_1 and C_2 be two distinct circles in the complex plane. Prove that there exists a linear fractional transformation $f(z) = (az + b)/(cz + d)$ that maps C_1 and C_2 onto two circles C_1' and C_2' with the same finite radius.

    (Hint: Analyze the problem in four cases based on the intersection of the circles: (i) one includes the other in the interior, (ii) disjoint and exterior, (iii) intersecting at two points, and (iv) tangent at one point.)

11. Let C_1 and C_2 be two disjoint circles in the complex plane (one strictly inside the other). Prove that there exists a linear fractional transformation that maps C_1 and C_2 onto two concentric circles centered at the origin.

12. Let C_1 and C_2 be two circles that intersect at two distinct points z_a and z_b with an angle of intersection α $(0 < \alpha < \pi)$. Find a linear fractional transformation that maps the intersection region (the "lens" shape) onto an infinite angular sector (a wedge) of angle α with the vertex at the origin.

13. Let $z_1 = 0$, $z_2 = 1$, $z_3 = \infty$, and $z_4 = x \in \mathbb{C} \setminus \{0, 1\}$. Compute the cross ratio

$$[0, 1, \infty, x].$$

    What is the geometric meaning of your answer.

14. Let (z_1, z_2, z_3, z_4) and (w_1, w_2, w_3, w_4) be two quadruples of distinct points in $\bar{\mathbb{C}}$. Show that if

$$[z_1, z_2, z_3, z_4] = [w_1, w_2, w_3, w_4],$$

    then there exists a linear fractional transformation T such that $T(z_i) = w_i$ for all i.

15. Two points z and z^* are said to be *symmetric with respect to a circle* C if the line passing through the center of circle C and z also passes through z^*, and $|z - c||z^* - c| = R^2$ (where c is the center and R is the radius). Prove that linear fractional transformations preserve symmetry. That is, if z and z^* are symmetric with respect to a circle C, and T is an LFT, show that $T(z)$ and $T(z^*)$ are symmetric with respect to the image circle $T(C)$.

(Hint: Use the cross-ratio definition of symmetry: z, z^* are symmetric with respect to C passing through z_1, z_2, z_3 if and only if the cross-ratio $(z^*, z_1, z_2, z_3) = \overline{(z, z_1, z_2, z_3)}$. Use the fact that LFTs preserve cross-ratios.)

16. Using the cross ratio and linear fractional transformation, prove Ptolemy's theorem: four points A, B, C and D lie on a circle if and only if

$$\overline{AB} \cdot \overline{CD} + \overline{AD} \cdot \overline{BC} = \overline{AC} \cdot \overline{BD}.$$

17. Draw a circle with radius r centered at O. Pick two points A and B inside that circle so that $A \neq 0$ and $A \neq B$. On the line containing O and A, draw a point A^* so that $OA \cdot OA^* = r^2$, and A is between O and A. Similarly, on the line containing O and B, draw a point B^* so that $OB \cdot OB^* = r^2$, and B is between O and B. Show that A, B, A^* and B^* lie on a circle.

Chapter 3

Complex Exponential Function and Euler's formula

In this chapter we introduce the complex exponential function by defining it as an infinite series, extending its domain from the real number line to the complex plane.

3.1 Convergent sequences

Notation: We denote a *sequence* of complex numbers $z_1, z_2, z_3, \ldots$ is denoted by $(z_k)_{k=1}^{\infty}$ or simply $\{z_k\}$.

Definition 3.1.1. Given a complex sequence $(z_n)_{n=1}^{\infty}$, we say that z_n *converges* to a complex number $w \in \mathbb{C}$ if

$$\forall \epsilon > 0,\ \exists N,\ s.t.\ |z_n - w| < \epsilon,\ \forall n \geq N.$$

We write

$$\lim_{n\to\infty} z_n = w$$

if $(z_n)_{n=1}^{\infty}$ converges to the limit w.
A sequence that is *divergent* if it is not converging to any complex number in $\mathbb{C}$.

Geometrically, if a sequence is convergent, it is the same as saying that for any small ϵ,

the points in the complex sequence will eventually stay inside the open disk $D(w, \epsilon)$.

Example 3.1.1. The complex sequence $\{z_n\}$ defined by

$$z_n = \frac{1}{n} + \frac{i}{n^2}$$

converges to 0, because $|z_n| \to 0$ as $n \to \infty$.

Example 3.1.2. Compute $\lim_{n\to\infty} \frac{n}{n+i}$.

We can first make a guess that the limit should be 1, because when n is large, adding i to n has negligible effect. To make the argument rigorous, we write

$$\frac{n}{n+i} = \frac{n+i-i}{n+i} = 1 - \frac{i}{n+i}.$$

Then

$$\left|\frac{n}{n+i} - 1\right| = \left|\frac{i}{n+i}\right| = \frac{1}{\sqrt{n^2+1}} \to 0, \qquad \text{as } n \to \infty.$$

Therefore $n/(n+i) \to 1$ as $n \to \infty$.

The convergence of a complex sequence reduces to the real case through the following theorem.

Theorem 3.1.2. *A complex sequence $(z_n)_{n=1}^{\infty}$ converges if and only if both $(\mathrm{Re}(z_n))_{n=1}^{\infty}$ and $(\mathrm{Im}(z_n))_{n=1}^{\infty}$ converge.*

It is easy to show that a complex sequence is Cauchy if and only if the real and imaginary parts are real Cauchy sequences. As a result, the basic property of Cauchy sequence for real numbers extends to the complex case readily.

Definition 3.1.3. A sequence $(z_n)_{n=1}^{\infty}$ is called a *Cauchy sequence* if for all $\epsilon > 0$, there exists an integer N such that

$$|z_m - z_n| < \epsilon \quad \text{whenever } m, n \geq N.$$

The set of complex numbers inherits the completeness property of real numbers.

Theorem 3.1.4. *A complex sequence $(z_n)_{n=1}^{\infty}$ converges if and only if $(z_n)_{n=1}^{\infty}$ is Cauchy.*

Proof. The proof is a reduction to the real parts and the imaginary parts of the sequence by Theorem 3.1.2. □

If we consider the complex sequence as points on the Riemann sphere, then it may converge to the point at infinity, even though it does not converge to any point in the complex plane.

Definition 3.1.5. A sequence of complex numbers $\{z_k\}_{k=1}^{\infty}$ is said to *converge to the point at infinity* if for any positive number r, there exists a sufficiently large integer N such that $|z_k| > r$ for all $k \geq N$.

We say that a sequence z_k is *bounded* if there exists a real number M such that $|z_k| \leq M$ for all $k \geq 1$. A sequence that converges to ∞ cannot be bounded.

When the sequence $\{z_k\}_{k=1}^{\infty}$ converges to the point at infinity, we write $z_n \to \infty$ as $n \to \infty$, or

$$\lim_{n\to\infty} z_n = \infty.$$

Example 3.1.3. The sequence $(n(1+i))_{n=1}^{\infty}$ is divergent, but is converging to the point at infinity on the Riemann sphere.

Example 3.1.4. This example highlight a difference between sequence on the extended complex plane and sequence on the extended real number system. Let $\{z_k\}_{k=1}^{\infty}$ be a sequence of complex numbers such that

$$\lim_{n\to\infty} z_n = \infty.$$

Consider another sequence of complex numbers defined by $w_k = -z_k$. The sequence $\{w_k\}_{k=1}^{\infty}$ converges to the point at infinity as well,

$$\lim_{n\to\infty} w_n = \lim_{n\to\infty} (-z_n) = \infty.$$

The two sequences $(w_n)_{n\geq 1}$ and $(z_n)_{n\geq 1}$ have the same limit, even though they proceed in opposite direction.

We record some basic properties of complex sequences in the following theorem. The proof is the same as in the real case, and hence is omitted.

Theorem 3.1.6. *Let $(z_n)_{n=1}^{\infty}$ and $(w_n)_{n=1}^{\infty}$ be convergent complex sequences, with limits α and β, respectively.*

(a). $\lim_{n\to\infty}(az_n + bw_n) = a\alpha + b\beta$, *for complex constants a and b.*

(b). $\lim_{n\to\infty} z_n w_n = \alpha\beta$.

(c). $\lim_{n\to\infty} z_n/w_n = \alpha/\beta$, *if $\beta \neq 0$.*

(d). $\lim_{n\to\infty} |z_n| = |\alpha|$.

(e). $\lim_{n\to\infty}(z_n^*) = \alpha^*$.

3.2 Complex series

Definition 3.2.1. Given a sequence of complex numbers $(z_k)_{k=1}^{\infty}$, we define the value of the *infinite series* $\sum_{k=1}^{\infty} z_k$ by

$$\sum_{k=1}^{\infty} z_k \triangleq \lim_{n\to\infty} \sum_{k=1}^{n} z_k$$

if the limit exists.

As in calculus, the convergence of infinity series is the same as the convergence of sequence of partial sums.

Example 3.2.1. (Complex geometric series) Evaluate $\sum_{k=1}^{\infty}(0.5i)^k$.

For any finite n, we have

$$\sum_{k=1}^{n}(0.5i)^k = \frac{(0.5i)^{n+1} - 0.5i}{0.5i - 1}.$$

We take limit as $n \to \infty$,

$$\sum_{k=1}^{\infty}(0.5i)^k = \lim_{n\to\infty} \frac{(0.5i)^{n+1} - 0.5i}{0.5i - 1} = \frac{-0.5i}{0.5i - 1} = \frac{-1 + 2i}{5}.$$

We record a few elementary properties of complex series in the next theorem.

Theorem 3.2.2. *Let $\sum_{k=1}^{\infty} z_k$ denote a convergent complex series.*

(a). $\sum_{k=1}^{\infty}(z_k^*) = \left(\sum_{k=1}^{\infty} z_k\right)^*$.

(b). $\mathrm{Re}\left(\sum_{k=1}^{\infty} z_k\right) = \sum_{k=1}^{\infty} \mathrm{Re}(z_k)$.

(c). $\mathrm{Im}\left(\sum_{k=1}^{\infty} z_k\right) = \sum_{k=1}^{\infty} \mathrm{Im}(z_k)$.

We give a proof of the first part of this theorem. The proof of the remaining parts are similar.

Proof. Let L be the limit of $\sum_{k=1}^{\infty} z_k$. By definition, for any positive real number ϵ, there exists a positive integer N such that

$$\Big|\sum_{k=1}^{n} z_k - L\Big| \leq \epsilon$$

for all $n \geq N$. Because a complex number and its conjugate have the same absolute value, we can write

$$\Big|\sum_{k=1}^{n} z_k^* - L^*\Big| \leq \epsilon$$

for any $n \geq N$. Hence by the definition of limit,

$$\sum_{k=1}^{\infty} z_k^* = L^* = \big(\sum_{k=1}^{\infty} z_k\big)^*.$$

□

As in the real case, we have the linearity property for infinite series.

Theorem 3.2.3. *If $\sum_{k=1}^{\infty} z_k$ and $\sum_{k=1}^{\infty} w_k$ are convergent complex series, and α and β are complex constants, then*

$$\sum_{k=1}^{\infty}(\alpha z_k + \beta w_k) = \alpha \sum_{k=1}^{\infty} z_k + \beta \sum_{k=1}^{\infty} w_k.$$

The next two proposition provides two tests for convergence. The first one is a test for divergence, and the second one is a test for convergence.

Proposition 3.2.4. *(n-th term test) If $\sum_{k=1}^{\infty} z_k$ converges, then $|z_n| \to 0$ as $n \to \infty$. In other words, if $(|z_n|)_{n=1}^{\infty}$ does not converge to 0, then $\sum_{k=1}^{\infty} z_k$ does not converge.*

Proof. Suppose $\sum_{k=1}^{\infty} z_k$ converges to w. The sequence of partial sums $(\sum_{k=1}^{n} z_k)_{n=1}^{\infty}$ converges, and is thus a Cauchy sequence (Prop. 3.1.4 part (ii)). By the definition of Cauchy sequence, given any $\epsilon > 0$, there exists an integer N such that

$$\Big|\sum_{k=1}^{m} z_k - \sum_{k=1}^{n} z_k\Big| < \epsilon$$

for any $n, m \geq N$. By picking $m = n + 1$, we obtain $|z_{n+1}| < \epsilon$ for all $n \geq N$. □

Proposition 3.2.5. *(Absolute convergence test) Given a sequence of complex numbers $(z_k)_{k=1}^{\infty}$, if the real infinite series $\sum_{k=1}^{\infty} |z_k|$ converges, then $\sum_{k=1}^{\infty} z_k$ also converges.*

Proof. Suppose $\sum_{k=1}^{\infty} |z_k|$ converges. Since $|\operatorname{Re}(z_k)| \leq |z_k|$ for all $k \geq 1$, the real series $\sum_{k=1}^{\infty} |\operatorname{Re}(z_k)|$ converges. By the absolute convergence property for real series, $\sum_{k=1}^{\infty} \operatorname{Re}(z_k)$ converges. Similarly, $\sum_{k=1}^{\infty} \operatorname{Im}(z_k)$ is convergent. The convergence of $\sum_{k=1}^{\infty} z_k$ then follows by applying part (i) of Theorem. 3.1.4. □

Example 3.2.2. The series

$$\sum_{k=1}^{\infty} \frac{2 + i/k}{k^2 + 2ki} \tag{3.2.1}$$

is convergent because we can upper bound the magnitude of each term

$$\Big|\frac{2 + i/k}{k^2 + 2ki}\Big| \leq c/k^2$$

for some real constant c. Because the series $\sum_{k=1}^{\infty} k^{-2}$ is convergent, by Prop. 3.2.5, the series in (3.2.1) converges.

In view of Prop. 3.2.5, we make the following definition.

Definition 3.2.6. We say that a series $\sum_{k=1}^{\infty} z_k$ is *absolutely convergent* if $\sum_{k=1}^{\infty} |z_k|$ is convergent. If $\sum_{k=1}^{\infty} z_k$ converges but $\sum_{k=1}^{\infty} |z_k|$ does not converge, then we say that it *converges conditionally*.

Proposition 3.2.7. *(limit ratio test) Suppose z_k, for $k = 1, 2, 3, \ldots$ are complex numbers and $\lim_{k\to\infty} |z_{k+1}/z_k|$ exists and is equal to L.*

(a). If $L > 1$, then $\sum_{k=1}^{\infty} z_k$ diverges;

(b). If $L < 1$, then $\sum_{k=1}^{\infty} z_k$ converges absolutely.

Proof. First consider the case $L > 1$. By the meaning of converging to L, which is larger than 1 by assumption, there exists an integer N such that $|z_{k+1}/z_k| > 1$ for all $k \geq N$. Therefore

$$|z_N| < |z_{N+1}| < |z_{N+2}| < \cdots$$

By applying the n-th term test, we see that $\sum z_k$ is divergent.

Next consider the case $L < 1$. Let ϵ be a small positive real number such that $L < 1 - \epsilon < 1$. By the definition of convergence, there exists an integer N such that

$$\left|\frac{z_{k+1}}{z_k}\right| < 1 - \epsilon < 1$$

for all $k \geq N$. Then, we have

$$\begin{aligned} |z_{N+1}| &< |z_N|(1-\epsilon) \\ |z_{N+2}| &< |z_{N+1}|(1-\epsilon) < |z_N|(1-\epsilon)^2, \end{aligned}$$

and in general

$$|z_{N+k}| < |z_N|(1-\epsilon)^k$$

for $k \geq 1$. Therefore, for each positive number m, we have

$$\sum_{j=0}^{m} |z_{N+j}| \leq |z_N| \sum_{j=0}^{m} (1-\epsilon)^j.$$

Since $\sum_{j=0}^{\infty}(1-\epsilon)^j$ is finite, the sequence

$$\Big(\sum_{j=0}^{m} |z_{N+j}|\Big)_{m=0}^{\infty}$$

is a bounded and monotonically increasing sequence. Hence the series $\sum_{j=0}^{\infty} |z_j|$ is convergent. By Prop. 3.2.5, the infinite series $\sum_{j=0}^{\infty} z_{N+j}$ is also convergent. This proves that $\sum_{k=1}^{\infty} z_k$ is convergent. $\square$

We quote two properties of absolutely convergent series.

Proposition 3.2.8. *(Cauchy product) If $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ are absolutely convergent series, then*

$$\Big(\sum_{k=0}^{\infty} a_k\Big)\Big(\sum_{k=0}^{\infty} b_k\Big) = \sum_{k=0}^{\infty} c_k,$$

where

$$c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_{k-1} b_1 + a_k b_0.$$

Proposition 3.2.9. *If a series converges absolutely, then a series obtained by rearranging the terms converges and converges to the same limit.*

The proofs can be found in [Rudin, Theorem 3.50] and [Rudin, Theorem 3.55]. Although the corresponding theorems in [Rudin] are stated for real series, but the proofs works for the complex case in a straightforward manner.

3.3 Convergence of power series

We study the convergence of power series in this section. For ease of notation, we assume that the center of the power series is the origin. All results below hold in the general case when the center is not the origin.

Proposition 3.3.1. *Suppose $\sum_{k=0}^{\infty} a_k z^k$ converges at $z_0 \neq 0$, then it converges absolutely for all z with $|z| < |z_0|$.*

Proof. By the n-th term test (Prop. 3.2.4) we know that $|a_k z_0^k| \to 0$ as $k \to \infty$. Fix $\epsilon > 0$. there is a sufficiently large N such that

$$|a_k z_0^k| < \epsilon, \qquad \forall k \geq N.$$

Hence, for all $k \geq 0$, we have

$$|a_k z_0^k| \leq \max(\epsilon, |a_0|, |a_1 z_0|, |a_2 z_0^2|, \ldots, |a_{N-1} z_0^{N-1}|) \triangleq C.$$

We fix a complex number z with $|z| < |z_0|$. Let ρ be the ratio $|z|/|z_0|$. For each k, re-write $|a_k z^k|$ as

$$|a_k z^k| = |a_k z_0^k| \frac{|z|^k}{|z_0|^k} \leq C\rho^k,$$

We note that the constant C depends on the value of $|z|$.

Because $\rho < 1$ and $\sum_k C\rho^k$ is a convergent geometric series, we can apply comparison test and conclude that $\sum_k a_k z^k$ is convergent absolutely (See Prop. 3.2.5). □

We can re-write Prop. 3.3.1 as

Proposition 3.3.2. *If $\sum_k a_k z^k$ diverges at a point z_1, then $\sum_k a_k z^k$ diverges for all $|z| > |z_1|$.*

Proof. Suppose $\sum_k a_k z^k$ diverges at z_1. We want to show that if $\sum_k a_k z^k$ converges at any point z_2 with modulus $|z_2| > |z_1|$, we would obtain a contradiction. By the previous proposition, if such a z_2 exists, the power series must converge at $z = z_1$, and this directly contradicts the assumption that $\sum_k a_k z^k$ diverges at z_1. □

In view of the previous propositions, we can see that if a power series converges in a region, the region has a circular shape.

Definition 3.3.3. Given a power series $\sum_{k=0}^{\infty} a_k z^k$, we will call

$$R \triangleq \sup\left\{|z| : \sum_k a_k z^k \text{ converges}\right\} \tag{3.3.1}$$

the *radius of convergence* of $\sum_{k=0}^{\infty} a_k z^k$. The region $|z| < R$ is called the *region of convergence*.

By Theorem 3.3.1, the power series is convergent at all points z in the open disk $|z| < R$, and R is the largest radius such that the series converges for all points in the disk $|z| < R$. If $\sum_k a_k z^k$ converges for all z, then we write $R = \infty$, and say that the radius of convergence is infinite. If $\sum_k a_k z^k$ converges only at $z = 0$, then $R = 0$.

Example 3.3.1. The power series that define the complex functions $\exp(z)$, $\sin(z)$, $\cos(z)$, $\sinh(z)$ and $\cosh(z)$ have radius of convergence $R = \infty$.

Example 3.3.2. The power series $\sum_{k=0}^{\infty} z^k$ diverges for all z on the unit circle, but converges for all $|z| < 1$. Specifically, the k-th term in the series is z^k. When z lies on the unit circle, $|z|$ is equal to 1, had hence $|z|^k$ is also equal to 1. The k-th term test says that the power series $\sum_k z^k$ does not converge whenever $|z| = 1$.

The the convergence of the series $\sum_{k=0}^{\infty} z^k$ in $|z| < 1$ is not uniform. We note that the partial sum has a closed-form expression:

$$\sum_{k=0}^{n} z^k = \frac{1 - z^{n+1}}{1 - z},$$

which converges to $\frac{1}{1-z}$ for $|z| < 1$. However, the difference

$$\frac{1 - z^{n+1}}{1 - z} - \frac{1}{1 - z} = \frac{z^{n+1}}{1 - z}.$$

For each integer n, the supremum of $z^{n+1}/(1 - z)$ over $|z| < 1$ equals ∞. By the criterion in (5.5.1), the convergence of $\sum_{k=0}^{\infty} z^k$ is not uniform in $|z| < 1$. But the convergence is uniform if we restrict the domain to $|z| < r$ for any positive constant r that is strictly less than 1.

Example 3.3.3. The power series $\sum_{k=1}^{\infty} \frac{1}{k} z^k$ diverges at $z = 1$, but converges for all other points on the unit circle. An outline of proof is in the exercise.

Example 3.3.4. The power series $\sum_{k=1}^{\infty} \frac{1}{k^2} z^k$ converges at all points on the unit circle.

When $z = 1$, the series $\sum_{k=1}^{\infty} 1/k^2$ is convergent (converging to $\pi^2/6$). For other point z on the unit circle, the power series $\sum_{n=1}^{\infty} z^2/k^2$ converges by Theorem 5.5.4.

Example 3.3.5. (Lacunary series) The function $f(z)$ defined by

$$f(z) = \sum_{k=0}^{\infty} z^{2^k}$$

diverges when z is a 2^n-th root of unity, for all $n \geq 0$.

An explicit expression for calculating the radius of convergence is given by the Hadamard formula.

Theorem 3.3.4 (Hadamard formula for radius of convergence)**.** *Given a complex power series $\sum_{k=0}^{\infty} a_k z^k$, the radius of convergence can be computed by*

$$R = \frac{1}{\limsup_k |a_k|^{1/k}}.$$

Proof. Suppose z is a complex number with $|z| < R$. Pick a real number R_1 between $|z|$ and R, i.e., $|z| < R_1 < R$. By the definition of R in the theorem,

$$\frac{1}{R_1} > \limsup |a_k|^{1/k}.$$

Using the property of limsup, there exists a sufficiently large N such that for all $k \geq N$,

$$|a_k|^{1/k} < 1/R_1 \qquad \Rightarrow \qquad |a_k| < 1/R_1^k.$$

This yields

$$|a_k z^k| < \frac{|z|^k}{R_1^k}.$$

Since $|z|^k/R_1^k < 1$, the geometric series $\sum_k |z|^k/R_1^k$ is convergent. By comparison test, $\sum_k a_k z^k$ is convergent.

Now suppose that $|z| > R$. Pick a real number R_2 such that $|z| > R_2 > R$. Taking reciprocal,

$$\frac{1}{R_2} < \limsup |a_k|^{1/k}.$$

By the defining property of limsup,

$$\frac{1}{R_2} < |a_k|^{1/k} \quad \text{for infinitely many } k.$$

Hence $|a_k z^k| > \left|\frac{z}{R_2}\right|^k > 1$ for infinitely many k. By the nth-term test (Prop. 3.2.4), the power series diverges whenever $|z| > R$. □

3.4 Complex exponential function

Definition 3.4.1. A *complex power series* centered at the origin is a series in the form $\sum_{k=0}^{\infty} a_k z^k$, where $a_k \in \mathbb{C}$.

The power series in the next theorem is the most important power series.

Theorem 3.4.2. *For any complex number $z \in \mathbb{C}$, the power series $\sum_{k=0}^{\infty} z^k/k!$ converges. Moreover, it converges absolutely.*

Proof. It follows from the limit ratio test. The ratio of two consecutive terms has absolute value

$$\frac{\left|\frac{z^{k+1}}{(k+1)!}\right|}{\left|\frac{z^k}{k!}\right|} = \frac{|z|}{k+1}.$$

For any fixed complex number z, the absolute value $|z|$ is fixed, and hence the above ratio has limit zero. By part (b) of Proposition 3.2.7, the power series converges absolutely. $\square$

We can also use Theorem 3.3.4 to show that the radius of convergence is infinite. By Stirling's approximation, we have

$$k! \sim \sqrt{2\pi k} k^k e^{-k},$$

so that

$$(k!)^{1/k} \sim k/e.$$

The k-th root of the k-th coefficient approaches 0. By Hadamard formula, the radius of convergence is infinite. The power series converge for all points in the complex plane.

Definition 3.4.3. Define the *complex exponential function* $\exp(z)$ by the complex power series

$$\exp(z) \triangleq \sum_{k=0}^{\infty} \frac{z^k}{k!} = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$$

for $z \in \mathbb{C}$. An alternate notation is e^z.

When z is a real number, the above definition is the same as the power series of the exponential function with real domain. For example, we have $\exp(0) = 1$, and $\exp(1)$ is the constant e. This power series has infinite radius of convergence.

We establish below a fundamental property of the function $\exp(z)$.

Theorem 3.4.4. *For any complex numbers z_1 and z_2,*

$$\exp(z_1 + z_2) = \exp(z_1)\exp(z_2).$$

Proof. We know that both series $\sum_{k=0}^{\infty} z_1^k/k!$ and $\sum_{k=0}^{\infty} z_2^k/k!$ converge absolutely. We can apply Prop. 3.2.8 to $a_k = z_1^k/k!$ and $b_k = z_2^k/k!$. This gives

$$\exp(z_1)\exp(z_2) = \sum_{k=0}^{\infty} c_k$$

where c_k is defined by

$$c_k \triangleq \sum_{\ell=0}^{k} \frac{z_1^\ell}{\ell!}\frac{z_2^{k-\ell}}{(k-\ell)!}$$

for $k \geq 0$. By binomial theorem, we can simplify it to

$$c_k = \sum_{\ell=0}^{k} \frac{z_1^\ell}{\ell!}\frac{z_2^{k-\ell}}{(k-\ell)!} = \frac{1}{k!}\sum_{\ell=0}^{k}\binom{k}{\ell} z_1^\ell z_2^{k-\ell} = \frac{1}{k!}(z_1+z_2)^k.$$

Hence,

$$\exp(z_1)\exp(z_2) = \sum_{k=0}^{\infty}\frac{(z_1+z_2)^k}{k!} \triangleq \exp(z_1+z_2).$$

□

Using this theorem we can derive some immediate properties

Theorem 3.4.5.

(a). For any $z \in \mathbb{C}$, $\exp(z) \neq 0$, and $e^{-z} = (e^z)^{-1}$.

(b). If $z = x + iy$, then $\exp(z) = e^x e^{iy}$.

(c). For real number y, we have $|e^{iy}| = 1$.

Proof. To see that $e^z \neq 0$ for all $z \in \mathbb{C}$, we take $z_1 = z$ and $z_2 = -z$ in Theorem 3.4.4,

$$e^z e^{-z} = e^{z-z} = e^0 = 1.$$

Since 0 times any complex number w is equal to 0, we cannot have $e^z = 0$, otherwise it would contradict the above equality, which says that there exists a complex number, namely e^{-z}, such that e^z time e^{-z} is equal to a nonzero value. We can also deduce from this equality that the multiplicative inverse of e^z is equal to e^{-z}.

For any real numbers x and y, we let $z_1 = x$ and $z_2 = iy$ and apply Theorem 3.4.4 to get

$$e^{x+iy} = e^x e^{iy}.$$

The factor e^x is a real number because x is real. On the other hand, e^{iy} lies on the unit circle, because

$$|e^{iy}|^2 = e^{iy}(e^{iy})^* = e^{iy}e^{-iy} = e^{iy-iy} = e^0 = 1.$$

We have used part 1 of Theorem 3.2.2 to replace $(e^{iy})^*$ by e^{-iy}. This proves that e^{iy} has modulus 1. □

3.5 Euler's formula

From Theorem 3.4.5, we know that e^{iy} lies on the unit circle for any real number y. The precise location is given by Euler's formula.

Theorem 3.5.1 (Euler's formula)**.** *For any real number θ, we have*

$$e^{i\theta} = \cos(\theta) + i\sin(\theta). \tag{3.5.1}$$

Proof. The proof follows from re-arranging the terms in the power series that defines $e^{i\theta}$. This is permissible because

$$e^{i\theta} = 1 + \frac{i\theta}{1} + \frac{i^2\theta^2}{2!} + \frac{i^3\theta^3}{3!} + \frac{i^4\theta^4}{4!} + \frac{i^5\theta^5}{5!} + \cdots$$

is an absolutely convergent power series. By separating the real and imaginary parts by terms re-arrangement, we obtain

$$e^{i\theta} = \Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\Big) + i\Big(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\Big)$$

the first term is precisely the power series expansion of $\cos(\theta)$, and the second term is i times the power series expansion of $\sin(\theta)$. □

Combining Theorems 3.4.5 and 3.5.1, we can evaluate the complex exponential function in terms of the real exponential function and the real sine and cosine functions

Theorem 3.5.2.

$$e^{x+iy} = e^x(\cos(y) + i\sin(y)). \tag{3.5.2}$$

Remark. Some books, such as [BrownChurchill] defines the complex exponential function by (3.5.2). This approach usually defines holomorphic function first, and then verify that (3.5.2) is an example of holomorphic function. A drawback of this order of definition is that Euler's formula is then true by definition. In this book, we took the path of defining the complex exponential function by power series. With this approach, Euler's identity is a magical formula. Nevertheless, we will see later (in Example **??**) that there is only one complex function that is compatible with the real exponential function. Any definition of complex exponential function will yield the same result.

Remark. The Euler's formula is a generalization of DeMoivre formula. We can recover DeMoivre formula from Euler's formula by considering $e^{in\theta}$, which can be written as

$$e^{in\theta} = \cos(n\theta) + i\sin(n\theta).$$

By Prop. 3.4.4, we can also write it as

$$e^{in\theta} = (e^{i\theta})^n = (\cos\theta + i\sin\theta)^n.$$

Example 3.5.1. We can compute $e^{\sqrt{2}+i}$ by

$$e^{\sqrt{2}+i} = e^{\sqrt{2}}(\cos(1) + i\sin(1)).$$

Example 3.5.2. When $z = \pi i$, we obtain

$$e^{\pi i} = \cos(\pi) + i\sin(\pi) = -1.$$

From Euler's formula, we see that the complex exponential function is a periodic function with complex period $2\pi i$.

Theorem 3.5.3. *For any $z \in \mathbb{C}$, we have $\exp(z + 2\pi ik) = \exp(z)$ for all integers k.*

Example 3.5.3. Consider the infinite series

$$\sum_{k=0}^{\infty} \frac{\cos k\theta}{2^k}$$

as a function of real number θ. This series is convergent by applying the comparison test and comparing with $\sum_{k=0}^{\infty} 1/2^k$. We can evaluate the exact value by defining $z \triangleq \cos\theta + i\sin\theta$ and considering the series as the real part of the complex series

$$\sum_{k=0}^{\infty} \frac{z^k}{2^k}.$$

This is a complex geometric series, and the limit is

$$\frac{1}{1-\frac{z}{2}}.$$

The remaining task is to obtain the real part of $(1-z/2)^{-1}$. To this end, we write $(1-z/2)^{-1}$ as

$$\begin{aligned}\frac{2}{2-\cos\theta - i\sin\theta} &= \frac{2}{2-\cos\theta - i\sin\theta}\cdot\frac{2-\cos\theta+i\sin\theta}{2-\cos\theta+i\sin\theta}\\ &= \frac{4-2\cos\theta+2i\sin\theta}{5-4\cos\theta}.\end{aligned}$$

Therefore,

$$\sum_{k=0}^{\infty}\frac{\cos k\theta}{2^k} = \frac{4-2\cos\theta}{5-4\cos\theta}.$$

This is a periodic function of θ.

3.6 Application to alternating current

In an alternating-current (AC) circuit, voltages and currents vary sinusoidally in time. A typical voltage source can be written as

$$V(t) = V_0\cos(\omega t+\phi),$$

where V_0 represents the amplitude, ω is the angular frequency, and ϕ is the phase. Using Euler's formula,

$$e^{i(\omega t+\phi)} = \cos(\omega t+\phi) + i\sin(\omega t+\phi),$$

we can regard this voltage as the real part of a complex exponential:

$$V(t) = \mathrm{Re}\big(V_0 e^{i(\omega t+\phi)}\big).$$

The complex quantity

$$V_0 e^{i\phi}$$

is called a *phasor*. Although the physical voltage is always a real-valued function, the complex exponential form is often easier to manipulate.

Figure 3.1 illustrates the corresponding voltage phasor in the complex plane.

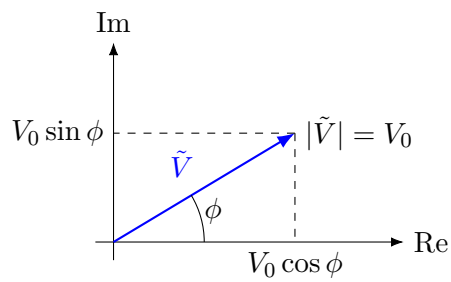


Figure 3.1: Voltage phasor $\tilde{V} = V_0 e^{i\phi}$ in the complex plane.

Example 3.6.1. Consider n sinusoidal functions with the same frequency ω,

$$A_k \cos(\omega t + \theta_k),$$

where amplitudes A_k and phase θ_k, for $k = 1, 2, \ldots, n$, are real numbers. The sum of them is another sinusoidal function with the same frequency ω. To find the amplitude and phase of the sum, we can simply add the n complex numbers $A_k e^{i\theta_k}$, for $k = 1, 2, \ldots, n$,

$$\sum_{k=1}^{n} A_k e^{i\theta_k} \triangleq z_0.$$

The modulus and argument of z_0 is then the amplitude and the phase of the resulting sinusoidal function

$$\sum_{k=1}^{n} A_k \cos(\omega t + \theta_k) = \mathrm{Re}\left(e^{i\omega t} \sum_{k=1}^{n} A_k e^{i\theta_k}\right) = \mathrm{Re}(e^{i\omega t} z_0) = |z_0| \cos(\omega t + \arg(z_0)).$$

Summary

- The basic properties of complex sequence and series are the same as real sequence and real series.

- (Absolute convergence of power series) If a power series $\sum_{k=0}^{\infty} a_k z^k$ converges at $z_0 \neq 0$, then it converges absolutely for all z with $|z| < |z_0|$.
- The region of convergence of power series is circular in shape. The radius of convergence is defined as
$$R \triangleq \sup\left\{|z| : \sum_k a_k z^k \text{ converges}\right\}.$$
- Hadamard formula for the radius of convergence:
$$R = \frac{1}{\limsup_k |a_k|^{1/k}}.$$
- "A function $f(z)$ converges to ∞" means that $1/f(z)$ converges to 0.
- Complex exponential function is defined by extending the power series of the real exponential function to complex variable. This power series is absolutely convergent for all complex numbers.
- A formula for computing complex exponential function: $e^{x+iy} = e^x(\cos y + i \sin y)$.
- Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$, for all real number θ.

Exercises

1. Determine whether the following sequences are convergent (converging to the point at infinity is allowed). If yes, what is the limit?

   (a) $\lim_{n\to\infty} \frac{n^2+i}{in^2+2}$, (b) $\lim_{n\to\infty} \sin(i\pi n)$, (c) $\lim_{n\to\infty} \left(\frac{z}{n}\right)^n$.

2. Show that
$$\sum_{k=0}^{n} \cos^2(k\theta) = \frac{n+1}{2} + \frac{\sin((n+1)\theta)}{\sin\theta}\cos(n\theta)$$
   for real number θ that is not an integral multiple of $\pi/2$.

3. Evaluate the followings and express your answer in the form $a + bi$.

   (a) $e^{3i\pi}$, (b) $e^{1+\pi i/6}$, (c) $\exp(\exp(\pi i/8))$, (d) $e^{412\pi i/6}$.

4. Let $(z_n)_{n\geq 1}$ and $(w_n)_{n\geq 1}$ be two convergent sequences of complex numbers. Prove that

$$\lim_{n\to\infty} \frac{z_{n-1}w_0 + z_{n-2}w_1 + \cdots + z_0 w_{n-1}}{n} = \big(\lim_{n\to\infty} z_n\big)\big(\lim_{n\to\infty} w_n\big).$$

5. (A problem from Polya) Given two series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, their Cauchy product is defined as the series $\sum_{n=0}^{\infty}$, with coefficient c_n given by

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

   (a) Suppose that $\sum_{n=0}^{\infty} a_n$ is absolutely convergent. Prove that if $\sum_{n=0}^{\infty}$ is convergent (but not necessarily converging absolutely), then the series $\sum_{n=0}^{\infty} c_n$ converges.

   (b) (hard) Show that if the Cauchy product of $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ is convergent whenever $\sum_{n=0}^{\infty} b_n$ is convergent, then $\sum_{n=0}^{\infty} a_n$ is necessarily an absolutely convergent series.

6. Prove the following identities about binomial coefficients.

$$\sum_{\substack{0\leq k\leq n \\ k\equiv 0 \bmod 4}} \binom{n}{k} = 2^{n-2} + 2^{(n-2)/2}\cos\big(\frac{n\pi}{4}\big)$$

$$\sum_{\substack{0\leq k\leq n \\ k\equiv 1 \bmod 4}} \binom{n}{k} = 2^{n-2} + 2^{(n-2)/2}\sin\big(\frac{n\pi}{4}\big)$$

$$\sum_{\substack{0\leq k\leq n \\ k\equiv 3 \bmod 4}} \binom{n}{k} = 2^{n-2} - 2^{(n-2)/2}\sin\big(\frac{n\pi}{4}\big)$$

7. (π formulae)

   (a) Compute the product $(2+i)(3+i)$ and hence prove

$$\frac{\pi}{4} = \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}.$$

   (b) Using similar technique, show that $\pi/4$ can be computed by

$$\frac{\pi}{4} = 4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{239}.$$

   (Hint: use $(5-i)^4(1+i)$)

(c) Use the result in part 2 and the power series expansion Use the power series expansion

$$\tan^{-1}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

to estimate the value of π.

8. The objective of this exercise is to prove

$$\sum_{k=0}^{n-1} \frac{1}{\cos^2(\pi k/n)} = n^2.$$

for odd n.

(a) Suppose that $x_1, x_2, \ldots, x_n$ are the roots of polynomial $p(z) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$. Assuming all the roots are nonzero, show that

$$\sum_{k=1}^{n} \frac{1}{x_k^2} = \frac{a_1^2 - 2a_0 a_2}{a_0^2}.$$

(b) For integer $m \geq 1$, define the m-th Chebyshev polynomial $T_m(x)$ by the relation

$$\cos(m\theta) = T_m(\cos\theta).$$

Show that $\cos(\pi k/n)$, for $0 \leq k \leq 2n-1$ are the roots of $T_{2n}(x) - 1$.

(c) Using parts (a) and (b), derive

$$\sum_{k=0}^{2n-1} \frac{1}{\cos^2(\pi k/n)} = 2n^2.$$

Chapter 4

Complex Trigonometric Function and Log Function

We complexify the elementary functions in calculus.

4.1 Trigonometric functions

We extend the definition of trigonometric function and hyperbolic trigonometric functions to the complex plane.

Definition 4.1.1. For $z \in \mathbb{C}$, define the *complex sine* and *complex cosine* functions by

$$\sin(z) \triangleq \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!},$$
$$\cos(z) \triangleq \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}.$$

By the limit ratio test (Prop. 3.2.7), both of them converge absolutely on the whole complex plane. The power series reduce to the real trigonometric and real when z is a real number. We can express the Euler's formula as

Theorem 4.1.2. *For $z \in \mathbb{C}$,*

$$e^{iz} = \cos(z) + i\sin(z). \tag{4.1.1}$$

Proof. The proof is basically the same as the proof of Theorem 3.5.1. We just need to apply the definition of $\cos(z)$ and $\sin(z)$ for complex number z.

Starting from the power series expansion of e^{iz}

$$e^{iz} = 1 + iz - \frac{z^2}{2!} - \frac{iz^3}{3!} + \frac{z^4}{4!} + \frac{iz^5}{5!} - \frac{z^6}{6!} + \cdots$$

because of absolute convergence, we can re-arrange the terms and separate the terms without i and the terms with i,

$$\begin{aligned} e^{iz} &\triangleq \sum_{n=0}^{\infty} \frac{(iz)^n}{n!} \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!} \\ &= \cos(z) + i\sin(z). \end{aligned}$$

The last equality is true by the definition of complex sine and complex cosine functions. □

Using Theorem 4.1.2, we can express sine function and cosine function in terms of the complex exponential function.

Theorem 4.1.3. *For any $z \in \mathbb{C}$,*

$$\begin{aligned} \cos(z) &= \frac{e^{iz} + e^{-iz}}{2} \\ \sin(z) &= \frac{e^{iz} - e^{-iz}}{2i}. \end{aligned}$$

Proof. The complex cosine function is an even function, i.e., $\cos(-z) = \cos(z)$, because the power series that define $\cos(z)$ consists of even powers. Similarly, because all terms in the power series that define $\sin(z)$ have odd powers, the complex sine function is an odd function, i.e., $\sin(-z) = -\sin(z)$. This gives

$$e^{-iz} = \cos(-z) + i\sin(-z) = \cos(z) - i\sin(z) \tag{4.1.2}$$

By adding (4.1.2) to (4.1.1), we get

$$e^{iz} + e^{-iz} = 2\cos(z).$$

By subtracting (4.1.2) from (4.1.1), we get

$$e^{iz} - e^{-iz} = 2i\sin(z).$$

□

The result in Theorem 4.1.3 holds for any complex numbers. In particular, when we restrict z to a real number θ, this yields

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}, \tag{4.1.3}$$

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \tag{4.1.4}$$

Remark. Some people take the two identities in Theorem 4.1.3 as the definition of complex sine and complex cosine, and then deduce Definition 4.1.1 as a corollary. Which definitions we adopt is a matter of taste. The resulting theory is the same.

Example 4.1.1. Evaluate $\sin(i)$. By Theorem 4.1.3,

$$\sin(i) = \frac{e^{i\cdot i} - e^{-i\cdot i}}{2i} = \frac{e^{-1} - e^{1}}{2i} = i\frac{e^{1} - e^{-1}}{2} = i\sinh(1).$$

The trigonometric functions in the numpy library can compute complex number. We can verify the calculations in this example as follows.

Example 4.1.2. For any real number a, we can evaluate $\cos(\pi + ai)$ by using the formula in Theorem 4.1.3,

$$\cos(\pi + ai) = \frac{e^{i(\pi+ai)} + e^{-i(\pi+ai)}}{2} = \frac{e^{-a+i\pi} + e^{a-i\pi}}{2} = -\frac{e^{-a} + e^{a}}{2} = -\cosh(a).$$

Some trigonometric identities in the real case continue to hold when the variables are complex numbers.

Theorem 4.1.4. *For any complex numbers z and w, the following hold:*

(a). $\cos^2 z + \sin^2 z = 1$.

(b). $\sin(z+w) = \sin z \cos w + \sin w \cos z$.

(c). $\cos(z+w) = \cos z \cos w - \sin z \sin w$.

Proof. We prove the first part. The proofs of the rest are similar.

$$\begin{aligned} L.H.S. &= \cos^2 z + \sin^2 z \\ &= \left(\frac{e^{iz}+e^{-iz}}{2}\right)^2 + \left(\frac{e^{iz}-e^{-iz}}{2i}\right)^2 \\ &= \frac{1}{4}[(e^{iz}+e^{-iz})^2 - (e^{iz}-e^{-iz})^2] \\ &= \frac{1}{4}[e^{2iz}+2+e^{-2iz} - e^{2iz}+2-e^{-2iz}] \\ &= 1 = R.H.S. \end{aligned}$$

□

The complex hyperbolic sine and cosine functions are defined similarly by power series.

Definition 4.1.5. For $z \in \mathbb{C}$, The *complex hyperbolic sine* and *complex hyperbolic cosine* functions are defined as

$$\sinh(z) \triangleq \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!},$$
$$\cosh(z) \triangleq \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!}.$$

The hyperbolic trigonometric functions are closely related to the complex sine and cosine functions.

Theorem 4.1.6. *For all complex number $z \in \mathbb{C}$,*

$$\sin(iz) = i\sinh(z),$$
$$\cos(iz) = \cosh(z).$$

Proof. The first identity can be proved by considering the power series expansion of $\sin(iz)$,

$$\sin(iz) \triangleq \sum_{k=0}^{\infty}(-1)^k \frac{(iz)^{2k+1}}{(2k+1)!} = i\sum_{k=0}^{\infty}(-1)^k \frac{i^{2k}z^{2k+1}}{(2k+1)!} = i\sum_{k=0}^{\infty}\frac{z^{2k+1}}{(2k+1)!} = i\sinh(z).$$

The proof of the second identity is similar. □

In particular, for any real number x, we have

$$\sin(ix) = i\sinh(x), \quad \text{and } \cos(ix) = \cosh(x).$$

Theorem 4.1.7. *For any complex number $z = x + iy$, we have*

$$\cos(z) = \cos x \cosh y - i \sin x \sinh y,$$
$$\sin(z) = \sin x \cosh y + i \cos x \sinh y.$$

4.2 Log function

The complex exponential function has a complex period $2\pi i$; that is, for any $z \in \mathbb{C}$,

$$e^{z+2\pi k i} = e^z, \quad \text{for } k \in \mathbb{Z}.$$

As a result, the inverse function of e^z is multi-valued.
Notation: We will use the notation $\ln x$ for the natural log function with positive real number as input.

Example 4.2.1. Find $z \in \mathbb{C}$ such that $e^z = 1 + i$.

The complex number $1 + i$ in polar coordinates is

$$\sqrt{2}(\cos(\pi/4) + i\sin(\pi/4)).$$

We want to find all complex numbers $x + iy$ such that

$$e^x(\cos y + i \sin y) = \sqrt{2}(\cos(\pi/4) + i \sin(\pi/4)).$$

By comparing the moduli and arguments, we have

$$\begin{aligned} x &= \frac{1}{2}\ln(2) \\ y &= \pi/4 + 2\pi k, \quad \text{for } k = 0, \pm 1, \pm 2, \ldots \end{aligned}$$

Hence

$$\log(1+i) = \frac{1}{2}\ln(2) + i(\pi/4 + 2\pi k)$$

for $k \in \mathbb{Z}$.

In general, given a complex number $r(\cos\theta + i\sin\theta)$ in polar form, the complex log of $r(\cos\theta + i\sin\theta)$ can take values

$$\ln r + i(\theta + 2\pi k)$$

for any integer k. In general, we have the following

Definition 4.2.1. For nonzero $w \in \mathbb{C}$, the *complex log function* is a multi-function defined as

$$\log(w) \triangleq \ln|w| + i(\arg(w) + 2\pi k)$$

where $k = 0, \pm 1, \pm 2, \ldots$

The function $\arg(w)$ in Definition 4.2.1 is the multi-valued argument function. If we specify a range for the angle by take the principal argument (Definition 1.4.3), we have a uniquely defined function.

Definition 4.2.2. The function

$$\mathrm{Log}(w) \triangleq \ln|w| + i\,\mathrm{Arg}(w),$$

is called the *principal value* or the *principal branch* of complex log function. We will denote the principal log function by $\mathrm{Log}(z)$.

The complex log function extends the domain of the real log function to the whole complex plane except the origin. We note that it is not possible to define the $\log(0)$ even if we extend the number system to complex numbers, because the complex exponential function is never equal to zero. One way to see it is by the following general fact

$$\forall z \in \mathbb{C}, \quad e^z e^{-z} = e^{z-z} = e^0 = 1.$$

If we assume $e^z = 0$ for some z, then we would obtain $0 \cdot 0 = 1$, which cannot be true. This means that $e^z \neq 0$ for all $z \in \mathbb{C}$.

Using the complex log function we can compute the log of a negative number.

Example 4.2.2. Compute the principal log of -10, using a principal argument function with range $[0, 2\pi)$.

We first write -10 as

$$-10 = 10(\cos\pi + i\sin\pi)$$

If $e^{x+iy} = 10(\cos\pi + i\sin\pi)$, we must have $10 = e^x$, giving $x = \ln 10$. The imaginary part y equals π, which is the unique argument in the range $[0, 2\pi)$. Therefore, the principal log of -10 is

$$\mathrm{Log}(-10) = \ln(10) + i\pi.$$

4.3 Power function

We can define the complex power function via the complex log function.

Definition 4.3.1. Given a nonzero complex number z and a complex number w, we define the *complex power* z^w by

$$z^w \triangleq e^{w\log(z)}$$

where $\log(z)$ is the complex log function.

Example 4.3.1. Compute 2^i.

$$\begin{aligned} 2^i \triangleq \exp(i\log 2) &= \exp(i(\ln 2 + i2\pi k)) \\ &= \exp(-2\pi k + i\log 2) \\ &= e^{-2\pi k}(\cos(\ln 2) + i\sin(\ln 2)), \end{aligned}$$

where k can take any integer as its value. We can consider the value when $k = 0$ as the principal value. If we want to write down just one answer, we can write

$$2^i = \cos(\ln 2) + i\sin(\ln 2).$$

We can use the complex power function to compute the n-th roots of a complex numbers. We can compare this method with the DeMoivre formula in (1.4.1).

Example 4.3.2. Calculate the cube roots of $2i$ by $(2i)^{1/3} \triangleq \exp(\log(2i)/3)$.

We compute

$$\begin{aligned} \log(2i) &= \ln 2 + i\big(\pi/2 + 2\pi k\big), \qquad \text{for } k \in \mathbb{Z} \\ \frac{\log(2i)}{3} &= \frac{\ln 2}{3} + i\big(\pi/6 + 2\pi k/3\big). \end{aligned}$$

Exponentiating both sides, we get

$$\exp\big(\frac{\log(2i)}{3}\big) = \sqrt[3]{2}\Big(\cos\big(\frac{\pi}{6} + \frac{2\pi k}{3}\big) + i\sin\big(\frac{\pi}{6} + \frac{2\pi k}{3}\big)\Big)$$

for $k = 0, 1, 2$. There are three answers, all with magnitude $\sqrt[3]{2}$.

The next example has complex base and complex exponents at the same time.

Example 4.3.3. Compute i^i by calculating $e^{i\log i}$.

$$\begin{aligned} \exp(i\log i) &= \exp\Big(i(\log 1 + i(\frac{\pi}{2} + 2\pi k))\Big) \\ &= \exp\big(-(\frac{\pi}{2} + 2\pi k)\big), \end{aligned}$$

where $k \in \mathbb{Z}$. All possible values are real numbers. By taking $k = 0$, we can say that the principal value of i^i is $e^{-\pi/2}$.

4.4 Inverse of complex trigonometric functions

Inverse trigonometry function can be computed by solving quadratic equation.

Example 4.4.1. Find all complex numbers z such that $\cos(z) = 2$.

Recall from Theorem 4.1.3 that we can express $\cos(z)$ as

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2}.$$

Let $w = e^{iz}$. This amounts to solve

$$\frac{w + w^{-1}}{2} = 2.$$

This is a quadratic equation $w^2 - 4w + 1 = 0$. The roots are $2 \pm \sqrt{3}$. Hence

$$e^{iz} = 2 \pm \sqrt{3}.$$

By taking the complex log function, we obtain

$$z = -i\big[\ln(2 \pm \sqrt{3}) + 2\pi i k\big] = -i \ln(2 \pm \sqrt{3}) + 2\pi k,$$

where $k \in \mathbb{Z}$.

Example 4.4.2. Solve $\cot(z) = 2i$ for $z \in \mathbb{C}$.

This is equivalent to solving

$$\frac{e^{iz} + e^{-iz}}{2} \frac{2i}{e^{iz} - e^{-iz}} = 2i.$$

By letting $w = e^{iz}$, we can simplify it to

$$\begin{aligned} \frac{w + w^{-1}}{w - w^{-1}} &= 2 \\ \frac{w^2 + 1}{w^2 - 1} &= 2. \end{aligned}$$

The values for w is $\pm\sqrt{3}$. Then, we can obtain z by taking complex log

$$z = -i \log(\pm\sqrt{3}) = -i(\ln \sqrt{3} + i\pi k) = \pi k - i \ln \sqrt{3},$$

where k is any integer.

4.5 Visualization of complex function using phase portrait

Phase portrait, which is also known as domain coloring, plots the argument of a complex function on a plane, by associating the numbers in the range $[0, 2\pi)$ with the color spectrum. The brightness corresponds to the magnitude of the complex function. A dark point means the function is zero. A bright color indicates the area where function magnitude is large. We take the phase portrait of the identity function as the reference (Fig. 4.1).

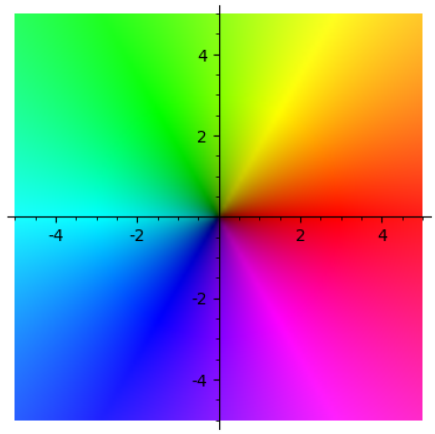


Figure 4.1: Phase portrait of $f(z) = z$

The phase portrait of some complex functions are shown in Fig. 4.2. The picture for the function $f(z) = 1/z$ has a bright point near the origin, because this function has large magnitude near the origin. We also note that as we go around the origin, we see the color in the reverse order. It is because the function $f(z) = 1/z$ multiply the argument by -1. In the phase portrait of $f(z) = z^2$, we see a zero of order 2 at the origin. As we go around the origin, we meet all the colors twice.

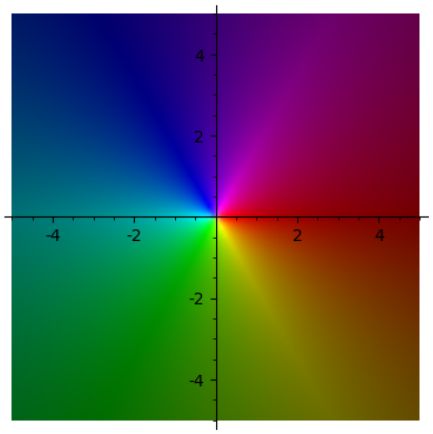
4.6 Numerical calculations

Python support complex trigonometric functions. For example, we can check that $\sin(i) = i\sinh(1)$ by

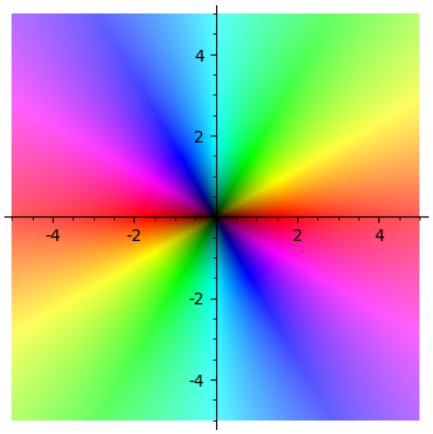
```
In [0]: from numpy import sin, sinh
In [1]: sin(1j)
Out[1]: 1.1752011936438014j

In [2]: sinh(1)
Out[2]: 1.1752011936438014
```

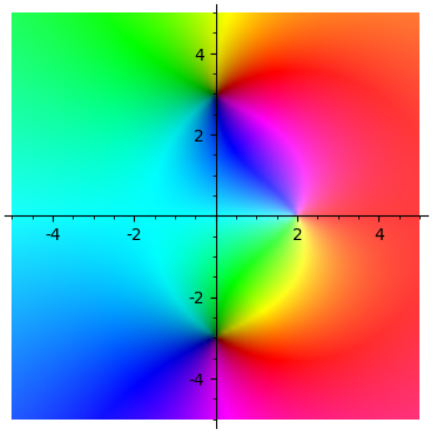
The arccos function in Python is able to compute the complex arc cosine function. We can check that it returns $-i\log(2+\sqrt{3})$ as the answer.



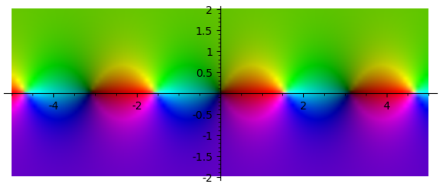
(a) Phase portrait of $f(z) = 1/z$



(b) Phase portrait of $f(z) = z^2$



(c) Phase portrait of
$f(z) = (z^2 + 9)/(z - 2)$



(d) Phase portrait of $f(z) = \tan(z)$

Figure 4.2: Phase portraits.

```
In [0]: from numpy import arccos, log
In [1]: arccos(2+0j)
Out[1]: -1.3169578969248166j

In [2]: -log(2+sqrt(3))*1j
Out[2]: (-0-1.3169578969248166j)
```

The other solution is $-i\log(2-\sqrt{3})$.

```
In [3]: -log(2-sqrt(3))*1j
Out[3]: 1.3169578969248164j
In [4]: cos(-log(2-sqrt(3))*1j)
Out[4]: (1.9999999999999996-0j)
In [5]: cos(-log(2+sqrt(3))*1j)
Out[5]: (1.9999999999999998-0j)
```

If we directly ask Python to compute $\log(-10)$, we will get a warning, saying that there is a type mismatch.

```
In [0]: from numpy import log
In [1]: log(-10)
RuntimeWarning: invalid value encountered in log
Out[1]: nan
```

One way to compute it works is to manually convert the input to a complex number.

```
In [2]: log( complex(-10) )
Out[2]: (2.302585092994046+3.141592653589793j)
```

An alternate way is to use the complex log function in the cmath library. It automatically converts the input value to the complex number data type.

```
In [1]: from cmath import log
In [2]: log(-10)
Out[2]: (2.302585092994046+3.141592653589793j)
```

The following Python commands compute the principal value of 2^i.

```
In [0]: from numpy import cos, sin, log
In [1]: 2**1j
```

```
Out[1]: (0.7692389013639721+0.6389612763136348j)

In [2]: cos(log(2))+1j*sin(log(2))
Out[2]: (0.7692389013639721+0.6389612763136348j)
```

The value of i^i is a real number $e^{-\pi/2}$.

```
In [0]: from numpy import exp, pi
In [1]: 1j**1j
Out[1]: (0.20787957635076193+0j)

In [2]: exp(-pi/2)
Out[2]: 0.20787957635076193
```

Relation between complex log function and arctan function.

```
In [0]: from numpy import log, arctan

In [1]: -1j*log(3**(1/2))
Out[1]: -0.5493061443340548j

In [2]: arctan(1/2j)
Out[2]: -0.5493061443340549j
```

4.7 Summary

- The complex version of $\sin(z)$, $\cos(z)$, $\sinh(z)$, and $\cosh(z)$ are defined by power series, extending the domain to the whole complex plane.
- The Euler's formula $e^{iz} = \cos(z) + i\sin(z)$ holds for all complex number $z \in \mathbb{C}$.
- The complex exponential function is related to the complex trigonometric functions by
$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2}, \quad \text{and } \sin(z) = \frac{e^{iz} - e^{-iz}}{2i}.$$
- The complex log function is a multi-function:
$$\log(z) \triangleq \ln|z| + i(\arg(z) + 2\pi k),$$

defined for $z \neq 0$, where $k = 0, \pm 1, \pm 2, \ldots$

- The complex power is defined in terms of log function,

$$z^w \triangleq e^{w \log(z)},$$

and hence is a multi-function.

Exercises

1. Evaluate $\sin(z)$, $\cos(z)$, $\tan(z)$ for the following values of z:

   (a) $1 - i$, (b) $1 + i$ (c) $\frac{\pi}{4} i$.

2. In each part, find all complex number z that satisfy the equation.

   (a) $z^{\sqrt{2}} = 2i$ (b) $\cos(z) = -4$ (c) $e^z = 1 + \sqrt{3}i$

   (d) $z^i = 3$ (e) $\ln(z) = \pi i$

3. Find a solution to the following equation

$$i^z = 1 + i,$$

   and verify your answer with Python.

4. Define

$$\tanh z \triangleq \frac{\sinh z}{\cosh z} \quad (\text{for } \cosh z \neq 0), \qquad \coth z \triangleq \frac{\cosh z}{\sinh z} \quad (\text{for } \sinh z \neq 0),$$

   Prove that

   (a) $\tanh(iz) = i \tan z$,

   (b) $\tan(iz) = i \tanh z$,

   (c) $\coth(iz) = -i \cot z$,

   (d) $\cot(iz) = -i \coth z$.

5. Solve

$$\cos z + i \sin z = 1/\sqrt{3}.$$

6. In this exercise we derive some trigonometric identities using complex roots of unity.

   (a) By expressing $\tan(\pi/5)$ and $\sin(\pi/10)$ in terms of the 5-th root of unity $e^{2\pi i/5}$, show that

$$\tan\left(\frac{\pi}{5}\right) = \sqrt{5 - 2\sqrt{5}}$$
$$\sin\left(\frac{\pi}{10}\right) = \frac{\sqrt{5} - 1}{4}.$$

   (b) By using the 7-th root of unity $e^{2\pi i/7}$, prove

$$\tan\left(\frac{\pi}{7}\right) + \tan\left(\frac{2\pi}{7}\right) + \tan\left(\frac{3\pi}{7}\right) = \sqrt{7}.$$

7. Derive the following formulae for inverse trigonometric functions.

   (a) $\arcsin(z) = -i\log(\sqrt{1 - z^2} + iz)$.

   (b) $\arccos(z) = -i\log(z + i\sqrt{1 - z^2})$.

   (c) $\arctan(z) = \frac{1}{2i}\log\frac{1+iz}{1-iz}$.

8. Show that the zeros of $\sin(z)$ and $\cos(z)$ are located on the real axis.

   (a) $\sin z = 0$ if and only if $z = k\pi$ for some integer k.

   (b) $\cos z = 0$ if and only if $z = (\pi/2) + k\pi$ for some integer k.

Chapter 5

Topology of the Complex Plane and Continuity

5.1 Point-set topology in the (extended) complex plane

The complex plane has the same topology as the Euclidean plane $\mathbb{R}^2$. As a result, the convergence concepts and limits in the real case and the complex case are basically the same. The concepts of point-set topology in $\mathbb{R}^2$ can be directly imported to the complex case. For example, open disc is defined as follows.

Definition 5.1.1. The *open disk* of radius r centered at z_0 is the set

$$D(z_0, r) \triangleq \{z \in \mathbb{C} : |z - z_0| < r\}.$$

The collection of all open disks $D(z, r)$, with z varying over all points in $\mathcal{C}$, and $r > 0$, is the base of the standard topology on $\mathbb{C}$.

Definition 5.1.2. A subset A in $\mathbb{C}$ is called an *open set* for every point $z_0 \in A$, we can find a sufficiently small radius r so that $D(z_0, r)$ is a subset of A. A set B is said to be *closed* if the complement of B in $\mathbb{C}$ is open, i.e., for any point w_0 that is not in B, there is radius r such that $D(w_0, r)$ and B are mutually disjoint.

Example 5.1.1. The upper half-plane $\mathrm{Im}(z) > 0$ is open. Given any point z in the upper

half-plane, we can enclose it by an open disk $D(z, |z|/2)$, which lies inside the upper half-plane.

Example 5.1.2. The set $D(z_0, r)$ is open according to Definition 5.1.2. Pick any point $w \in D(z_0, r)$. The distance between w and the closest point on the circle $|z - z_0| = r$ is $r - |w - z_0|$. The set $D(w, (r - |w - z_0|)/2)$ lies inside the set $D(z_0, r)$, hence $D(z_0, r)$ is an open set.

It is not difficult to show that (i) the empty set is open, (ii) the whole complex plane is open, (iii) arbitrary union of open sets is open, and (iv) intersection of finitely many open sets is open. These are the axioms of open sets in topology. In general, given a set X and a collection $\mathcal{T}$ of subsets of X that satisfies these four axioms, the pair $(X, \mathcal{T})$ is called a *topology space*.

In the extended complex plane, we also need a notion of open disks centered at the point at infinity. A convenient way to formalize this is to use the inversion map $w = f(z) = 1/z$. Under this transformation, a small disk around the origin in the w-plane,

$$\{w \in \mathbb{C} : |w| < \epsilon\},$$

corresponding to a neighborhood of ∞ in the z-plane. Thus an open disk centered at infinity is any set of the form

$$\{\, z \in \mathbb{C} : |z| > R \,\} \cup \{\infty\},$$

for some radius $R > 0$. Geometrically speaking, a point with very large modulus is close to the point at infinity.

Remark. In this book, the term "*neighborhood of a point*" will mean a set S whose interior contains the given point. In other words, a set S is a neighborhood of point p if we can find a small disc centered at p that is completely inside S. A neighborhood of a point needs not be an open set. We will use the term "open neighborhood" to refer to a neighborhood that is an open set.

The concept of accumulation point, closure, interior, and isolated point are defined in the same way as in Euclidean plane. We briefly review them in the remainder of this section.

Definition 5.1.3. A set S is called *closed* if its complement $S^c = \mathbb{C} \setminus S$ is open. A point z in $\mathbb{C}$ is called an *accumulation point* of a set A if, for all $r > 0$, we can find a point $z \in D(z_0, r) \cap A$ that is different from z_0.

We can prove that a set A is closed if and only if A contains all of its accumulation points.

Definition 5.1.4. We define the *closure* of a set S as the union of S and the accumulation points of S, and use the notation $\overline{S}$ to refer to the closure of S.

Example 5.1.3. the closure of open disk $D(z_0, r)$ is the set of complex numbers z that satisfy the inequality $|z - z_0| \leq r$. We will denote a closed disk centered at z_0 with radius r by $\overline{D}(z_0, r)$.

Example 5.1.4. Given two real numbers a and b with $a < b$, the annulus $\{z : a \leq |z| \leq b\}$ is a closed set. The accumulation points lie on the circle $|z| = a$ and $|z| = b$.

Definition 5.1.5. Given a set $S \subseteq \mathbb{C}$, a point $z \in S$ is called an *isolated point* of S if we can find a radius $r > 0$ such that $D(z, r) \cap S = \{z\}$. A set S is said to be *discrete* all points in S are isolated.

Geometrically, the points in a discrete set are separated from each other by some positive distance. It can be shown that a discrete set is closed, but do not have any accumulation point.

Example 5.1.5. The set of points $\{1/n : n \in \mathbb{N}_{>0}\}$ is discrete. However, the set of points $\{1/n : n \in \mathbb{N}_{>0}\} \cap \{0\}$ is not discrete, because the origin is an accumulation point.

Example 5.1.6. (Point lattice in complex plane) Let w_1 and w_2 be two nonzero complex numbers that are linear independent as vectors. The set of points

$$\{aw_1 + bw_2 : a, b \in \mathbb{Z}\}$$

is a discrete set.

5.2 Limit and continuity of complex function

Definition 5.2.1. (Limit of function) Let f be a complex function defined on a set $S \subseteq \mathbb{C}$, and let z_0 be a point in the closure of S. We say that $f(z)$ *converges* to a complex number w as $z \to z_0$ if for each $\epsilon > 0$, we can find $\delta > 0$ such that for all $z \in S$ satisfying $0 < |z - z_0| < \delta$, we have $|f(z) - w| < \epsilon$.

We note that the value of f at z_0 is irrelevant in the above definition. In the case when z_0 is an accumulation point of S but not in S, then the function value $f(z_0)$ is undefined. However, if the limit exists, the value of the limit is determined uniquely by the value of function $f(z)$ for z close z_0 (Exercise 1).

If we take the point at infinity into account, we define the limit of a function as follows:

1. When z_0 is finite and the limit w is ∞, $\lim_{z\to z_0} f(z) = \infty$ means
$$\lim_{z\to z_0} \frac{1}{f(z)} = 0.$$
2. When $z_0 = \infty$ and w is finite, $\lim_{z\to\infty} f(z) = w$ means
$$\lim_{z\to 0} f(1/z) = w.$$
3. When both z_0 and w is ∞, $\lim_{z\to\infty} f(z) = \infty$ means
$$\lim_{z\to 0} \frac{1}{f(1/z)} = 0.$$

Definition 5.2.2. (Function continuity) A function $f : S \to \mathbb{C}$ defined on a set $S \subseteq \mathbb{C}$ is said to be *continuous at a point* $z_0 \in S$ if, for any $\epsilon > 0$, there exists $\delta > 0$ such that
$$(z \in S \text{ and } |z - z_0| < \delta) \;\Rightarrow\; |f(z) - f(z_0)| < \epsilon.$$
A function f is said to be *continuous* in a domain S if f is continuous at every point in S.

The definition of continuity at a point is the same as requiring that (i) $f(z_0)$ exists, (ii) $\lim_{z\to z_0} f(z)$ exists, and (iii) $\lim_{z\to z_0} f(z) = f(z_0)$.

We note that by the definition of continuity at a point, a function is always continuous at an isolated point.

Continuity at the point at infinity is defined by: given a function $f : S \to \mathbb{C}$, we say that f is continuous at the point at infinity if for all $\epsilon > 0$, there exists $R > 0$ such that

$$(z \in S \text{ and } |z| > R) \;\Rightarrow\; |f(z) - f(\infty)| < \epsilon.$$

In the next theorem, the value of w_0, z_0 nad ζ may be the point at infinity. This is a result on the Riemann sphere.

Theorem 5.2.3. *(Limit of composite function) Let $f(z)$ and $g(w)$ denote complex functions Suppose $f(z)$ converges to w_0 as $z \to z_0$, $g(w)$ converges to ζ as $w \to w_0$, and g is continuous at w_0. Then,*

$$\lim_{z \to z_0} g(f(z)) = \zeta.$$

If the point at infinity is allowed in the domain of f and g, then the proof of the above theorem requires the consideration of several cases, depending on whether z_0, w_0 and ζ are finite or ∞.

By consider the real and imaginary part separately, we can prove the following

Theorem 5.2.4. *A complex function f is continuous if and only if the real and imaginary parts are continuous.*

The proof of this theorem is straightforward and is omitted.

Another way to understand continuity is by means of sequence. If a complex function f is continuous at z_0, and if $(z_k)_{k=1}^{\infty}$ is a sequence of complex numbers converging to z_0, then the image points $(f(z_k))_{k=1}^{\infty}$ should converge to $f(z_0)$. We can approach z_0 from any direction, and the limit of the sequence should equal $f(z_0)$.

The limit for complex function behaves in the same as the limit of vector function. Hence, complex limit has the same properties as in multi-variable calculus.

Example 5.2.1. Show that $f(z) = 1/z$ is continuous in the domain $\mathbb{C} \setminus \{0\}$.

Suppose $z = x + iy$ and $z \neq 0$.

$$\frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$$

The real part of $f(z)$ is $x/(x^2+y^2)$ and the imaginary part is $-y/(x^2+y^2)$. Both of them are continuous functions in the domain $\mathbb{C} \setminus \{0\}$. Hence $f(z)$ is continuous by the previous theorem.

Example 5.2.2. We compute the limit of the linear fractional transformation $f(z) = \frac{z+i}{z-8}$ as z approaches ∞. We first write the function as

$$\frac{z+i}{z-8} = \frac{z-8+8+i}{z-8} = 1 + \frac{8+i}{z-8}.$$

Since we are considering $z \to \infty$, the fractional $(8+i)/(z-i)$ will tend to 0, and the limit is 1.

If we want to verify that the answer is indeed equal to 1 by definition, we need to bound

$$\left|\frac{z+i}{z-8} - 1\right| = \left|\frac{8+i}{z-8}\right| = \frac{\sqrt{65}}{|z-8|}.$$

Using the reverse triangular inequality, we know that $|z-8| \geq |z| - 8$. Let ϵ be a positive real number. If we set

$$r = \frac{\sqrt{65}}{\epsilon} + 8,$$

then for all z with modulus $|z| > r$, we have

$$\frac{\sqrt{65}}{|z-8|} \leq \frac{\sqrt{65}}{|z|-8} \leq \frac{\sqrt{65}}{r-8} = \epsilon.$$

This proves that

$$\lim_{z\to\infty} \frac{z+i}{z-8} = 1.$$

Another way to understand continuity is by means of sequence. If a complex function f is continuous at z_0, and if $(z_k)_{k=1}^{\infty}$ is a sequence of complex numbers converging to z_0, then the image points $(f(z_k))_{k=1}^{\infty}$ should converge to $f(z_0)$. We can approach z_0 from any direction, and the limit of the sequence should equal $f(z_0)$.

The limit for complex function behaves in the same as the limit of vector function. Hence, complex limit has the same properties as in multi-variable calculus.

The next theorem is an expected result, because the analogous result holds for real functions.

Theorem 5.2.5. *The composition of two continuous complex functions is continuous.*

Let $f(z)$ be a complex-valued function mapping from a set $A \subseteq (C)$ to $\mathbb{C}$. We define the pre-image of a set $S \subseteq \mathbb{C}$ under a function $f(z)$ by

$$f^{-1}(S) \triangleq \{z \in A : f(z) \in S\}.$$

Using this notation, we can characterize continuity as:

Theorem 5.2.6. *A complex function $f : A \to \mathbb{C}$ is continuous if and only if for all open set $U \subseteq \mathbb{C}$, the pre-image $f^{-1}(U)$ is an open set in A.*

Proof is left as an exercise (Exercise 3).

5.3 Paths and parametric curves

Definition 5.3.1. A *path/parametric curve* is defined as a continuous function γ, that maps a closed interval $[a, b]$ to the complex numbers $\mathbb{C}$:

$$\gamma : [a, b] \to \mathbb{C}. \tag{5.3.1}$$

The *trajectory* or *locus* of a path is the set of points that is equal to the image of a path $\gamma : [a, b] \to \mathbb{C}$. We denote the trajectory of γ by $|\gamma|$.

We call $\gamma(a)$ and $\gamma(b)$ the initial point and end point, respectively. We say that the curve represented by γ goes from $\gamma(a)$ to $\gamma(b)$. Since γ is continuous, the locus of a parametric curve is a compact set.

Given two distinct complex numbers z_1 and z_2, the line segment from z_1 to z_2 can be represented parametrically by

$$\gamma(t) = z_1(1 - t) + z_2 t$$

for $t \in [0, 1]$. Alternately, we may represent the same trajectory by

$$\tilde{\gamma}(t) = z_1(1 - t^2) + z^2 t^2$$

for $t \in [0, 1]$. How a curve is parameterized is part of the data of parametric curve. As a result, $\gamma(t)$ and $\tilde{\gamma}(t)$ are considered two different parametric curve, even though the underlying trajectory is the same. We will call a path a *line segment* from z_1 to z_2 if the start

points and end points are z_1 and z_2, respectively, and the locus of the path consists of the points $z_1(1-t)+z_2t$, for $t \in [0,1]$.

A circular path centered at z_0 with radius r and counter-clockwise orientation can be represented in polar form:

$$\gamma(t) = z_0 + re^{i\theta} \tag{5.3.2}$$

for $\theta \in [0, 2\pi]$.

Path concatenation. Given two paths $\gamma_1 : [a_1, b_1] \to \mathbb{C}$, and $\gamma_2 : [a_2, b_2] \to \mathbb{C}$, if $\gamma_1(b) = \gamma_2(a)$, we can concatenate the two paths. The combined path

$$\gamma : [a_1, b_1 + b_2 - a_2] \to \mathbb{C}$$

is defined as

$$\gamma(t) \triangleq \begin{cases} \gamma_1(t) & \text{for } t \in [a_1, b_1] \\ \gamma_2(t + b_1 - a_1) & \text{for } t \in [b_1, b_1 + b_2 - a_2]. \end{cases}$$

We denote the *concatenated* path by $\gamma_1 + \gamma_2$. Even though we use the symbol "+" to stand for path concatenation, it is important to note that path addition is not commutative. Moreover, if the end point of γ_1 and the start point of γ_2 do not coincide, then the sum of γ_1 and γ_2 are not defined.

However, if the concatenation of m paths $\gamma_1, \ldots, \gamma_m$ is well defined, i.e., when the end point of γ_i is the start point of γ_{i+1}, then the path summation is associative, and we will denote the overall concatenated path as

$$\gamma_1 + \gamma_2 + \cdots + \gamma_m.$$

Path negation. The *negation* or the *reverse* of a path $\gamma : [a, b] \to \mathbb{C}$ is the path with the same locus with reverse direction, denoted by $-\gamma$,

$$(-\gamma)(t) = \gamma(a + b - t) \quad \text{for } t \in [a, b].$$

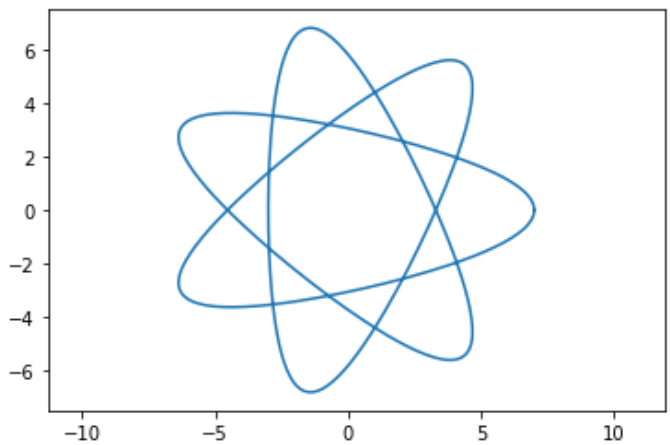


Figure 5.1: Parametric plot of $\gamma(t) = 5e^{i3t} + 2e^{-i4t}$ for $0 \leq t \leq 2\pi$.

Definition 5.3.2. The parametric curve in (5.3.1) is said to be

(a) *piece-wise linear* or *polygonal* if it is the concatenation of finitely many line segments.

(b) *simple* if there is no self-intersection, i.e., $\gamma(t_1) \neq \gamma(t_2)$ for any $a \leq t_1 < t_2 \leq b$, except when $t_1 = a$ and $t_2 = b$.

(c) *closed* if the start point is the same as the end point, i.e., $\gamma(a) = \gamma(b)$.

Example 5.3.1. We can obtain very symmetrical curves using parameterization in the form

$$\gamma(t) = ae^{ikt} + be^{i\ell t}$$

for $t \in [0, 2\pi]$. For example, if we choose $a = 5$, $b = 2$, $k = 3$ and $\ell = -4$, the corresponding curve is drawn in Fig. 5.1. We note that this is a closed curve with 7-fold symmetry. This is a closed but not simple path.

5.4 Connected set

Definition 5.4.1. Two nonempty sets A and B in $\mathbb{C}$ are said to be *separated* if $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$.

In other words, A and B are separated set if the accumulation points of A are not in B, and the accumulation points of B are not in A.

Example 5.4.1. The two open disks $D(-1,1)$ and $D(1,1)$ are separated, even though the origin is a common accumulation point. However, the open disk $D(-1,1)$ and the closed disk $\overline{D}(1,1)$ are not separated.

Definition 5.4.2. An nonempty set $X \subseteq \mathbb{C}$ is said to be *disconnected* if X can be written as the union of two separated sets. A set is said to be *connected* if it is not disconnected.

In other words, a connected set cannot be written as the union of two separated sets.

Example 5.4.2. The square lattice $\{x + iy : x, y \in \mathbb{Z}\}$ is disconnected. In fact, any discrete set defined in Definition 5.1.5 is disconnected.

We introduce a few more concepts in topology. Let X be nonempty subset in $\mathbb{C}$. If we regard X be a metric space by itself, then the distance function pertaining to X is just the restriction of the distance function of $\mathbb{C}$, namely the absolute value. However, an open set in X may not be be open as a subset of $\mathbb{C}$. In view of this, an open set in X will be referred to as a *relative open set*, a closed set in X is called a *relative closed set*. The *relative topology* of X is the collection

$$\mathcal{T}_X \triangleq \{X \cap U : \text{open set } U \text{ in } \mathbb{C}\}.$$

It is straight forward to show that the sets in $\mathcal{T}_X$ satisfy the four axioms of topology, so that $(X, \mathcal{T}_X)$ is a topological space.

For example, if X is the real axis in $\mathbb{C}$, the relative open sets X are the same as usual open sets in $\mathbb{R}$.

Example 5.4.3. From real analysis, open interval (a, b) and closed interval are connected.

Theorem 5.4.3. *Let X be a nonempty subset of $\mathbb{C}$. The following are equivalent.*

(a) X is disconnected, i.e., X can be written as the union of two separated nonempty sets.

(b) X can be written as the union of two disjoint and nonempty open sets (in the relative topology of X).

In particular, X is connected if and only if it cannot be written as the union of two disjoint and nonempty open sets in the relative topology of X.

Proof. We first show that (1) implies (2). Suppose X is disconnected. Then, by definition, there exist nonempty sets A and B such that

$$X = A \cup B, \qquad A \cap B = \emptyset,$$

and A and B are separated, that is,

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset,$$

where the closures are taken in $\mathbb{C}$.

Since $X = A \cup B$ and $B \cap \overline{A} = \emptyset$, we have

$$X \cap \overline{A} = A.$$

But $X \cap \overline{A}$ is exactly the closure of A in the relative topology of X. Hence A is relatively closed in X. Similarly, from $A \cap \overline{B} = \emptyset$ we obtain

$$X \cap \overline{B} = B,$$

so B is relatively closed in X as well.

Now A and B are complements of each other in X, so $A = X \setminus B$ and $B = X \setminus A$. Since A and B are relatively closed, their complements are relatively open. Therefore A and B are both open in the relative topology of X. We have thus written X as the union of two disjoint, nonempty, relatively open sets. This proves (2).

We now show that (2) implies (1). Suppose X can be written as $X = U \cup V$, where U and V are disjoint, nonempty sets that are open in the relative topology of X. Then U and

V are also relatively closed in X, because

$$X \setminus U = V, \qquad X \setminus V = U,$$

and complements of relatively open sets are relatively closed.

Since U is relatively closed in X, its closure in X is U itself. We write $\overline{U}^X$ to represent the relative closure of U in X. We have $\overline{U}^X = U$. But the closure in X is given by $\overline{U}^X = X \cap \overline{U}$, where $\overline{U}$ denotes the closure in $\mathbb{C}$. Hence

$$X \cap \overline{U} = U.$$

Now suppose, for contradiction, that there exists a point $z \in \overline{U} \cap V$. Then $z \in X$ (since $V \subseteq X$), and $z \in \overline{U}$, so $z \in X \cap \overline{U} = U$. This contradicts the assumption that U and V are disjoint. Therefore

$$\overline{U} \cap V = \emptyset.$$

A similar argument shows that $U \cap \overline{V} = \emptyset$. Thus U and V are separated sets in the sense of Definition 5.4.1.

We have thus written X as the union of two separated nonempty sets U and V. This proves that X is disconnected. □

Theorem 5.4.4. *If $f(z)$ is a continuous function on a connected set X, the image $f(X)$ is also connected.*

Proof. Let Y denote the image of X under the function f, and suppose that Y is not connected. We can find two sets U and V which is open in the relative topology of Y such that $Y = U \cup V$. By the continuity of f, the preimage $f^{-1}(U)$ and $f^{-1}(V)$ are open (in the relative topology of X). This contradict that X is connected. □

Definition 5.4.5. A function $f : X \to \mathbb{C}$ is said to be *locally constant* if every point $z \in X$ is contained in a neighborhood $U \subseteq X$ such that the restriction of f on U is a constant function.

In general, a locally constant function may not be a constant function, and it depends on whether the set X is connected.

Example 5.4.4. Let X be the union of two open spheres $D(0,1)$ and $D(2,1)$. For any complex constants a and b, the function

$$f_{a,b}(z) = \begin{cases} a & \text{if } |z| < 1 \\ b & \text{if } |z-2| < 1 \end{cases}$$

is locally constant. The function $f_{a,b}$ is not constant if $a \neq b$.

Theorem 5.4.6. *A locally constant function is continuous.*

Proof. Let X be a nonempty subset of $\mathbb{C}$ (equipped with the relative topology), and let $f : X \to \mathbb{C}$ be a locally constant function. We want to show that f is continuous. By definition of continuity in terms of topology, it suffices to show that for every open set V in $\mathbb{C}$, the pre-image $f^{-1}(V)$ is a relative open set in X.

Let V be an open set in $\mathbb{C}$, and suppose that $f^{-1}(V)$ is nonempty. Take any point $z_0 \in f^{-1}(V)$. Then $f(z_0) \in V$. Since f is locally constant, there exists a neighborhood $U \subseteq X$ of z_0 such that the restriction of f on U is a constant function. Denote this constant value by c. In particular, $c = f(z_0) \in V$.

Therefore, for every $z \in U$, we have $f(z) = c \in V$, and hence $U \subseteq f^{-1}(V)$. This shows that every point $z_0 \in f^{-1}(V)$ is contained in a neighborhood U (relative open set in X) such that $U \subseteq f^{-1}(V)$.

Consequently, $f^{-1}(V)$ is a union of relative open sets in X, and hence $f^{-1}(V)$ is itself a relative open set in X. Since V was an arbitrary open set in $\mathbb{C}$, we conclude that f is continuous. □

Theorem 5.4.7. *Let X be an nonempty subset of $\mathbb{C}$. With the relative topology of X, the followings are equivalent.*

(a) Every locally constant function on X is a constant function.

(b) A subset in X that is both open and closed is either the empty set or the set X.

(c) X cannot be written as a union of two disjoint and nonempty open sets (in the relative topology of X).

(d) X is connected .

Proof. ((a)⇒(b)) Suppose S is a nonempty subset of X such that S and $X \setminus S$ are both open in the relative topology of X. We want to show that $S = X$.

Consider the indicator function

$$\mathbf{1}_S(z) = \begin{cases} 1 & \text{if } z \in S, \\ 0 & \text{if } z \notin S. \end{cases}$$

We claim that it is a locally constant function. We first consider the case $z \in S$. Since S is open in the relative topology of X, we can find a set U that is open in $\mathbb{C}$ such that $S = U \cap X$. This implies that $z \in U$, and we can thus find an open disk $D(z, r)$ with a small enough radius r such that $D(z, r) \subseteq U$. Then the restriction of $\mathbf{1}_S$ to $D(z, r) \cap X$ is constantly equal to 1.

For the case $z \in X \setminus S$, we can apply a similar argument to show that there is an open set V in $\mathbb{C}$ such that $z \in V$ and $V \cap X \subseteq X \setminus S$. Hence the restriction of $\mathbf{1}_S$ to $V \cap X$ is constantly equal to 0. This completes the proof of the claim that $\mathbf{1}_S$ is locally constant.

By the hypothesis in part (a), $\mathbf{1}_S$ is a constant function. The constant must be equal to 1 as S is nonempty. Therefore $X \setminus S$ must be the empty set, and so $S = X$.

((b)⇒(c)) Suppose X can be written as the union of A and B, for some disjoint sets A and B, such that both A and B are nonempty and open (in the relative topology of X). Then A and B are both open and closed sets in X.

Since A and B are both nonempty, by the assumption in part (b) we must have $A = X$ and $B = X$. But this violates the assumption that A and B are disjoint. This gives a contradiction. Therefore, X cannot be written as the union of two disjoint and nonempty open sets.

((c)⇒(a)) Suppose $f : X \to \mathbb{C}$ is a locally constant function on X. Since X is nonempty, we can choose a point $z_0 \in X$. Denote the pre-image

$$U \triangleq f^{-1}\big(\{f(z_0)\}\big).$$

The set U is a subset of X such that all points in U have the same f-value as z_0. Obviously $z_0 \in U$ and U is not empty. Moreover, U is open in X because f is locally constant (the pre-image of a point under a locally constant function is a union of relative open sets).

If U is not equal to X, then consider

$$V \triangleq f^{-1}\big(\mathbb{C} \setminus \{f(z_0)\}\big).$$

The set V is an open subset of X (again by local constancy of f), and it is disjoint from U by construction. Furthermore, $U \cup V = X$ and V is nonempty (otherwise f would already be constant with value $f(z_0)$). This contradicts the condition in part (c) that X cannot be written as the union of two disjoint and nonempty open sets. Hence, U must be equal to X, and f is constant on X.

((c)$\Leftrightarrow$(d)) By definition, a subset X of $\mathbb{C}$ (equipped with the relative topology) is said to be *connected* if it cannot be written as the union of two disjoint and nonempty open sets. This is exactly the statement in part (c) by Theorem 5.4.3. Therefore, the condition in part (d) is equivalent to the condition in part (c). □

As a corollary, we have

Theorem 5.4.8. *The restriction of a locally constant function to a connected component of the domain is a constant function.*

We next investigate path-connected set.

Definition 5.4.9. A set X in $\mathbb{C}$ is said to be

(a) *connected* if it satisfies one of the equivalent conditions in Theorem 5.4.7,

(b) *path-connected* if for any two points z_1 and z_2 in S, there is a path in X from z_1 to z_2.

A example of path-connected set is depicted in Fig. 5.2.

Theorem 5.4.10. *If a subset X of $\mathbb{C}$ is path-connected, then it is connected (in the relative topology of X).*

Proof. Suppose X is path-connected. To show that X is connected, we verify condition (c) of Theorem 5.4.7.

Assume that X can be written as

$$X = A \cup B,$$

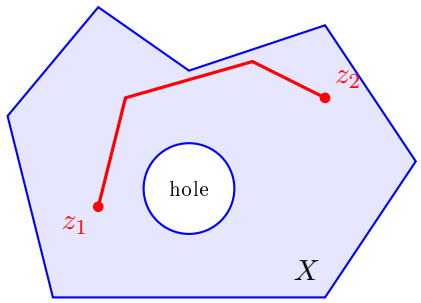


Figure 5.2: A path-connected set

where A and B are disjoint, nonempty sets that are open in the relative topology of X. Choose points $z_1 \in A$ and $z_2 \in B$. Since X is path-connected, there exists a path

$$\gamma : [0,1] \to X$$

such that $\gamma(0) = z_1$ and $\gamma(1) = z_2$.

Consider the sets

$$S \triangleq \gamma^{-1}(A), \qquad T \triangleq \gamma^{-1}(B).$$

Because A and B are open in X and γ is continuous, both S and T are open in $[0,1]$ (with the relative topology). They are disjoint, and

$$[0,1] = S \cup T,$$

since every point of the path lies in either A or B. Moreover, S and T are both nonempty: $0 \in S$ because $\gamma(0) = z_1 \in A$, and $1 \in T$ because $\gamma(1) = z_2 \in B$.

Thus we have written the interval $[0,1]$ as a union of two disjoint, nonempty, open subsets. But it is known from real analysis that the interval $[0,1]$ is connected, so such a decomposition is impossible.

This contradiction shows that no such sets A and B can exist. Therefore X cannot be written as the union of two disjoint nonempty open sets, and condition (c) of Theorem 5.4.7 holds. Hence X is connected. □

Convex set is a special kind of connected set.

Definition 5.4.11. A set S in $\mathbb{C}$ is said to be *convex* if for any two points z_1 and z_2 in S, the line segment between z_1 and z_2 is in S.

For example, a disk and a half-plane are convex, but a punctured disk and an annulus is not convex.

Theorem 5.4.12. *Let X be a nonempty open set in $\mathbb{C}$. Then X is path-connected if it is connected. Moreover, we can always choose a polygonal path that connects two points in X.*

Proof. Let X be a nonempty open and connected subset of $\mathbb{C}$. Fix a point $z_0 \in X$. We define

$$Y \triangleq \{z \in X : \text{there exists a polygonal path in } X \text{ from } z_0 \text{ to } z\}.$$

Clearly $z_0 \in Y$, so Y is nonempty. We will show belowthat Y is both open and closed in X.

(Y is open in X) Let $z \in Y$. By definition, there is a polygonal path in X from z_0 to z. Since X is open, there exists $r > 0$ such that the open disk $D(z, r)$ is contained in X.

Take any $w \in D(z, r)$. Because $D(z, r)$ is convex, the line segment from z to w lies entirely in $D(z, r)$, and hence in X. Concatenating the polygonal path from z_0 to z with this line segment from z to w gives a polygonal path in X from z_0 to w. Therefore $w \in Y$, and we have shown that $D(z, r) \subseteq Y$. Thus, for each $z \in Y$, there is a neighborhood of z in X contained in Y, so Y is open in X.

(Y is closed in X) We show that $X \setminus Y$ is open in X. Let $z \in X \setminus Y$. Since X is open, there exists $r > 0$ such that $D(z, r) \subseteq X$.

We claim that $D(z, r) \cap Y = \emptyset$. Suppose, for contradiction, that there exists $w \in D(z, r) \cap Y$. Then, by definition of Y, there is a polygonal path in X from z_0 to w. As before, the line segment from w to z lies entirely in $D(z, r)$, and hence in X. Concatenating the polygonal path from z_0 to w with this line segment from w to z gives a polygonal path in X from z_0 to z. This implies $z \in Y$, contradicting our assumption that $z \in X \setminus Y$. Therefore $D(z, r) \subseteq X \setminus Y$, so every point of $X \setminus Y$ has a neighborhood in X contained in $X \setminus Y$. Hence $X \setminus Y$ is open in X, and so Y is closed in X.

(Conclusion) We have shown that Y is nonempty, open, and closed in X. Since X is connected, the only subsets of X that are both open and closed are $\emptyset$ and X itself. Thus we

must have $Y = X$. Consequently, for any two points $z_1, z_2 \in X$, there is a polygonal path in X from z_1 to z_2 (take the polygonal path from z_0 to z_1 and from z_0 to z_2, and concatenate appropriately). In particular, X is path-connected, and the path can always be chosen to be polygonal. $\square$

Definition 5.4.13. A nonempty, open and connected set in $\mathbb{C}$ is referred to as a *domain* or *region*.

The followings are examples of domain.

- Open disk $D(z_0, r)$.
- Punctured disk $D(z_0, r) \setminus \{z_0\}$.
- The complex plane $\mathbb{C}$.
- Annulus $\{z \in \mathbb{C} : a < |z| < b\}$.
- Open half-plane $\mathrm{Re}(\alpha z) > 0$, where α is a complex constant.

5.5 Uniform convergence and continuity of power series

The point-wise limit of a sequence of real-valued continuous functions needs not be a continuous function. A simple example is the sequence of function x^k, for $x \in [0, 1]$. For x that is strictly less than 1, the function x^k converges to 0 as $k \to \infty$, but for $x = 1$, the limit is 1 as $k \to \infty$.

We can guarantee that the limit function is continuous if the convergence is uniform.

Definition 5.5.1. A sequence of function $f_k(z)$ is said to be *uniformly converging* to a limit function $f(z)$ in a domain D if for any $\epsilon > 0$, we can find a sufficiently large index $N(\epsilon)$, which does not depend on z, such that $|f_k(z) - f(z)| \leq \epsilon$ for all $k \geq N(\epsilon)$ and for all $z \in D$.

We can show that $f_k(z)$ converge uniformly to $f(z)$ for $z \in D$ if and only if

$$\lim_{k\to\infty} \left(\sup_{z\in D} |f_k(z) - f(z)| \right) = 0. \tag{5.5.1}$$

Theorem 5.5.2. *Let $f_k(z)$ be a sequence of continuous functions defined in a domain D, for $k = 1, 2, \ldots$. If $f_k(z)$ converges to $f(z)$ uniformly in D, then $f(z)$ is continuous.*

Proof. Pick a point z_0 in the domain D. For all $z \in D$ and for all $k \geq 1$, we have

$$|f(z) - f(z_0)| \leq |f(z) - f_k(z)| + |f_k(z) - f_k(z_0)| + |f_k(z_0) - f(z_0)|,$$

by triangle inequality. Since $f_k(z)$ converges uniformly, given any $\epsilon > 0$, we can find two indices N_1 and N_2 such that $|f(z) - f_k(z)| \leq \epsilon/3$ for all $k \geq N_1$, and $|f_k(z_0) - f(z_0)| \leq \epsilon/3$ for all $k \geq N_2$. Let k_0 be an index larger than $\max(N_1, N_2)$. The middle term $|f_{k_0}(z) - f_{k_0}(z_0)|$ can be made arbitrarily small. More precisely, we can find $\delta > 0$ such that $|f_{k_0}(z) - f_{k_0}(z_0)| \leq \epsilon/3$ whenever $|z - z_0| \leq \delta$.

We put all the estimates together. Given any $\epsilon > 0$, there is a $\delta > 0$ such that

$$|f(z) - f(z_0)| \leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

if $|z - z_0| \leq \delta$. This is exactly the definition of continuity at z_0. □

When applied to a series of functions, the condition of uniform convergence becomes: for all $\epsilon > 0$, there exists an integer $N(\epsilon)$ that depends only on ϵ but not on the value of z, such that

$$\Big|\sum_{k=1}^{n} g_k(z) - \sum_{k=1}^{\infty} g_k(z) z^k\Big| < \epsilon \tag{5.5.2}$$

for all $n \geq N(\epsilon)$ and for all z in a domain D. The number $N(\epsilon)$ should be a function of ϵ only independent of z.

Weierstrass M test is a convenient method to show that a series of functions converge uniformly.

Theorem 5.5.3 (Weierstrass M test)**.** *Suppose $(g_k(z))_{k=1}^{\infty}$ is a sequence of function defined on a domain D. If we can find positive real numbers M_k, for $k \geq 1$, such that*
(i) $|g_k(z)| \leq M_k$ for all $z \in D$, and
(ii) $\sum_{k=1}^{\infty} M_k$ converges,
then $\sum_{k=1}^{\infty} g_k(z)$ converges uniformly in the domain D.

Proof. We first show that the sequence of function $g_k(z)$ converges. For any two indices $m < n$, we have

$$\Big|\sum_{k=m+1}^{n} g_k(z)\Big| \leq \sum_{k=m+1}^{n} |g_k(z)| \leq \sum_{k=m+1}^{n} M_k \tag{5.5.3}$$

for all $z \in D$. Since $\sum_{k=1}^{\infty} M_k$ converges, given any $\epsilon > 0$, we can find an index N such that $\displaystyle\sum_{k=m+1}^{n} M_k$ is less than ϵ for all m and n larger than N. For this choice of N, we have

$$\Big|\sum_{k=m+1}^{n} g_k(z)\Big| \leq \epsilon$$

for each $z \in D$. By Cauchy's criterion for convergence (Theorem 3.1.4), the limit of $g_k(z)$ as $k \to \infty$ exists for all z. We denote the limit function as $g(z)$.

Next, we show that the convergence is uniform. By triangle inequality, for all indices m and n,

$$\Big|\sum_{k=1}^{n} g_k(z) - g(z)\Big| \leq \Big|\sum_{k=1}^{n} g_k(z) - \sum_{k=1}^{m} g_k(z)\Big| + \Big|\sum_{k=1}^{m} g_k(z) - g(z)\Big|. \tag{5.5.4}$$

Given any $\epsilon > 0$, using the argument in Equation (5.5.3), we can find a larger index N_ϵ, which only depends on ϵ, such that $|\sum_{k=1}^{n} g_k(z) - \sum_{k=1}^{m} g_k(z)| \leq \epsilon/2$ for all $m, n \geq N$ and for all $z \in D$.

Moreover, since $g_k(z)$ converges to $g(z)$, there exists an integer $M(z, \epsilon)$, which depends on both z and ϵ, such that $|\sum_{k=1}^{m} g_k(z) - g(z)| \leq \epsilon/2$ for all $m \geq M(z, \epsilon)$. For each $z \in D$, we choose an integer $m(z)$ that is larger than $\max(M(z, \epsilon), N_\epsilon)$ (using axiom of choice). Then for all $z \in D$, we can conclude from (5.5.4) that

$$\begin{aligned}\Big|\sum_{k=1}^{n} g_k(z) - g(z)\Big| &\leq \Big|\sum_{k=1}^{n} g_k(z) - \sum_{k=1}^{m(z)} g_k(z)\Big| + \Big|\sum_{k=1}^{m(z)} g_k(z) - g(z)\Big| \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.\end{aligned}$$

for all $n \geq N_\epsilon$. This proves that g_k converges to g uniformly as $k \to \infty$. □

The following theorem shows that absolute convergence implies uniform convergence for power series.

Theorem 5.5.4. *Suppose $\sum_{k=0}^{\infty} a_k z^k$ converges absolutely at $z = z_0$, then $\sum_{k=0}^{\infty} a_k z^k$ converges uniformly for $|z| \leq |z_0|$.*

Proof. For $k \geq 0$, let $M_k \triangleq |a_k||z_0|^k$. By assumption, $\sum_{k=0}^{\infty} M_k$ converges. In addition, we have

$$|a_k z^k| \leq |a_k||z_0|^k = M_k$$

for all z in the closed disk $|z| \leq |z_0|$. The result follows from Weierstrass M-test in Theorem 5.5.3. □

Theorem 5.5.5. *The function defined by a power series is continuous in the region of convergence.*

Proof. Suppose R is the radius of convergence of a power series $\sum_{k=0}^{\infty} a_k z^k$. Let z be a complex number with modulus $|z|$ strictly less than R. Since $|z| < R$, we can find a real number r between $|z|$ and R, e.g. we can take $r = (|z| + R)/2$. The power series $\sum_k a_k z^k$ is converging at $z = r$, and hence by Theorem 5.5.4, the power series converges uniformly for $|z| \leq r$. Finally, we observe that the partial sum $\sum_{k=0}^{n} a_k z^k$ are polynomials for each natural number n, and is thus continuous. We see that the function defined by this power series is continuous at z. Since this argument works for all $|z| < R$, the function is continuous in the open disk $|z| < R$. □

We thus see that the functions defined by power series are all continuous.

Summary

- The complex plane has the same topological properties of the Euclidean plane $\mathbb{R}^2$.
- A function $f(z)$ is continuous at z_0 if (i) $f(z_0)$ exists, (ii) $\lim_{z \to z_0} f(z)$ exists, and (iii) $\lim_{z \to z_0} f(z) = f(z_0)$.
- The continuity of a complex function at the point at infinity can be seen from the continuity of $f(1/z)$ at $z = 0$.
- A path in the complex plane is represented parametrically by a continuous function $\gamma : [a, b] \to \mathbb{C}$.

- A set X is path-connected if any two points in X are connected by a path in X.
- A set X is connected if it cannot be written as a union of two disjoint subsets that are relatively open in X.
- When X is connected, the only sets that are open and closed at the same time are the open set and X itself.
- A domain is a nonempty, open and connected set in $\mathbb{C}$.
- When X is nonempty and open, path-connectedness is equivalent to connectedness.
- A series $\sum_{k=1}^{\infty} g_k(z)$ converges uniformly to $g(z)$ in a domain D if for any $\epsilon > 0$, there exists N such that $|\sum_{k=1}^{n} g_k(z) - g(z)| \leq \epsilon$ for all $n \geq N$ and for all $z \in D$.
- The limit of a sequence of continuous functions is continuous if the convergence is uniform.
- Weierstrass M test is a tool that proves uniform convergence.
- A function defined by power series is continuous in its region of convergence.

Exercises

1 If z_0 is an accumulation point of a set $S \subseteq \mathbb{C}$ and $\lim_{z \to z0} f(z) = w$, then the limit is unique.

2 Prove the usual rules for computing with limits of functions. Suppose $f(z)$ and $g(z)$ are complex function with limits $\lim_{z \to z_0} f(z) = w_1$ and $\lim_{z \to z_0} g(z) = w_2$. Show that

(a) $\lim_{z \to z_0}(f(z) + g(z)) = w_1 + w_2$;

(b) $\lim_{z \to z_0}(f(z) - g(z)) = w_1 - w_2$;

(c) $\lim_{z \to z_0}(f(z)g(z)) = w_1 w_2$;

(d) $\lim_{z \to z_0}(f(z)/g(z)) = w_1/w_2$ (when $w_2 \neq 0$).

3 Prove Theorem 5.2.6.

4 Prove that the trajectory of a path is compact.

5 Show that the condition in (5.5.1) is equivalent to the definition of uniform convergence in Definition 5.5.1.

Chapter 6

Complex Differentiability

In this chapter we investigate the meaning that a complex function is differentiable at a point.

6.1 Differentiable real functions from multi-variable calculus

When a real-valued function $f(x)$ is differentiable at a point x_0, we can approximate the function near x_0 by a linear function

$$L(x) = f(x_0) + f'(x_0)(x - x_0).$$

The value of $f'(x_0)$ has the geometric meaning of slope in the graph of $f(x)$.

Now consider a real-valued function $f(x, y)$ in variables x and y. If $f(x, y)$ is differentiable at a point (x_0, y_0), we can approximate $f(x, y)$ near (x_0, y_0) by a linear function

$$L(x, y) = f(x_0, y_0) + \alpha(x - x_0) + \beta(y - y_0),$$

where $\alpha = f_x(x_0, y_0)$ and $\beta = f_y(x_0, y_0)$ are the partial derivatives of $f(x, y)$ with respect to x and y, respectively, at the point (x_0, y_0). We can write the linear function $L(x, y)$ as

$$L(x, y) = f(x_0, y_0) + \begin{bmatrix} f_x(x_0, y_0) & f_y(x_0, y_0) \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}.$$

The vector $\begin{bmatrix} f_x(x_0, y_0) & f_y(x_0, y_0) \end{bmatrix}$ is called the gradient of f at (x_0, y_0). We can express

the linear approximation in terms of the differences, $\Delta_x \triangleq x - x_0$ and $\Delta_y \triangleq y - y_0$,

$$L(x_0 + \Delta_x, y_0 + \Delta_y) = f(x_0, y_0) + \begin{bmatrix} f_x(x_0, y_0) & f_y(x_0, y_0) \end{bmatrix} \begin{bmatrix} \Delta_x \\ \Delta_y \end{bmatrix}.$$

Next, we consider a vector-valued function

$$\mathbf{F}(x, y) = (u(x, y), v(x, y)),$$

which is also called a *vector field*. Differentiation of vector field is about linear approximation.

Definition 6.1.1. A vector-valued function $\mathbf{F}(x, y) \triangleq (u(x, y), v(x, y))$ is said to be *real differentiable* at a point (x_0, y_0) in the domain of $\mathbf{F}$ if there exists a 2×2 matrix A such that

$$\begin{bmatrix} u(x_0 + \Delta x, y_0 + \Delta y) \\ v(x_0 + \Delta x, y_0 + \Delta y) \end{bmatrix} \doteq \begin{bmatrix} u(x_0, y_0) \\ v(x_0, y_0) \end{bmatrix} + A \cdot \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \tag{6.1.1}$$

when $(\Delta x, \Delta y)$ is small.

The symbol "$\doteq$" means that the difference between the left and right side of (6.1.1) converges to zero faster than the linear term, i.e.,

$$\lim_{(\Delta x, \Delta y) \to (0,0)} \frac{\|\text{Difference between L.H.S and R.H.S. of (6.1.1)}\|}{\sqrt{\Delta x^2 + \Delta y^2}} = 0. \tag{6.1.2}$$

The matrix A is called the *total derivative* of $\mathbf{F}$ at (x_0, y_0), if $\mathbf{F}$ is real differentiable at (x_0, y_0).

It is known that if $\mathbf{F}(x, y)$ is real differentiable at (x_0, y_0), the partial derivatives of $u(x, y)$ and $v(x, y)$ exists at (x_0, y_0), and the total derivative equals

$$\begin{bmatrix} u_x(x_0, y_0) & u_y(x_0, y_0) \\ v_x(x_0, y_0) & v_y(x_0, y_0) \end{bmatrix}.$$

The entries in the 2×2 matrix are the partial derivatives of u and v evaluated at (x_0, y_0) [Rudin, Theorem 9.17]. It is important to note that the mere existence of the partial derivatives of $\mathbf{F}$ is not sufficient to guarantee that (6.1.1) holds. The existence of of partial derivatives is a necessary condition of real differentiability.

In general, we have the following sufficient condition for real differentiability, which is easier to check [Rudin, Theorem 9.21].

Theorem 6.1.2. *A vector field* $\mathbf{F}(x, y) = (u(x, y), v(x, y))$ *is real differentiable at a point* (x_0, y_0) *if*

(i). partial derivatives u_x, u_y, v_x *and* v_y *exists in a neighborhood of* (x_0, y_0), *and*

(ii). the partial derivatives u_x, u_y, v_x *and* v_y *are continuous at* (x_0, y_0) *in this neighborhood.*

Example 6.1.1. The function $\mathbf{F}(x, y)$ defined by $(x^2y, x + y)$ is real differentiable. Partial derivatives of the first component $u(x, y) = x^2y$ and the second component $v(x, y) = x + y$ exist, and are all continuous. Therefore F is real differentiable, and we have

$$\begin{bmatrix} (x_0 + \Delta x)^2(y_0 + \Delta y) \\ x_0 + \Delta x + y_0 + \Delta y \end{bmatrix} \doteq \begin{bmatrix} x_0^2 y_0 \\ x_0 + y_0 \end{bmatrix} + \begin{bmatrix} 2x_0y_0 & x_0^2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \tag{6.1.3}$$

for any x_0 and y_0.

A classic example that has partial derivatives but not differentiable.

Example 6.1.2. Let $F(x, y)$ be the vector field defined by

$$\mathbf{F}(x, y) \triangleq \begin{cases} \left(\frac{x^2y}{x^2+y^2}, \frac{xy^2}{x^2+y^2}\right) & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Let u and v denote the first and the second components of $\mathbf{F}(x, y)$ The partial derivatives u_x, y_y, v_x, v_y at the origin exist and are all equal to 0, but $\mathbf{F}$ is not real differentiable at the origin. It is not differentiable because if we approach the origin along the straight line $x = y$, the value of $F(x, x)$ equals $(x/2, x/2)$, so that

$$\lim_{x \to 0} \frac{\|\mathbf{F}(x, x)\|}{\sqrt{2x^2}} = \frac{1}{2},$$

contradicting the defining condition in (6.1.2) that the limit should be 0.

6.2 Complex differentiability and Cauchy-Riemann condition

After the revision of differentiation in multi-variable calculus, we now consider a complex function $f(z)$ as a two-dimensional vector field,

$$f(x+iy) = u(x,y) + iv(x,y).$$

and study $f(z)$ as if it is a vector field. We motivate the notion of complex differentiability by an example.

Example 6.2.1. Consider the function

$$f(z) = z^3 = (x+iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3).$$

The real and imaginary parts are $u(x,y) = x^3 - 3xy^2$ and $v(x,y) = 3x^2y - y^3$, respectively. The partial derivatives of u and v are

$$\begin{aligned} u_x(x,y) &= 3x^2 - 3y^2, \\ u_y(x,y) &= -6xy, \\ v_x(x,y) &= 6xy, \\ v_y(x,y) &= 3x^2 - 3y^2. \end{aligned}$$

Suppose we fix a base point $(x_0, y_0) = (2,1)$. The linear approximation in (6.1.1) at $(2,1)$ can be written as

$$\begin{bmatrix} u(2+\Delta x, 1+\Delta y) \\ v(2+\Delta x, 1+\Delta y) \end{bmatrix} \doteq \begin{bmatrix} 2 \\ 11 \end{bmatrix} + \begin{bmatrix} 9 & -12 \\ 12 & 9 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}.$$

For general $z = x + iy$, the 2×2 matrix consisting of the partial derivatives is

$$\begin{bmatrix} 3x^2 - 3y^2 & -6xy \\ 6xy & 3x^2 - 3y^2 \end{bmatrix}.$$

The total derivative is in the form as in the matrix representation of complex number (Sec. 1.2.2). We can use simplify the linear approximation in terms of complex variable,

$$\begin{aligned} f(z+\Delta z) &= f(z) + (3x^2 - 3y^2 + i(6xy)) \cdot (\Delta x + i\Delta y) \\ &= f(z) + (3z^2)\Delta z. \end{aligned}$$

Definition 6.2.1. Let $f(z)$ be a complex function defined in a domain D. Given a point $z_0 \in D$, we say that f is *complex differentiable at z_0* if there exists a complex constant w_0 such that

$$f(z_0 + \Delta z) \doteq f(z_0) + w_0 \cdot \Delta z. \tag{6.2.1}$$

The constant w_0 is called the derivative of $f(z)$ at z_0, and is denoted by $f'(z_0)$.

The symbol $\doteq$ in (6.2.1) means

$$f(z_0 + \Delta z) = f(z_0) + w_0 \cdot \Delta z + o(\Delta z),$$

where $o(\Delta z)$ represents a function that satisfies

$$\lim_{\Delta z \to 0} \frac{|o(\Delta z)|}{|\Delta z|} = 0. \tag{6.2.2}$$

In some textbooks, the following equivalent condition is adopted as the definition of complex differentiability.

Theorem 6.2.2. *Let $f(z)$ be a complex function defined an a domain D and z_0 is a point in D. Then $f(z)$ is complex differentiable at z_0 if and only if the limit*

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} \tag{6.2.3}$$

exists.

Proof. ($\Rightarrow$) Suppose $f(z)$ is complex differentiable at z_0, then the complex constant w_0 is precisely the limit in (6.2.2). We can see this by re-writing the fraction in (6.2.2),

$$\frac{|o(\Delta z)|}{|\Delta z|} = \frac{|f(z_0 + \Delta z) - f(z_0) - w_0 \cdot \Delta z)|}{|\Delta z|} \overset{(a)}{=} \left| \frac{f(z_0 + \Delta z) - f(z_0) - w_0 \cdot \Delta z)}{\Delta z} \right|$$

The assumption that this has limit zero as $\Delta z \to 0$ is the same as saying

$$\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0) - w_0 \cdot \Delta z}{\Delta z} = 0,$$

which is then equivalent to

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} = w_0.$$

($\Leftarrow$) The steps in the previous paragraph are all reversible. Starting from the assumption that the limit $\lim_{h\to 0}\frac{f(z_0+h)-f(z_0)}{h}$ exists and equal to w_0, we can deduce

$$\lim_{\Delta z\to 0}\frac{|f(z_0+\Delta z)-f(z_0)-w_0\cdot\Delta z)|}{|\Delta z|}=0.$$

This proves that the two definitions are equivalent to each other. $\square$

We note that the step (a) in the previous proof is the following elementary fact about complex modulus:

$$\left|\frac{z_1}{z_2}\right|=\frac{|z_1|}{|z_2|}.$$

Immediately from Definition 6.2.1, we have a necessary condition for complex differentiability.

Theorem 6.2.3 (Cauchy-Riemann condition)**.** *Suppose $f(x+iy)=u(x,y)+iv(x,y)$ is complex differentiable at $z_0=x_0+iy_0$ (see Definition 6.2.1), where z_0 is in the domain of $f(z)$. Then the partial derivatives u_x, u_y, v_x and v_y at z_0 exist, and they satisfy*

$$u_x(x_0,y_0)=v_y(x_0,y_0),\quad \textit{and}\quad u_y(x_0,y_0)=-v_x(x_0,y_0).$$

The concept of complex differentiability is thus the same as

$$\boxed{\text{Complex differentiability} = \text{Real differentiability } + \text{Cauchy-Riemann condition}}$$

We formalize this in the form of a theorem.

Theorem 6.2.4. *Suppose $f(z)=u(x,y)+iv(x,y)$ is a complex function defined on a domain D which contains $z_0=x_0+iy_0$. Then, the limit*

$$\lim_{h\to 0}\frac{f(z_0+h)-f(z_0)}{h}$$

exists if and only if

(i). the vector function $(u(x,y),v(x,y))$ is real-differentiable at (x_0,y_0),

(ii). the partial derivatives satisfy the Cauchy-Riemann equations $u_x=v_y$, $u_y=-v_x$ at (x_0,y_0).

We give a self-contained proof of this theorem in the appendix of this chapter.

Example 6.2.2. For any complex constant a, the function $f(z) = az$ is complex differentiable at every point z_0 in $\mathbb{C}$, i.e., it is complex differentiable at every point in the complex plane, and the derivative is equal to a. We can see this by checking the definition of complex derivative.

$$\left|\frac{a(z_0+h) - az_0}{h} - a\right| = 0.$$

We have $\frac{a(z_0+h)-az_0}{h} = a$ for all nonzero $h \in \mathbb{C}$. If we take limit, the limit is equal to a. We have thus proved that the complex derivative of the linear function $f(z) = az$ is equal to a constant a.

Likewise, we can see that the derivative of a translation function $z \mapsto z + b$ is complex differentiable, and the derivative is equal to 1.

Example 6.2.3. Let $f(z) = 1/z$ be the inversion function. We will show that this function is complex differentiable at all nonzero z_0 from definition. For $z_0 \neq 0$,

$$\begin{aligned}\frac{\frac{1}{z_0+h} - \frac{1}{z_0}}{h} &= \frac{1}{h}\Big(\frac{z_0 - (z_0+h)}{(z_0+h)z_0}\Big) \\ &= -\frac{1}{z_0(z_0+h)}.\end{aligned}$$

When $h \to 0$, the limit of $-\frac{1}{z_0(z_0+h)}$ is $-1/z_0^2$. Therefore $f(z) = 1/z$ is complex differentiable for $z \in \mathbb{C} \setminus \{0\}$, and the complex derivative is $-1/z_0^2$.

The function in Example 6.1.1 is real differentiable everywhere but not complex differentiable. As a matter of fact, the function in Example 6.1.1 fails the Cauchy-Riemann equation for all real numbers x_0 and y_0, and thus is not complex differentiable anywhere. The following is a primary example that is real differentiable function but not complex differentiable.

Example 6.2.4. The conjugate function $f(z) = \bar{z}$ is not complex differentiable anywhere. It is because $u_x = 1$ and $v_y = -1$, and $u_x \neq v_y$ at any point in $\mathbb{C}$. It fails the Cauchy-Riemann equations everywhere.

Since checking real differentiability by definition is not easy in general, we can use Theorem 6.1 to provide a sufficient condition for complex differentiability.

Theorem 6.2.5 (A sufficient condition for complex differentiability)**.** *A complex function f is complex differentiable at z_0 if*

(i). The partial derivatives u_x, u_y, v_x, v_y exists in a neighborhood of z_0;

(ii). u_x, u_y, v_x, v_y are continuous at z_0;

(iii). Cauchy-Riemann equations are satisfied at z_0.

As in the real case, complex differentiability implies continuity.

Theorem 6.2.6. *If $f(z)$ is complex differentiable at z_0, then $f(z)$ is continuous at z_0.*

Proof. Suppose $f(z)$ is complex differentiable at z_0. By the definition of complex differentiability, we can find a complex number w_0 such that

$$f(z_0 + h) = f(z_0) + w_0 \cdot h + o(h),$$

where h is a complex variable and $o(h)$ is a function with the property that $|o(h)/h| \to 0$ as $h \to 0$. If we do not divide by h, the function $o(h)$ converges to zero as well, because

$$\lim_{h\to 0} o(h) = \lim_{h\to 0} \frac{o(h)}{h} h = \lim_{h\to 0} \frac{o(h)}{h} \cdot \lim_{h\to 0} h = 0 \cdot 0 = 0.$$

Therefore

$$\lim_{h\to 0} f(z_0 + h) = f(z_0) + \lim_{h\to 0} w_0 \cdot h + \lim_{h\to 0} o(h) = f(z_0).$$

This matches the definition of continuity of a function at a point in Definition 5.2.2. □

The following formulae are useful in calculating complex derivative.

Theorem 6.2.7. *If a complex function $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at $z_0 = x_0 + iy_0$, the complex derivative at z_0 can be computed by*

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0), \tag{6.2.4}$$

or

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0). \tag{6.2.5}$$

Proof. Let f be a complex function that is complex differentiable at $z_0 = x_0 + iy_0$. The limit in (6.2.3) that defines the complex derivative does not depend on how we approach the point z_0. We can approach z_0 horizontally or vertically, and the results must be the same if the function is complex differentiable.

Let $\Delta z = \Delta x$ and take $\Delta x \to 0$.

$$\begin{aligned}
&\lim_{\Delta x \to 0} \frac{u(x_0 + \Delta x, y_0) + iv(x_0 + \Delta x, y_0) - u(x_0, y_0) - iv(x_0, y_0)}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x} \\
&= \frac{\partial u}{\partial x}(x_0, y_0) + i \frac{\partial v}{\partial x}(x_0, y_0).
\end{aligned}$$

This proves that

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

Next suppose $\Delta z = i\Delta y$ and take $\Delta y \to 0$.

$$\begin{aligned}
&\lim_{\Delta y \to 0} \frac{u(x_0, y_0 + \Delta y) + iv(x_0, y_0 + \Delta y) - u(x_0, y_0) - iv(x_0, y_0)}{i\Delta y} \\
&= \lim_{\Delta y \to 0} \frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{i\Delta y} + i \lim_{\Delta y \to 0} \frac{v(x_0, y_0 + \Delta y) - v(x_0, y_0)}{i\Delta y} \\
&= -i \frac{\partial u}{\partial y}(x_0, y_0) + \frac{\partial v}{\partial y}(x_0, y_0).
\end{aligned}$$

We obtain

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

□

Example 6.2.5. The identity function $f(z) = z$ is complex differentiable at all points in $\mathbb{C}$, and the derivative is constantly equal to 1. We can check it using (6.2.3). For any $z_0 \in \mathbb{C}$,

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \lim_{h \to 0} \frac{z_0 + h - f(z_0)}{h} = \lim_{h \to 0} 1 = 1.$$

Example 6.2.6. Consider the square function

$$f(z) = z^2 = (x + iy)^2 = x^2 - y^2 + i(2xy).$$

We check that the Cauchy-Riemann condition are satisfied:

$$u_x = 2x = v_y$$
$$u_v = -2y = -v_x$$

Since the partial derivatives are continuous for all x and y, by Theorem 6.2.5, the square function $z \mapsto z^2$ is complex differentiable. The derivative is

$$2x + i(2y) = 2z$$

by (6.2.4).

Example 6.2.7. The complex exponential function

$$e^z = e^x \cos y + ie^x \sin y = u(x, y) + iv(x, y)$$

is complex differentiable. We compute the partial derivatives of the real and imaginary parts:

$$u_x = e^x \cos y, \qquad u_y = -e^x \sin y, \qquad v_x = e^x \sin y, \qquad v_y = e^x \cos y.$$

The partial derivatives satisfy the Cauchy-Riemann equation, and are all continuous. Therefore, by Theorem 6.2.5, the complex exponential function is complex differentiable at every point on the complex plane. Applying (6.2.4), the derivative is

$$f'(z) = u_x + iv_x = e^x \cos y + ie^x \sin y = e^z.$$

Following the same strategy as in the previous example, we can prove that $\sin z$, $\cos z$, $\sinh z$, $\cosh z$ are complex differentiable at all points on $\mathbb{C}$.

Theorem 6.2.8. *Suppose f is complex differentiable at z_0 and g is complex differentiable at $f(z_0)$. Then the composition function $g \circ f$ is complex differentiable at z_0.*

Proof. We leverage the fact from real analysis that the composition of real differentiable functions are real differentiable. Let $u(x, y)$ and $v(x, y)$ be the real and imaginary parts of $f(z)$. The assumption that $f(z)$ is differentiable at $z_0 = x_0 + iy_0$ means that we have the linear approximation

$$\begin{bmatrix} u(x_0 + \Delta x, y_0 + \Delta y) \\ v(x_0 + \Delta x, y_0 + \Delta y) \end{bmatrix} \doteq \begin{bmatrix} u(x_0, y_0) \\ v(x_0, y_0) \end{bmatrix} + \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix},$$

for some real constants a and b. Suppose $f(z_0) = r_0 + is_0$, and $g(z)$ is complex differentiable at $f(z_0)$. Denote the real and imaginary parts of $g(z)$ by $\mu(r,s)$ and $\nu(r,s)$. We approximate g near (r_0, s_0) by

$$\begin{bmatrix} \mu(r_0 + \Delta r, s_0 + \Delta s) \\ \nu(r_0 + \Delta r, s_0 + \Delta s) \end{bmatrix} \doteq \begin{bmatrix} \mu(r_0, s_0) \\ \nu(r_0, s_0) \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} \cdot \begin{bmatrix} \Delta r \\ \Delta s \end{bmatrix},$$

for some constants c and d.

By [Rudin, Theorem 9.15], The composed function $g \circ f$ is real differentiable at (x_0, y_0),

$$\begin{bmatrix} \mathrm{Re}(g \circ f)(x_0 + \Delta x, y_0 + \Delta y) \\ \mathrm{Im}(g \circ f)(x_0 + \Delta x, y_0 + \Delta y) \end{bmatrix} \doteq \begin{bmatrix} \mathrm{Re}(g \circ f)(x_0, y_0) \\ \mathrm{Im}(g \circ f)(x_0, y_0) \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} \cdot \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}.$$

The product of the two matrices are in the from

$$\begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix}.$$

Hence, the Cauchy-Riemann condition is satisfied for the composition $g(f(z))$ at z_0. By Theorem 6.2.4, $g(f(z))$ is complex differentiable at z_0. □

6.3 Complex differential form and Wirtinger derivatives

We give a characterization of complex differentiability using differential form. The main reference of this section are [Spivak, Chapter 5] and [ChernChenLam, Chapter 1].

In calculus, a differential dx is usually regarded as an infinitesimal quantity, representing a very small change in the x variable. The *total differential* of a bi-variate function $f(x,y)$ is

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

However, the treatment of differential form in calculus is usually not very rigorous.

We study complex differentiability by first considering multi-variable real functions and give a more rigorous definition of differential dx and dy. Then we extend it to the complex case and give a concrete meaning of the symbol dz. In the remainder of this section we will borrow some notation from real manifolds. However, we do not need the full generality but only need the special case of dimension 2 and when the manifold is flat.

6.3.1 Vector field and differential form

We first consider the Euclidean space $\mathbb{R}^2$ in the real case.

We define the tangent space at a given a point $\mathbf{p}$ in $\mathbb{R}^2$ as the set of all vectors that emanate from $\mathbf{p}$. The center point is represented by a position vector $\mathbf{p}$. To emphasize the starting point of a tangent vector, we write

$$(\mathbf{p}, \mathbf{v})$$

to represent a tangent vector emanating from the point $\mathbf{p}$. As an alternate notation, we will also write $\mathbf{v}_\mathbf{p}$ to mean a vector $\mathbf{v}$ with initial point $\mathbf{p}$.

The first component, $\mathbf{p}$, indicates that $\mathbf{p}$ is the initial point. We can add two vectors in $T_\mathbf{p}$ as follows,

$$(\mathbf{p}, \mathbf{v}_1) + (\mathbf{p}, \mathbf{v}_2) \triangleq (\mathbf{p}, \mathbf{v}_1 + \mathbf{v}_2)$$

and multiply by a scalar

$$a(\mathbf{p}, \mathbf{v}) \triangleq (\mathbf{p}, a\mathbf{v}).$$

However, if $\mathbf{p}$ and $\mathbf{q}$ are two distinct vector, the addition of $(\mathbf{p}, \mathbf{u})$ and $(\mathbf{q}, \mathbf{v})$ are not defined. Dot product of two vectors in $T_\mathbf{p}$ is done in the natural way

$$(\mathbf{p}, \mathbf{u}) \cdot (\mathbf{p}, \mathbf{v}) \triangleq \mathbf{u} \cdot \mathbf{v}. \tag{6.3.1}$$

(See Fig. 6.1)

Definition 6.3.1. Given a point $\mathbf{p}$, the *tangent space at* $\mathbf{p}$ is defined as the set

$$T_\mathbf{p} \triangleq \{(\mathbf{p}, \mathbf{v}) : \mathbf{v} \in \mathbb{R}^2\}.$$

For each point $\mathbf{p}$, the tangent space $T_\mathbf{p}$ is a two-dimensional vector space. We will use $\mathbf{e}_1$ and $\mathbf{e}_2$ to denote the standard basis. We will also use the shorter notation $\mathbf{v}_\mathbf{p}$ instead of $(\mathbf{p}, \mathbf{v})$. The following is an example of computations in terms of the basis vectors $(\mathbf{e}_1)_\mathbf{p}$ and $(\mathbf{e}_2)_\mathbf{p}$,

$$2(4\mathbf{e_1} - \mathbf{e_2})_\mathbf{p} + (\mathbf{e_1} + 3\mathbf{e_2})_\mathbf{p} = (9\mathbf{e_1} + \mathbf{e_2})_\mathbf{p}.$$

Definition 6.3.2. A *smooth vector field* is a mapping that associates, for each point $\mathbf{p}$, a tangent vector $\mathbf{v}_\mathbf{p}$, in a continuous and smooth manner.

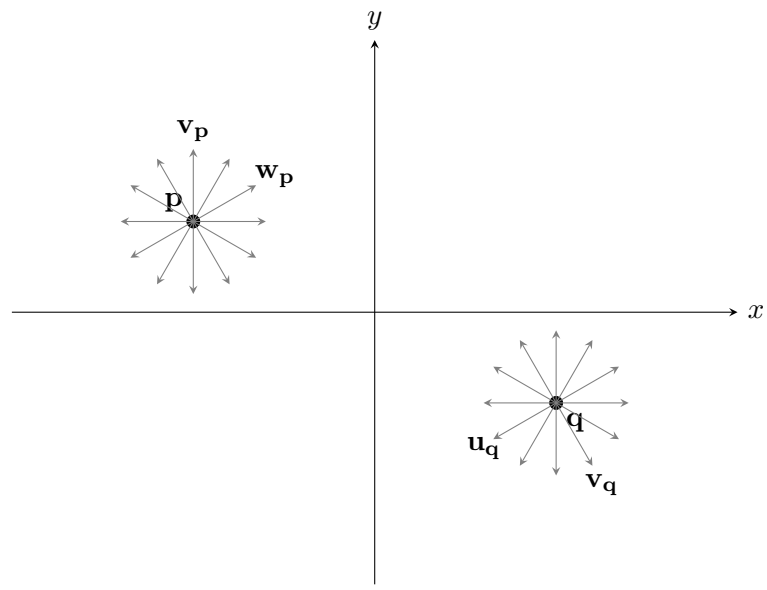


Figure 6.1: Tangent space at point $\mathbf{p}$ and tangent space at $\mathbf{q}$

For instance, we can represent a vector in each tangent space using a formula, such as

$$(a(x,y)\mathbf{e}_1 + b(x,y)\mathbf{e}_2)_{(x,y)} \tag{6.3.2}$$

for $(x,y) \in \mathbb{R}^2$, where $a(x,y)$ and $b(x,y)$ are smooth functions of x and y. In simple terms, the expression above says that at the point (x,y), we select the vector $a(x,y)\mathbf{e}_1 + b(x,y)\mathbf{e}_2$ from the associated tangent space $T_{(x,y)}$. To further simplify notation, very often we skip the subscript $_{(x,y)}$. In physics notation, the two unit vectors are commonly denoted as $\vec{i}$ and $\vec{j}$. We can write

$$a(x,y)\vec{i} + b(x,y)\vec{j},$$

representing a vector emanating from the point (x,y), with horizontal and vertical components equal to $a(x,y)$ and $b(x,y)$, respectively.

We now turn to the dual of vector space.

Definition 6.3.3. Given a real vector space V, a *covector* or *linear functional* is a linear function that takes a vector from V as input and return a real number.

When V is $\mathbb{R}^2$, a typical example of covector is

$$g_{a,b}(x, y) = ax + by,$$

where a and b are fixed parameters of the linear functional $g_{a,b}$. For example the function

$$(x, y) \mapsto x + y$$

is a covector that returns the sum of the two coordinates, the function

$$(x, y) \mapsto 2x - y$$

is another covector. But the function

$$(x, y) \mapsto xy$$

is not a covector, because it is not a linear function.

In general, a linear functional $f(x, y)$ can be expressed as a dot product $(a, b) \cdot (x, y)$ for some constants a and b.

The set of all covectors of a vector space has the structure of vector space. We can add two covectors $g(x, y)$ and $h(x, y)$ according to the rule

$$(g + h)(x, y) = g(x, y) + h(x, y),$$

and multiply a covector g by a scalar constant a to obtain an new covector ag,

$$(ag)(x, y) = a \cdot g(x, y).$$

We will call this vector space the *dual space* of V. The notation of dual space is V^*.

When V is isomorphic to $\mathbb{R}^2$, the dual space has dimension 2. We can take ϕ_1 and ϕ_2, defined by

$$\phi_1(x\mathbf{e}_1 + y\mathbf{e}_2) \triangleq x, \tag{6.3.3}$$

$$\phi_2(x\mathbf{e}_1 + y\mathbf{e}_2) \triangleq y. \tag{6.3.4}$$

as a basis of V^*.

The next result shows that ϕ_1 and ϕ_2 span the vector space $(\mathbb{R}^2)^*$.

Theorem 6.3.4. *Let ϕ_1 and ϕ_2 be the covectors (6.3.3) and (6.3.4).*

(i). A linear combination $a\phi_1 + b\phi_2$, where a and b are real constants, is a covector in $(\mathbb{R}^2)^*$.

(ii). any covector in $(\mathbb{R}^2)^*$ can be written in the form $a\phi_1 + b\phi_2$ for some real constants a and b.

(iii). ϕ_1 and ϕ_2 are linearly independent over $\mathbb{R}$.

Proof. For any two constants a and b, the function $a\phi_1 + b\phi_2$ maps vector $x\mathbf{e}_1 + y\mathbf{e}_2$ to

$$a\phi_1(x\mathbf{e}_1 + y\mathbf{e}_2) + b\phi_2(x\mathbf{e}_1 + y\mathbf{e}_2) = ax + by.$$

This is a linear function from $\mathbb{R}^2$ to $\mathbb{R}$.

Next, consider a linear function $\theta : \mathbb{R}^2 \to \mathbb{R}$. We define $a \triangleq \theta(\mathbf{e}_1)$ and $b \triangleq \theta(\mathbf{e}_2)$. For any vector $x\mathbf{e}_1 + y\mathbf{e}_2$, we have

$$\theta(x\mathbf{e}_1 + y\mathbf{e}_2) = x\theta(\mathbf{e}_1) + y\theta(\mathbf{e}_2) = ax + by.$$

We see that θ is equal to the linear combination $a\phi_1 + b\phi_2$ of ϕ_1 and ϕ_2.

To prove linear independence, suppose we can find two realk constants α and β such that $\alpha\phi_1 + \beta\phi_2$ is the zero function. Then we must have both α and β equal 0. We can see $\alpha = 0$ by evaluating at $\mathbf{e}_1$,

$$\alpha\phi_1(\mathbf{e_1}) + \beta\phi_2(\mathbf{e_1}) = \alpha,$$

which must be equal to zero because $\alpha\phi_1 + \beta\phi_2$ is the zero function. Similarly, by evaluating at $\mathbf{e}_2$, we obtain $\beta = 0$. □

Definition 6.3.5. The *cotangent space* at $\mathbf{p}$, denoted as $T^*_{\mathbf{p}}$, is defined as the dual space of $T_{\mathbf{p}}$. A *differential form* is a continuous and smooth selection of covector at each cotangent space.

A cotangent space at $\mathbf{p}$ consists of all linear functionals that take a tangent vector at $\mathbf{p}$ as input and return a real number.

Theorem 6.3.6. *For any point* $\mathbf{p}$*, the cotangent space* $T^*_{\mathbf{p}}$ *has dimension 2.*

An important source of linear functional is directional derivatives of a multi-variable real function. Let $f(x, y)$ be a real differentiable function. Consider a specific point $\mathbf{p} = (x_0, y_0)$ and a tangent vector $\mathbf{v}_{\mathbf{p}} = (r\mathbf{e}_1 + s\mathbf{e}_2)_{\mathbf{p}}$ at $\mathbf{p}$.

The directional derivative of f at $\mathbf{p}$ in the direction of $\mathbf{v}_{\mathbf{p}}$ is given by

$$\lim_{h\to 0} \frac{f(x_0 + rh, y_0 + sh) - f(x_0, y_0)}{h} = \frac{\partial f(x_0, y_0)}{\partial x} r + \frac{\partial f(x_0, y_0)}{\partial y} s.$$

We define the *gradient* of function f at $\mathbf{p}$ by

$$(\nabla f)_{\mathbf{p}} = (f_x(x_0, y_0), f_y(x_0, y_0))_{\mathbf{p}}.$$

We can express the directional derivative as a dot product $(\nabla f)_{\mathbf{p}} \cdot \mathbf{v}_{\mathbf{p}}$. This define a linear functional $\phi_f : T_{\mathbf{p}} \to \mathbb{R}$ by

$$\phi_f(\mathbf{v}_{\mathbf{p}}) = (\nabla f)_{\mathbf{p}} \cdot \mathbf{v}_{\mathbf{p}}. \tag{6.3.5}$$

This leads to the next definition.

Definition 6.3.7. Given a real-valued differentiable function $f(x, y)$, the *differential* of f, denoted by df, is the differential form that assigns to each point $\mathbf{p}$ the linear functional ϕ_f defined in (6.3.5).

At each point $\mathbf{p}$, the differential $(df)_{\mathbf{p}}$ can be written as the linear combination

$$(df)_{\mathbf{p}} = f_x(\mathbf{p}) \cdot \phi_1 + f_y(\mathbf{p}) \cdot \phi_2.$$

The coefficients $f_x(\mathbf{p})$ and $f_y(\mathbf{p})$ are precisely the components of the gradient ∇f evaluated at $\mathbf{p}$.

The differential forms ϕ_1 and ϕ_2 form a basis of the cotangent space at each point. They are traditionally denoted by dx and dy, respectively.

Definition 6.3.8. The differential form dx is obtained by applying the previous definition to the function $f(x, y) = x$, and dy is obtained by taking $f(x, y) = y$.

Thus, for any point $\mathbf{p}$, the form $dx_{\mathbf{p}}$ is the linear functional that takes a tangent vector $\mathbf{v}_{\mathbf{p}}$ and returns the x-component of $\mathbf{v}$. Likewise, $dy_{\mathbf{p}}$ returns the y-component of $\mathbf{v}$. The symbols dx and dy represent *constant* differential forms: their action on tangent vector is the same at every point.

With these conventions, we may write the differential df of any differentiable function f as

$$df = f_x dx + f_y dy. \tag{6.3.6}$$

The way df acts on the tangent varies from point to point because the coefficient functions f_x and f_y vary.

It is important to remember that both sides of the equation represent functions (specifically, differential form). They produce numerical values only after being evaluated it at a specific point and applied to a specific tangent vector.

Example 6.3.1. Let $f(x, y) = x^2y^3$. The gradient of f is

$$\nabla f = (2xy^3, 3x^2y^2).$$

At the point $\mathbf{p} = (-1, 2)$, the function f gives rise to a linear functional

$$\phi_f(\mathbf{v}) = (-16, 12) \cdot \mathbf{v}$$

If we change the point to $\mathbf{p}' = (1, 1)$, then the linear functional defined by f at $\mathbf{p}'$ is

$$\phi_f(\mathbf{v}) = (2, 3) \cdot \mathbf{v}.$$

The differential form of f is

$$df = 2xy^3\, dx + 3x^2y^2\, dy.$$

In calculus textbooks, it is often stated that we can assign any values to dx and dy, suggesting their linearly independence. This perspective makes sense when we treat dx and dy as numbers. However, after defining differentials as functions, we are able to assert that dx and dy are linearly independent just like in linear algebra.

Theorem 6.3.9. *At any point $\mathbf{p}$ in $\mathbb{R}^2$, the two differentials $(dx)_{\mathbf{p}}$ and $(dy)_{\mathbf{p}}$ are linearly independent, and form a basis of the cotangent space $T^*_{\mathbf{p}}$.*

Proof. Let $\mathbf{p}$ be a point in the domain of f. Suppose there exist constants a and b such that $a \cdot (dx)_{\mathbf{p}} + b \cdot (dy)_{\mathbf{p}}$ is equal to the zero covector in the cotangent space $T^*_{\mathbf{p}}$; this means that if we supply any input to this function, the output is always equal to 0. We apply this functional to the basis vector $\mathbf{e}_1$ to get

$$(a \cdot (dx)_{\mathbf{p}} + b \cdot (dy)_{\mathbf{p}})(\mathbf{e}_1) = a \cdot 1 + b \cdot 0 = a,$$

which should be equal to zero as $a \cdot (dx)_{\mathbf{p}} + b \cdot (dy)_{\mathbf{p}}$ is a zero function. Likewise, by evaluating the functional $a \cdot (dx)_{\mathbf{p}} + b \cdot (dy)_{\mathbf{p}}$ to $\mathbf{e}_2$, we obtain $b = 0$.

From Theorem 6.3.4, we know that the dual space T^*_p has dimension 2. Any two linearly independent linear functionals form a basis of $T^*_{\mathbf{p}}$. Since $(dx)_{\mathbf{p}}$ and $(dy)_{\mathbf{p}}$ are linearly independent, they form a basis of the dual space $T^*_{\mathbf{p}}$. □

Example 6.3.2. Let $f(x, y) = e^{x^2} \sin(y)$. The differential of f is

$$df = 2xe^{x^2} \sin(y)\, dx + e^{x^2} \cos(y)\, dy.$$

At the point $\mathbf{p} = (1, \pi)$, $(df)_{\mathbf{p}}$ is the linear functional

$$(df)_{(1,\pi)} = -e \cdot \phi_2.$$

This is a function that take a tangent vector with initial point $(1, \pi)$ and return $(-e)$ times the second coordinate of the tangent vector.

In general, a differential form can be written in the form of

$$a(x, y)dx + b(x, y)dy \tag{6.3.7}$$

for some real-valued functions $a(x, y)$ and $b(x, y)$. (Compare with (6.3.2))

Theorem 6.3.10. *For any two differentiable functions $f(z)$ and $g(z)$, we have the following computational rules for differential:*

(i). $d(f + g) = df + dg$,

(ii). $d(af) = a \cdot df$, *for a constant* a,

(iii). $d(fg) = f \cdot dg + g \cdot df$.

Proof. We use the formula for computing differential form df and dg in (6.3.6). Let $\mathbf{p}$ be a point in the domain in the intersection of the domains of f and g. To prove part (i), we add $(df)_{\mathbf{p}}$ and $(df)_{\mathbf{p}}$ to get

$$\begin{aligned}(df)_{\mathbf{p}} + (df)_{\mathbf{p}} &= (f_x(\mathbf{p})dx + f_y(\mathbf{p})dy) + (g_x(\mathbf{p})dx + g_y(\mathbf{p})dy) \\ &= (f_x(\mathbf{p}) + g_x(\mathbf{p})\,dx + (f_y(\mathbf{p}) + g_y(\mathbf{p}))\,dy \\ &= (f+g)_x(\mathbf{p})\,dx + (f+g)_y(\mathbf{p})\,dy \\ &= (d(f+g))_{\mathbf{p}}.\end{aligned}$$

This proves the additive rule in part (i). The proof of parts (ii) and (iii) are similar and omitted. □

We complexify the cotangent space by allowing complex numbers as the coefficients. This means that when forming a linear combination $a(dx)_{\mathbf{p}} + b(dy)_{\mathbf{p}}$, the value of a and b can be complex numbers[1].

As an example, consider the complex differential idy. It takes a vector $(r,s)_{\mathbf{p}}$ as input and returns the value $i \cdot s$. Here, i represents the imaginary unit. This shows that the complex differential idy assigns complex numbers to tangent vectors, allowing for a more general and flexible framework.

The symbol "$(dz)_{\mathbf{p}}$" represents a function, that takes a vector, say $(\alpha,\beta)_{\mathbf{p}}$ with an initial point $\mathbf{p}$, and return the complex number $\alpha + i\beta$. Similarly, the function $(d\bar{z})_{\mathbf{p}}$ accepts a vector $(\alpha,\beta)_{\mathbf{p}}$ as input and return the complex number $\alpha - i\beta$.

In the followings, we will simplify notation by not writing out the subscript $\mathbf{p}$.

The next example analyze a complex function $f(z)$ by treating it as a multi-variable real function.

Example 6.3.3. Find the differential of the real and imaginary parts of $f(z) = z^2$.

The real and imaginary parts are $u(x,y) = x^2 - y^2$ and $v(x,y) = 2xy$. Their differentials are

$$\begin{aligned}du &= 2x\,dx - 2y\,dy \\ dv &= 2y\,dx + 2x\,dy.\end{aligned}$$

[1] This complexification process can be formally realized by tensoring $T^*_{\mathbf{p}}$ with $\mathbb{C}$ over $\mathbb{R}$. In mathematical symbols, the complex differential lives in the space $\mathbb{C} \otimes_{\mathbb{R}} T^*_{\mathbf{p}}$.

We can put du and dv together to form a complex differential,

$$\begin{aligned} du + idv &= (2x\,dx - 2y\,dy) + i(2y\,dx + 2x\,dy) \\ &= 2(x+iy)(dx+idy) \\ &= 2zdz. \end{aligned}$$

Motivated by this example, we define the differential of a complex-valued function f below.

Definition 6.3.11. Suppose function $f(z) = u(x,y) + iv(x,y)$ is real differentiable as a vector function. We define the *complex differential* df as

$$df \triangleq du + idv.$$

As two special cases, the complex differential of the function $f(z) = z$ and $f(z) = \bar{z}$ are called dz and $d\bar{z}$, respectively.

Definition 6.3.12. Define dz and $d\bar{z}$ to be the complex differential form whose component at a point $\mathbf{p}$ is

$$\begin{aligned} (dz)_{\mathbf{p}} &\triangleq (dx)_{\mathbf{p}} + i(dy)_{\mathbf{p}} \\ (d\bar{z})_{\mathbf{p}} &\triangleq (dx)_{\mathbf{p}} - i(dy)_{\mathbf{p}}. \end{aligned}$$

Despite the relationship between z and $\bar{z}$, it is important to note that the two complex differentials, dz and $d\bar{z}$, are linearly independent. This fact can be formulated in the following theorem.

Theorem 6.3.13. *The two complex differential forms dz and $d\bar{z}$ are linearly independent over $\mathbb{C}$.*

Proof. Suppose $w_1 dz + w_2 d\bar{z} = 0$ for some complex numbers w_1 and w_2. We want to show that $w_1 = w_2 = 0$.

We write $w_1 dz + w_2 d\bar{z} = 0$ in terms of dx and dy as

$$w_1(dx + idy) + w_2(dx - idy) = (w_1 + w_2)dx + i(w_1 - w_2)dy.$$

This differential form is equal to the differential form that always return zero value. We can evaluate it at the standard basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$, and the results should be zero. This leads to the following equations:

$$\begin{aligned} w_1 + w_2 &= 0 \\ i(w_1 - w_2) &= 0. \end{aligned}$$

Solving the system of linear equations, we easily get $w_1 = 0$ and $w_2 = 0$. □

6.3.2 Wirtinger derivatives

By the virtue of Theorem 6.3.13, we will take dz and $d\bar{z}$ as a basis of complex differential forms. The formulae for the change of basis are

$$dx = \frac{dz + d\bar{z}}{2}, \tag{6.3.8}$$

$$dy = \frac{dz - d\bar{z}}{2i}. \tag{6.3.9}$$

This change of basis allows us to work with dz and $d\bar{z}$, which are more convenient when dealing with complex functions and complex analysis.

From Definition 6.3.11, we now express the differential of a complex function

$$f(x + iy) = u(x, y) + iv(x, y)$$

using the basis dz and $d\bar{z}$.

$$\begin{aligned} df \triangleq du + idv &= (u_x dx + u_y dy) + i(v_x dx + v_y dy) \\ &= (u_x + iv_x)dx + (u_y + iv_y)dy \\ &= (u_x + iv_x)\Big(\frac{dz + d\bar{z}}{2}\Big) + (u_y + iv_y)\Big(\frac{dz - d\bar{z}}{2i}\Big). \end{aligned}$$

With the notation,

$$\begin{aligned} \frac{\partial f}{\partial x} &\triangleq u_x + iv_x \\ \frac{\partial f}{\partial y} &\triangleq u_y + iv_y, \end{aligned}$$

we can write

$$\begin{aligned} df &= \frac{\partial f}{\partial x}\Big(\frac{dz + d\bar{z}}{2}\Big) + \frac{\partial f}{\partial y}\Big(\frac{dz - d\bar{z}}{2i}\Big) \\ &= \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big)dz + \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big)d\bar{z}. \end{aligned}$$

This motivates the following definition.

Definition 6.3.14. Define

$$\begin{aligned} \frac{\partial f}{\partial z} &\triangleq \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big) \\ \frac{\partial f}{\partial \bar{z}} &\triangleq \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big). \end{aligned}$$

The operator $\frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big)$ is called the *$\bar{\partial}$ operator*.

The two operators $\frac{\partial}{\partial z}$ and $\frac{\partial}{\partial \bar{z}}$ are called *Wirtinger derivatives*. In terms of these notation, we can write

$$df = \frac{\partial f}{\partial z}dz + \frac{\partial f}{\partial \bar{z}}d\bar{z} \tag{6.3.10}$$

for any real differentiable function f.

We now relate the $\bar{\partial}$ operator to the Cauchy-Riemann equations.

Theorem 6.3.15. *Suppose complex-valued function $f(z) = u(x, y) + iv(x, y)$ is real differentiable as a vector function. Then*

$f(z)$ satisfies the Cauchy-Riemann condition if and only if $\partial f/\partial \bar{z} = 0$.

Moreover, when $f(z)$ is complex differentiable we have

$$\frac{\partial f}{\partial z} = f'(z),$$

where $\frac{\partial f}{\partial z}$ is defined in Definition 6.3.14.

This theorem says that when $f(z)$ is complex differentiable, we can simplify (6.3.10) to

$$df = f'(z)dz. \tag{6.3.11}$$

Proof. We begin with

$$\begin{aligned}\frac{\partial f}{\partial \bar{z}} \triangleq \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) &= \frac{1}{2}(u_x + iv_x + i(u_y + iv_y)) \\ &= \frac{1}{2}(u_x - v_y + i(v_x + u_y)).\end{aligned}$$

If $\partial f/\partial \bar{z}$ is zero, then the real and imaginary parts on the right-hand side of the equation must both be zero. This is precisely the Cauchy-Riemann condition. Therefore, by Theorem 6.2.4, the function $f(z)$ is complex differentiable.

Now, assuming that the Cauchy-Riemann equations hold, we can expand the expression for $\partial f/\partial z$ as follows:

$$\begin{aligned}\frac{\partial f}{\partial z} \triangleq \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big) &= \frac{1}{2}(u_x + iv_x - i(u_y + iv_y)) \\ &= \frac{1}{2}(u_x + v_y + i(v_x - u_y) \\ &= u_x + iv_x.\end{aligned}$$

By Theorem 6.2.7, we see that $\partial f/\partial z = f'(z)$. □

Example 6.3.4. Consider the function $f(z) = \mathrm{Re}(z) = x$. We compute the Wirtinger derivatives in Definition 6.3.14,

$$\begin{aligned}\frac{\partial x}{\partial z} &= \frac{1}{2}\Big(\frac{\partial x}{\partial x} - i\frac{\partial x}{\partial y}\Big) = \frac{1}{2} \\ \frac{\partial x}{\partial \bar{z}} &= \frac{1}{2}\Big(\frac{\partial x}{\partial x} + i\frac{\partial x}{\partial y}\Big) = \frac{1}{2}.\end{aligned}$$

After making a change of basis to dz and $d\bar{z}$, we can express the differential dx as

$$dx = \frac{1}{2}dz + \frac{1}{2}d\bar{z}.$$

The partial derivative of x with respect to $\bar{z}$ is $1/2$, which is never zero. Hence, it is not complex differentiable at any point.

Example 6.3.5. The function $f(z) = z \cdot \bar{z} = |z|^2 = x^2 + y^2$. The partial derivatives of f with respect to x and y are $2x$ and $2y$, respectively. The Wirtinger derivatives are

$$\begin{aligned}\frac{\partial f}{\partial z} &= \frac{1}{2}\big(2x - i2y\big) \\ \frac{\partial f}{\partial \bar{z}} &= \frac{1}{2}\big(2x + i2y\big).\end{aligned}$$

Hence,

$$\begin{aligned} d(|z|^2) &= (x - iy)dz + (x + iy)d\bar{z} \\ &= \bar{z}dz + zd\bar{z}. \end{aligned}$$

We see that $\bar{\partial} f = z$ and it is equal to 0 only when $z = 0$. As a result, the function $f(z) = |z|^2$ is complex differentiable only at the origin.

The $\bar{\partial}$ operator behaves like taking derivative with respect to $\bar{z}$. We can evaluate $\bar{\partial}\bar{z}$ and $\bar{\partial}z$ as follows,

$$\bar{\partial}\bar{z} = \frac{\partial \bar{z}}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big)(x - iy) = 1 \tag{6.3.12}$$

$$\bar{\partial}z = \frac{\partial z}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big)(x + iy) = 0. \tag{6.3.13}$$

On the other hand, we have

$$\frac{\partial \bar{z}}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big)(x - iy) = 0 \tag{6.3.14}$$

$$\frac{\partial z}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big)(x + iy) = 1. \tag{6.3.15}$$

Equations (6.3.12) to (6.3.15) suggest that z and $\bar{z}$ behave like independent variables, even though in fact they are not. For instance, using the product rule in Theorem 6.3.10 to compute the derivative of the product $z\bar{z}$ directly as

$$d(z\bar{z}) = \bar{z}dz + zd\bar{z}.$$

For a complex function $f(z)$ that is complex differential in a domain D, we have the following results. If we apply the $\bar{\partial}$ operator to $f(z)$, the resulting function $\frac{\partial f}{\partial \bar{z}}$ is identically zero in D. We can interpret this as "f is a function of z only, but not a function of $\bar{z}$."

6.4 Appendix: Fréchet Derivatives in Real and Complex Variables

The definition of in (6.1.1) and (6.2.1) characterize differentiability in terms of linear approximation. They are special cases of a more general notion called Fréchet differentiability.

This explains why the real and complex derivatives have many properties in common. In this appendix we give a short introduction to linear approximation in a general Banach space.

Fréchet Differentiability in Banach Spaces. Let X and Y be Banach spaces over $\mathbb{R}$ or $\mathbb{C}$. A function $F : X \to Y$ is *Fréchet differentiable* at a point $x_0 \in X$ if there exists a bounded linear map

$$DF(x_0) : X \to Y$$

such that

$$\lim_{\|h\|\to 0} \frac{\|F(x_0 + h) - F(x_0) - DF(x_0)[h]\|}{\|h\|} = 0.$$

The map $DF(x_0)$ is called the *Fréchet derivative* of F at x_0.

Uniqueness. If a Fréchet derivative exists, it is unique. Indeed, if A and B are two bounded linear maps satisfying the above limit, then

$$\frac{\|(A - B)h\|}{\|h\|} \to 0 \quad \text{as } h \to 0.$$

Since $A - B$ is linear and bounded, this implies $A = B$.

Real Fréchet Derivatives for Functions on $\mathbb{C}$. We regard $\mathbb{C}$ as a 2-dimensional real Banach space. A real-linear map $L : \mathbb{C} \to \mathbb{C}$ always has a unique decomposition

$$L(h) = ah + b\overline{h}, \qquad \text{for } a, b \in \mathbb{C}.$$

Thus, if $f : \mathbb{C} \to \mathbb{C}$ is Fréchet differentiable as a real map at z_0, then

$$Df(z)[h] = a(z)\, h + b(z)\, \overline{h}.$$

The coefficients $a(z)$ and $b(z)$ correspond to the Wirtinger derivatives:

$$a(z) = \frac{\partial f}{\partial z}(z), \qquad b(z) = \frac{\partial f}{\partial \overline{z}}(z).$$

Holomorphicity and Complex Linearity. A function $f : \mathbb{C} \to \mathbb{C}$ is complex differentiable at z_0 if and only if:

(i). it is Fréchet differentiable as a real map at z_0, and

(ii). its derivative is complex-linear, i.e. $b(z) = 0$.

In this case, $Df(z_0)[h] = f'(z)\,h$, and the Cauchy–Riemann equations are equivalent to the condition $b(z_0) = 0$.

Example 6.4.1. $f(z) = z^2$.

$$f(z+h) = z^2 + 2zh + h^2,$$

so

$$Df(z)[h] = 2z\,h.$$

This is complex-linear; hence f is complex differentiable everywhere.

Example 6.4.2. $f(z) = |z|^2 = z\overline{z}$.

$$f(z+h) = |z|^2 + z\overline{h} + \overline{z}h + |h|^2,$$

so

$$Df(z)[h] = z\overline{h} + \overline{z}h.$$

It is real-linear but not complex-linear unless $z = 0$.

Fréchet Derivatives in Complex Banach Spaces. If X and Y are complex Banach spaces, a map $F : X \to Y$ is *complex Fréchet differentiable* at x if:

(i). it is Fréchet differentiable as a real map, and

(ii). the derivative $DF(x)$ is complex-linear.

This generalizes holomorphicity to infinite-dimensional settings.

Example 6.4.3. Let $F : L^2([0,1]) \to \mathbb{C}$ be

$$F(f) = \int_0^1 f(t)^2\,dt.$$

Then

$$DF(f)[h] = 2\int_0^1 f(t)h(t)\,dt.$$

This operator is real-linear but not complex-linear with respect to the usual Hilbert space structure, so F is not complex differentiable as a map of complex Banach spaces.

6.5 Appendix: A necessary and sufficient condition for complex differentiability at a given point

($\Rightarrow$) Suppose $f(z)$ is complex differentiable at z_0. This means that

$$\lim_{h\to 0} \frac{f(z_0+h)-f(z_0)}{h}$$

exists. Let the limit be $a+ib$. When the magnitude of h is small, we can approximate $f(z_0+h)$ by

$$f(z_0+h) = f(z_0) + (a+ib)\cdot h + o(h), \tag{6.5.1}$$

where $o(h)$ represents a complex function of h that satisfies

$$\lim_{h\to 0} \frac{o(h)}{h} = 0.$$

By expressing $f(z)$ in terms of the real and imaginary parts $u(x,y)$ and $v(x,y)$, and $h = \Delta x + i\Delta y$, we can write (6.5.1) as

$$\begin{aligned} u(x_0+\Delta x, y_0+\Delta y) &= u(x_0,y_0) + a\Delta x - b\Delta y + \mathrm{Re}(o(\Delta x + i\Delta y)), \\ v(x_0+\Delta x, y_0+\Delta y) &= v(x_0,y_0) + a\Delta y + b\Delta x + \mathrm{Im}(o(\Delta x + i\Delta y)). \end{aligned}$$

In terms of matrix, we have

$$\begin{bmatrix} u(x_0+\Delta x, y_0+\Delta y) \\ v(x_0+\Delta x, y_0+\Delta y) \end{bmatrix} = \begin{bmatrix} u(x_0,y_0) \\ v(x_0,y_0) \end{bmatrix} + \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \begin{bmatrix} \mathrm{Re}(o(\Delta x + i\Delta y)) \\ \mathrm{Im}(o(\Delta x + i\Delta y)) \end{bmatrix}. \tag{6.5.2}$$

The length of the error term approaches to zero faster than $\sqrt{\Delta x^2 + \Delta y^2}$, because

$$\frac{\sqrt{\mathrm{Re}(o(\Delta x + i\Delta y))^2 + \mathrm{Im}(o(\Delta x + i\Delta y))^2}}{\sqrt{\Delta x^2 + \Delta y^2}} = \frac{|o(h)|}{|h|} = \left|\frac{o(h)}{h}\right|$$

converges to zero as $h \to 0$. This proves that the vector field $(u(x,y), v(x,y))$ is real differentiable at (x_0, y_0).

The fact that the 2×2 matrix in (6.5.2) is in the special form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ implies that the Cauchy-Riemann equations hold at (x_0, y_0).

($\Leftarrow$) In the reverse direction, we suppose that the vector field $(u(x,y), v(x,y))$ is real-differentiable at (x_0, y_0), and the Cauchy-Riemann equations are satisfied at (x_0, y_0). From the definition of real-differentiability, the quantity

$$\frac{1}{\sqrt{\Delta x^2 + \Delta y^2}} \left\| \begin{bmatrix} u(x_0 + \Delta x, y_0 + \Delta y) \\ v(x_0 + \Delta x, y_0 + \Delta y) \end{bmatrix} - \begin{bmatrix} u(x_0, y_0) \\ v(x_0, y_0) \end{bmatrix} - \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \right\|$$

approaches zero as the vector $(\Delta x, \Delta y)$ approaches the origin. We thus have a vector function $\mathbf{g}(\Delta x, \Delta y)$ with variables Δx and Δy, satisfying equation

$$\begin{bmatrix} u(x_0 + \Delta x, y_0 + \Delta y) \\ v(x_0 + \Delta x, y_0 + \Delta y) \end{bmatrix} = \begin{bmatrix} u(x_0, y_0) \\ v(x_0, y_0) \end{bmatrix} + \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \mathbf{g}(\Delta x, \Delta y) \tag{6.5.3}$$

and

$$\frac{\|\mathbf{g}(\Delta x, \Delta y)\|}{\sqrt{\Delta x^2 + \Delta y^2}} \to 0$$

as $\sqrt{\Delta x^2 + \Delta y^2} \to 0$.

Moreover, from the Cauchy-Riemann equation, we can write (6.5.3) as

$$\begin{bmatrix} u(x_0 + \Delta x, y_0 + \Delta y) \\ v(x_0 + \Delta x, y_0 + \Delta y) \end{bmatrix} = \begin{bmatrix} u(x_0, y_0) \\ v(x_0, y_0) \end{bmatrix} + \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \mathbf{g}(\Delta x, \Delta y)$$

for some real numbers a and b.

If we identify the vector function $\mathbf{g}(\Delta x, \Delta y)$ with a complex-valued function $e(\Delta x + i\Delta y)$ whose real and imaginary parts equal the first and second component of $\mathbf{g}(\Delta x, \Delta y)$, and $h = \Delta x + i\Delta y$, we get

$$\mathbf{g}(\Delta x, \Delta y) = \begin{bmatrix} \mathrm{Re}(e(\Delta x + i\Delta y) \\ \mathrm{Im}(e(\Delta x + i\Delta y)) \end{bmatrix}$$

and

$$f(z_0 + h) = f(z_0) + (a + ib) \cdot h + e(h).$$

We thus obtain

$$\frac{f(z_0 + h) - f(z_0)}{h} - (a + ib) = \frac{e(h)}{h},$$

Since $\lim_{h \to 0} |e(h)|/|h| = 0$, the limit

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists and is equal to $a + ib$.

Summary

- A domain/region is an open and connected set.
- A complex function $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at a point $z_0 = x_0 + iy_0$ if the vector-valued function $F(x, y) = (u(x, y), v(x, y))$ is real differentiable at (x_0, y_0), and the Cauchy-Riemann condition is satisfied at (x_0, y_0).
- Equivalently, $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at z_0 if the limit $\lim_{z \to z_0}(f(z_0 + h) - f(z_0))/h$ exists.
- A sufficient condition for a function $f(z)$ to be complex differentiable at z_0: (a) the partial derivatives u_x, u_y, v_x, v_y exist and are continuous in a neighborhood of z_0, (b) the Cauchy-Riemann condition is satisfied at z_0.
- If $f(z)$ is complex differentiable at z_0, we can compute $f'(z_0)$ by $u_x(x_0, y_0) + iv_x(x_0, y_0)$ or $v_y(x_0, y_0) - iu_y(x_0, y_0)$.
- Wirtinger derivatives:
$$\frac{\partial f}{\partial z} \triangleq \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big), \qquad \frac{\partial f}{\partial \bar{z}} \triangleq \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big).$$
- del-bar operator
$$\bar{\partial} \triangleq \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big).$$
- Cauchy-Riemann condition is equivalent to $\bar{\partial} f = 0$.
- When $f(z)$ is complex differentiable, $\frac{\partial f}{\partial z} = f'(z)$.
- The region of convergence of a power series is circular in shape. The radius of the convergence region is called the radius of convergence.
- Hadamard formula for the radius of convergence of a power series $\sum_{k=0}^{\infty} a_k z^k$:
$$R = \frac{1}{\limsup_k |a_k|^{1/k}}.$$
- The function $f(z)$ defined by $\sum_{k=0}^{\infty} a_k z^k$ is holomorphic in the region of convergence. The derivative $f'(z)$ can be computed by $\sum_{k=1}^{\infty} k a_k z^{k-1}$.

Exercise

1. Let $f(z) = \frac{z^9 - i}{z - i}$.

   (a) Find the limit of $f(z)$ as $z \to i$.

   (b) Let $g(z)$ be the function

$$g(z) = \begin{cases} f(z) & \text{if } z \neq i, \\ \text{answer in (a)} & \text{if } z = i. \end{cases}$$

   Compute the derivative of $g(z)$ at $z = i$.

2. Define a complex function $f(z)$ by

$$f(z) \triangleq \begin{cases} \frac{\bar{z}^2}{z} & \text{if } z \neq 0 \\ 0 & \text{if } z = 0. \end{cases}$$

   (a) Show that $f(z)$ is not complex differentiable at $z = 0$.

   (b) Verify that $f(z)$ satisfies the Cauchy-Riemann condition at $z = 0$.

3. Write $z = x + iy$. Determine whether the following functions $f(z)$ are complex differentiable at $z = 0$.

   (a) $x\sqrt{x^2 + y^2}$ (b) $x + y$ (c) $\sqrt{|xy|}$

4. Determine where f satisfies the Cauchy-Riemann equations and where it is complex differentiable.

   (a) $f(z) = x^2 + y^2 + i(2xy)$.

   (b)

$$f(z) = \frac{x - y}{x + y} + i\frac{xy}{x^2 + y^2}.$$

5. Find the values of a, b and c such that the function $ay^3 + ix^3 + bx^2y + icy^2x$ is complex differentiable for all $z = x + iy$.

6. (Cauchy-Riemann condition in polar coordinates) We write z in polar form $re^{i\theta}$. Prove the statements in in parts (a) and (b).

(a) The Cauchy-Riemann conditions for complex function $f(z) = u(r,\theta) + iv(r,\theta)$ in polar coordinates

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

(b) If $f(z)$ is complex differentiable, the complex derivative of f at $z_0 = r_0 e^{i\theta_0}$ is equal to $e^{-i\theta}(u_r + iv_r)$, evaluated at (r_0, θ_0).

7. Use the previous exercise to derive the complex derivative of $f(z)$.

   (a) $f(z) = \sqrt{z} = \sqrt{r}e^{i\theta/2}$, $f'(z) = \frac{1}{2f(z)}$.

   (b) $f(z) = \log z = \ln r + i\theta$, $f'(z) = \frac{1}{z}$.

8. Verify that $d(\log z) = \frac{dz}{z}$.

9. Show that the $\bar{\partial}$ operator in polar coordinates can be expressed as

$$\frac{e^{i\theta}}{2}\left(\frac{\partial}{\partial r} + \frac{i}{r}\frac{\partial}{\partial \theta}\right)$$

   assuming that $r \neq 0$.

10. Prove that

$$\partial\bar{\partial} = \frac{\Delta}{4},$$

   where Δ represents the Laplacian operator.

Exercise

1.

Chapter 7

Holomorphic Functions

7.1 Holomorphic functions and entire functions

It turns out that being differentiable at an isolated point (as defined in Definition 6.2.1) is not a very useful property. Much more structure can be seen if we require a function to be complex differentiable in an open neighborhood of a point.

Definition 7.1.1. A complex function $f(z)$ is said to be *holomorphic* at a point z_0 if there is a neighborhood of z_0 such that f is complex differentiable at every point in the neighborhood. A function is said to be *entire* if it is complex differentiable at every point in $\mathbb{C}$. We denote the complex derivative of f by $\frac{df}{dz}$ when f is holomorphic.

If we can show that a function is complex differentiable at every point in the domain of definition, then it is holomorphic in the domain.

Remark. In this notes, we will use the term "analytic" to refer to complex functions that are locally representable by complex power series. We will show later that this is equivalent to the condition of complex differentiable at every point in the domain.

Using Theorem 6.2.5, we can check that a complex function is holomorphic if it satisfies the Cauchy-Riemann equations and has continuous first partial derivatives throughout the domain of definition.

In Example 6.3.5 we see that the function $f(z) = x^2 + y^2$ is complex differentiable only at the origin. Since this function is only complex differentiable at a single point, it is not

classified as holomorphic function.

Example 7.1.1. The function $f(x+iy) = x+iy^2$ is continuous and smooth as a real function. We check that $\partial f/\partial \bar{z}$ equals

$$\begin{aligned}\frac{\partial f}{\partial \bar{z}} &= \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big)(x+iy^2)\\ &= \frac{1}{2} - y,\end{aligned}$$

which is zero when and only when $y = 1/2$. The function $x + iy^2$ is thus complex differentiable on the line $y = 1/2$.

Example 7.1.2. Define a complex function

$$f(z) = \begin{cases} \frac{x^3-y^3+i(x^3+y^3)}{x^2+y^2} & \text{if } x+iy \neq 0 \\ 0 & \text{if } x+iy = 0. \end{cases}$$

This function is continuous at $z = 0$, satisfies the Cauchy-Riemann equations at $z = 0$, but fail to be complex differentiable at $z = 0$.

We first show that $f(z)$ is continuous at $z = 0$. Take the absolute value of $f(z)$,

$$\begin{aligned}|f(z)| &= \frac{1}{x^2+y^2}|x^3 - y^3 + i(x^3+y^3)|\\ &= \frac{\sqrt{2(x^6+y^6)}}{x^2+y^2}.\end{aligned}$$

We simplify and upper bound $x^6 + y^6$ as

$$(x^2+y^2)(x^4 - x^2y^2 + y^4) \leq (x^2+y^2)(x^4 + 2x^2y^2 + y^4) = (x^2+y^2)^3.$$

We then get

$$|f(z)| \leq \sqrt{2(x^2+y^2)} = \sqrt{2}|z|$$

If we take $|z| \to 0$, we have $|f(z)| \to 0$. Therefore $f(z)$ approaches 0 as z approaches 0. The function $f(z)$ is a continuous function.

The real and imaginary parts of $f(z)$ are

$$u(x,y) = \frac{x^3-y^3}{x^2+y^2}, \qquad v(x,y) = \frac{x^3+y^3}{x^2+y^2},$$

Table 7.1: Summary of the examples

Not real differentiable	Real differentiable but not holomorphic	Holomorphic in a domain	Entire
Ex. 6.1.2, 7.1.2	Ex. 6.1.1, 6.2.4, 6.3.5, 7.1.1	Ex. 6.2.3	Ex. 6.2.6, 6.2.7

respectively. We next show that the Cauchy-Riemann equations hold at $z = 0$,

$$u_x(x,0) = \frac{\partial x}{\partial x} = 1, \qquad v_y(0,y) = \frac{\partial y}{\partial y} = 1,$$
$$u_y(0,y) = \frac{\partial(-y)}{\partial y} = -1, \qquad v_x(x,0) = \frac{\partial x}{\partial x} = 1.$$

Therefore, we have $u_x(0,0) = v_y(0,0)$ and $u_y(0,0) = -v_x(0,0)$.

However, this function is not complex differentiable at $z = 0$, (and hence is not real differentiable by Theorem 6.2.4). If this function were complex differentiable, the complex derivative would be

$$u_x(0,0) + iv_x(0,0) = 1 + i$$

by Corollary 6.2.7. But if we approach $z = 0$ in a straight line from the point $1 + i$, we will get

$$\lim_{t\to 0} \frac{1}{t(1+i)}(f(t+it) - f(0)) = \lim_{t\to 0} \frac{1}{t(1+i)}\Big(u(t,t) + iv(t,t)\Big) = \frac{1}{t(1+i)} i \frac{2t^3}{2t^2} = \frac{1+i}{2}.$$

Obviously $1 + i$ is not equal to $(1 + i)/2$. Therefore $f(z)$ is not complex differentiable at $z = 0$.

7.2 Term-wise differentiation of power series

In this section we give examples of complex differentialbe function. Using the property of uniform convergence, we can prove that the function $\sum_{k=0}^{\infty} a_k z^k$ is complex differentiable within the region of convergence.

Theorem 7.2.1 (Term-wise differentiation of power series)**.** *Suppose power series* $\sum_{k=0}^{\infty} a_k z^k$ *has radius of convergence* $R > 0$*.*

(a). The series $\sum_{k=1}^{\infty} k a_k z^{k-1}$ *has radius of convergence* R*.*

(b). The function $f(z) = \sum_{k=0}^{\infty} a_k z^k$ *is complex differentiable for all points in* $|z| < R$*, with complex derivative given by* $f'(z) = \sum_{k=1}^{\infty} k a_k z^{k-1}$*.*

Proof. We first use the Hadamard formula in Theorem 3.3.4 to show that $\sum_k k a_k z^{k-1}$ has radius of convergence equal to R. We note that the radius of convergence of $\sum_k k a_k z^{k-1}$ is the same as the radius of convergence of $\sum_k k a_k z^k$, and the radius of convergence of the latter series can be computed by

$$\frac{1}{\limsup_k |k a_k|^{1/k}} = \frac{1}{\limsup_k \sqrt[k]{k} |a_k|^{1/k}} = \frac{1}{\limsup_k |a_k|^{1/k}} = R.$$

In the second equality, we have used the fact that $\lim_{k\to\infty} \sqrt[k]{k} = 1$.

To prove the second part, we pick a complex number z_0 inside the region of convergence, i.e., $|z_0| < R$. Let δ be a real number such that $|z_0| + \delta < R$. The choice of δ is made to ensure that the open disc $\{z : |z - z_0| < \delta\}$ lies inside the region of convergence.

We set up some notation. For $n \geq 0$, let $f_n(z)$ denote the n-th partial sum

$$f_n(z) \triangleq \sum_{k=0}^{n} a_k z^k,$$

and let $f(z)$ denote the limit

$$f(z) \triangleq \sum_{k=0}^{\infty} a_k z^k.$$

By Theorem 5.5.4, $f_n(z)$ converges to $f(z)$ uniformly in the closed disk

$$D \triangleq \{z : |z| \leq |z_0| + \delta\}.$$

It is important to note that the radius of closed disk D depends on the location of z_0.

By Theorem 5.5.4, the series $\sum_{k=1}^{\infty} k a_k z^{k-1}$ converges absolutely and uniformly in the closed disk S.

For each $n \geq 0$, define

$$\varphi_n(h; z_0) \triangleq \frac{f_n(z_0+h) - f_n(z_0)}{h} = \sum_{k=0}^{n} a_k \frac{(z_0+h)^k - z_0^k}{h} \tag{7.2.1}$$

for nonzero h with $|h| \leq \delta$. When $n \to \infty$, $\varphi_n(h; z_0)$ converges to

$$\frac{f(z_0+h) - f(z_0)}{h}.$$

The convergence is uniform for $|h| \leq \delta$. Indeed, we can write the absolute value of the k-th term in in (7.2.1) as

$$\left| a_k \frac{(z_0+h)^k - z_0^k}{h} \right| = |a_k| \frac{|(z_0+h)^k - z_0^k|}{|h|}.$$

By using the identity

$$\alpha^k - \beta^k = (\alpha - \beta)(\alpha^{k-1} + \alpha^{k-2}\beta + \alpha^{k-3}\beta^2 + \cdots + \alpha\beta^{k-2} + \beta^{k-1}),$$

we can upper bound it by

$$\begin{aligned} \left| a_k \frac{(z_0+h)^k - z_0^k}{h} \right| &= |a_k| \Big(\sum_{j=0}^{k-1} |z_0+h|^j |z_0|^{k-1-j} \Big) \\ &\leq |a_k| k (|z_0| + \delta)^{k-1}. \end{aligned}$$

The last inequality follows from the fact that there are k terms in the summation, and each term has absolute value upper bounded by $(|z_0| + \delta)^{k-1}$. Because the series

$$\sum_{k=1}^{\infty} |a_k| k (|z_0| + \delta)^{k-1}$$

is convergent (by the first part of the theorem) and does not depend on the value of h, we can see that the convergence $\varphi(h; z_0)$ is uniform over h, by Weierstrass M-test (Theorem 5.5.3).

Consider the double limit

$$\lim_{n\to\infty} \Big(\lim_{h\to 0} \varphi_n(h; z_0) \Big).$$

For each n, $f_n(z)$ is a polynomial. Hence $\lim_{h\to 0} \varphi_n(h; z_0) = \sum_{k=1}^{n} k a_k z_0^{k-1}$. This gives

$$\lim_{n\to\infty} \Big(\lim_{h\to 0} \varphi_n(h; z_0) \Big) = \lim_{n\to\infty} \sum_{k=1}^{n} k a_k z_0^{k-1} = \sum_{k=1}^{\infty} k a_k z_0^{k-1}. \tag{7.2.2}$$

(Convergence of the last summation is justified by part 1.)

On the other hand, since φ_n converges uniformly for $|h| \leq \delta$, we can exchange the order of the two limits to get

$$\lim_{h\to 0}\left(\lim_{n\to\infty}\varphi_n(h;z_0)\right) = \lim_{h\to 0}\frac{f(z_0+h)-f(z_0)}{h}. \tag{7.2.3}$$

Since (7.2.2) and (7.2.3) are equal, the limit in (7.2.3) exists and is equal to the power series in (7.2.2). We thus obtain

$$f'(z_0) = \sum_{k=1}^{\infty} k a_k z_0^{k-1}.$$

This proves that a power series is complex differentiable, and we can differentiate it term-wise. □

Using Theorem 7.2.1, we can prove that

$$\frac{d}{dz}e^z = e^z.$$

We differentiate the power series term-wise,

$$\frac{d}{dz}\sum_{k=0}^{\infty}\frac{z^k}{k!} = \sum_{k=0}^{\infty} k\frac{e^{k-1}}{k!} = \sum_{k=1}^{\infty}\frac{e^{k-1}}{(k-1)!}.$$

After a change of index, we can see that this is the same as the power series of e^z. Similarly, we can derive

$$\frac{d}{dz}\sin(z) = \cos(z),$$
$$\frac{d}{dz}\cos(z) = -\sin(z).$$

By repeatedly applying Theorem 7.2.1, we can differentiate a power series arbitrarily many times.

Theorem 7.2.2. *Suppose the radius of convergence of power series $\sum_{k=0}^{\infty} a_k z^k$ is positive. For any positive integer j, the function $f(z)$ defined by a power series $\sum_{k=0}^{\infty} a_k z^k$ can be differentiated j-th times at any point in the interior of the region of convergence, and the j-th derivative can be computed by the power series*

$$f^{(j)}(z) = \sum_{k=j}^{\infty} k(k-1)(k-2)\cdots(k-j+1)a_k z^{k-j}.$$

In real analysis, we have example of infinitely differentiable functions that does not have power series expansion. The following real continuous function

$$f(x) \triangleq \begin{cases} 0 & \text{if } x \leq 0 \\ e^{-1/x^4} & \text{if } x > 0 \end{cases}$$

is infinitely smooth. However, the derivatives of all orders at $x = 0$ are all equal to zero. The power series expansion of $f(x)$ at $x = 0$ is the constant function 0, which is not the same as the original function $f(x)$. This function fails to have power series expansion at $x = 0$.

7.3 Differentiation rules and differentiation at the point at infinity

The usual rules for differentiation continue to hold for holomorphic functions. We note that another common notation for complex derivative is $\frac{df}{dz}$. This is a complex function whose value at z_0 is $f'(z_0)$.

The followings are the basic differentiation formulae.

Theorem 7.3.1. *Suppose $f(z)$ and $g(z)$ are holomorphic in a domain D.*

$$\begin{aligned} (f(z) + g(z))' &= f'(z) + g'(z), \\ (ag(z))' &= ag'(z), \text{ for constant } a \in \mathbb{C}, \\ (f(z)g(z))' &= f'(z)g(z) + f(z)g'(z), \\ (f(z)/g(z))' &= \frac{f'(z)g(z) - f(z)g'(z)}{g(z)^2}, \quad \text{provided that } g(z) \neq 0. \end{aligned}$$

If the co-domain of $g(z)$ is inside the domain of f, and g, f are both holomorphic, we have the chain rule

$$\frac{d}{dz} f(g(z)) = f'(g(z)) \cdot g'(z).$$

Proof. The chain rule is proved in Theorem 6.2.8. The first three differentiation rule are re-phrasing the analogous results in Theorem 6.3.10.

To prove the differentiation rule for division, we combine Example 6.2.3 and the chain rule to obtain the derivative of $g(z)^{-1}$.

$$\frac{d}{dz}(g(z)^{-1}) = -\frac{g'(z)}{g(z)^2}.$$

We then conclude by using the product rule,

$$\frac{d}{dz}\Big(\frac{f(z)}{g(z)}\Big) = \frac{d}{dz}(f(z)\cdot g(z)^{-1}) = f'(z)g(z)^{-1} + f(z)\frac{-g'(z)}{g(z)^2} = \frac{f'(z)g(z) - f(z)g'(z)}{g(z)^2}.$$

□

We illustrate the differentiation rules by some examples.

Example 7.3.1. Show that $(z^n)' = nz^{n-1}$ for all $z \in \mathbb{C}$.

We prove by induction. It is trivially true when $n = 1$. Suppose that it is true for $n = k$. When $n = k + 1$, we apply the product rule to see that

$$(z^k)' = (zz^{k-1})' = (1)z^{k-1} + z(k-1)z^{k-2} = kz^{k-1}.$$

Example 7.3.2. By combining the previous example with the sum rule, we see that polynomials are entire (complex differentiable at all point), and

$$\frac{d}{dz}\big(a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n\big) = a_1 + 2a_2 z + \cdots + na_n z^{n-1}.$$

Example 7.3.3. Show that $(\sin z)' = \cos z$.

We differentiate both sides of the identity $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$ to get

$$(\sin z)' = \Big(\frac{e^{iz} - e^{-iz}}{2i}\Big)' = \frac{(e^{iz} - e^{-iz})'}{2i} = \frac{ie^{iz} - (-i)e^{-iz}}{2i} = \frac{e^{iz} + e^{-iz}}{2} = \cos z.$$

To study the behavior of a function $f(z)$ at the point at infinity, we make a change of variable $w = 1/z$, and consider the function $g(w) = f(1/w)$ when w is close to the origin. The derivative of $f(z)$ at the point at infinity is defined as follows.

Definition 7.3.2. Given a complex function $f(z)$, we let $g(w) = f(1/w)$, and say that f is

- *continuous at the point at infinity* if $g(w)$ is continuous at $w = 0$,
- *complex differentiable at the point at infinity* if $g(w)$ is complex differentiable at $w = 0$,
- *holomorphic at the point at infinity* if $g(w)$ is holomorphic at $w = 0$.

Example 7.3.4. Consider the function $f(z) = (z+1)/(z-1)$. It is differentiable at ∞. The function $g(w) = f(1/w)$ is

$$g(w) = \frac{\frac{1}{w}+1}{\frac{1}{w}-1} = \frac{1+w}{1-w}.$$

As $w \to 0$, we have $g(w) \to 1$. The derivative of $g(w)$ is

$$g'(w) = \frac{2}{(1-w)^2}$$

At the point $w = 0$, the derivative of $g(w)$ is $g'(0) = 2$.

The previous example can be extended to general Möbius transformation $f(z) = (az + b)/(cz + d)$ with $c \neq 0$. Define

$$g(w) = \frac{a\frac{1}{w}+b}{c\frac{1}{w}+d} = \frac{wb+a}{wd+c}.$$

By the quotient rule, we obtain the derivative of $g(w)$,

$$g'(w) = \frac{b(wd+c) - d(wb+a)}{(wd+c)^2} = \frac{bc-da}{(wd+c)^2}.$$

The function $g(w)$ is holomorphic at $w = 0$, and the value of $g'(0)$ is $(bc - da)/c^2$. This proves that $f(z)$ is conformal at the point at infinity.

The complex number $-d/c$ is mapped to the point at infinity by the LFR $f(z) = (az + b)/(cz + d)$. We can study the behavior new the point $-d/c$ by considering the reciprocal $h(z) \triangleq 1/f(z)$. The function $h(z)$ is an LFR, and it maps $-d/c$ to the origin. We can compute the derivative of $h(z)$ as

$$h'(z) = \left(\frac{cz+d}{az+b}\right)' = \frac{c(az+b) - a(cz+d)}{(az+b)^2} = \frac{bc-ad}{(az+b)^2}$$

The value of $h'(z)$ at $z = -d/c$ is

$$h'(-d/c) = \frac{bc - ad}{(a(-d/c) + b)^2} = \frac{c^2(bc - ad)}{(-ad + bc)^2} = \frac{c^2}{bc - ad}$$

If the coefficient c is nonzero, we see that the function $f(z)$ has nonzero derivative at the point $-d/c$.

7.4 Chain rule for parametric curve

Definition 7.4.1. A parametric curve C represented by

$$z : [a, b] \to \mathbb{C}$$
$$z(t) = x(t) + iy(t)$$

is said to be *smooth* if $x(t)$ and $y(t)$ are continuously differentiable on $[a, b]$.

To emphasize the direction/orientation of the curve from $t = a$ to $t = b$, we sometime use the word "contour" instead of "path". If a curve can be divided into finitely many parts and each part is smooth, then we call it a *piece-wise smooth* curve, i.e., it is a concatenation of smooth curves. Be default, all curves in this note are piece-wise smooth curve.

Consider a curve $z(t)$ that passes through the point z_0 if and $z(t_0) = z_0$, for some t_0 in the range $[a, b]$. Suppose $f(z)$ is a complex function that is complex differentiable at z_0.

We want to study the rate of change at the point $z(t_0) = x_0 + iy_0$ on the curve $z(t)$. By considering the real and imaginary parts of

$$f(z(t)) = u(x(t), y(t)) + iv(x(t), y(t))$$

separately, we obtain, by the chain rule from real analysis,

$$\frac{d}{dt} u(x(t), y(t))\Big|_{t=t_0} = u_x(x_0, y_0) \cdot x'(t_0) + u_y(x_0, y_0) \cdot y'(t_0),$$

and

$$\frac{d}{dt} v(x(t), y(t))\Big|_{t=t_0} = v_x(x_0, y_0) \cdot x'(t_0) + v_y(x_0, y_0) \cdot y'(t_0).$$

The derivative of the vector function

$$\frac{d}{dt} \begin{bmatrix} u(x(t), y(t)) \\ v(x(t), y(t)) \end{bmatrix}$$

with respect to t, evaluated at $t = t_0$, can be expressed using matrix-vector multiplication

$$\begin{bmatrix} u_x(x_0, y_0) & u_y(x_0, y_0) \\ v_x(x_0, y_0) & v_y(x_0, y_0) \end{bmatrix} \begin{bmatrix} x'(t_0) \\ y'(t_0) \end{bmatrix}.$$

So far, we only use the hypothesis that the functions u, v, x and y are real-differentiable. If $f(z)$ is complex differentiable at $z_0 = x_0 + iy_0$, the above matrix is in the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ for some real numbers a and b, and we can express the calculations using complex multiplication:

$$\frac{d}{dt} f(z(t))\Big|_{t=t_0} = (a + bi)(x'(t_0) + iy'(t_0)).$$

The complex constant $a + bi$ is precisely the complex derivative $f'(z_0)$, and $x'(t_0) + iy'(t_0)$ is by definition the derivative $z'(t)$ evaluated at $t = t_0$.

In summary, we have established the following result.

Theorem 7.4.2. *If f is complex differentiable at $z_0 = z(t_0)$,*

$$\frac{d}{dt} f(z(t))\Big|_{t=t_0} = f'(z_0) \cdot z'(t_0). \tag{7.4.1}$$

When the complex derivative of $f(z)$ at z_0 is nonzero, the geometric action of $f(z)$ in a small vicinity of z_0 is a rotation followed by a scaling. In terms of differential form, we can express the same idea as

$$f(z(t))\, dt = f'(z(t))z'(t)\, dt. \tag{7.4.2}$$

7.5 Conformal property

Imagine we have two parametric curves $\gamma_1(t)$ and $\gamma_2(t)$ through z_0. Let the range of the parameter t in $\gamma_1(t)$ be $[a, b]$, where $a < b$, and $\gamma_1(t_1) = z_0$, for some t_1 in the range $[a, b]$. Likewise, let the range of t in $\gamma_2(t)$ be $[c, d]$, where $c < d$ and $\gamma_2(t_2) = z_0$, for some t_2 in $[c, d]$. Assuming both $\gamma_1'(t_1)$ and $\gamma_2'(t_2)$ are nonzero, the angle between the tangent line of γ_1 at z_0 and the tangent line of γ_2 at z_0 equals

$$\arg\left(\frac{\gamma_2'(t_2)}{\gamma_1'(t_1)}\right).$$

In Fig. 7.1, the angle between $\gamma_1(t)$ and $\gamma_2(t)$ is indicated by θ.

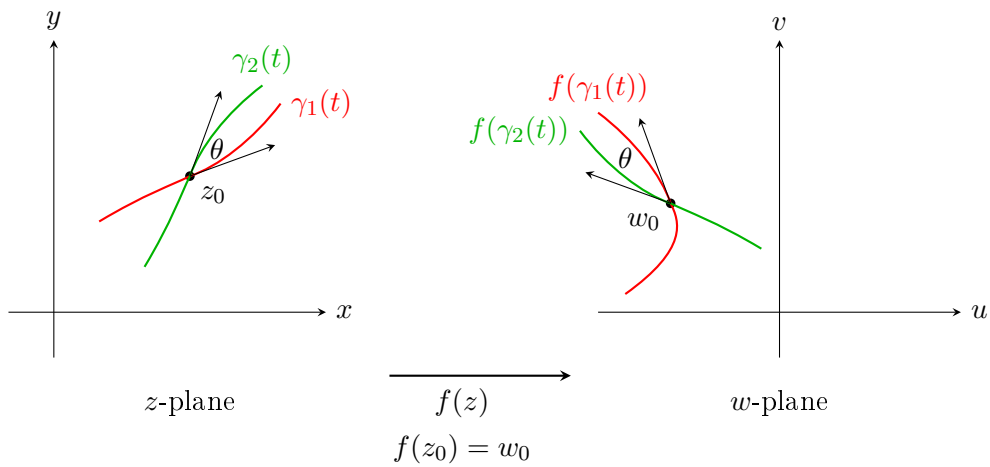


Figure 7.1: Angle-preserving property.

Suppose $f(z)$ is complex differentiable at a given point z_0 in $\mathbb{C}$, and $f'(z_0)$ is nonzero. If we apply the function f to $\gamma_1(t)$ and $\gamma_2(t)$, we obtain a pair of curves intersecting at the point $w_0 = f(z_0)$,

$$w_0 = f(\gamma_1(t_1)) = f(\gamma_2(t_2)).$$

We will show that f is *angle-reserving*, or *conformal* at z_0. This means that the angle between γ_1 and γ_2 at z_0 is the same as the angle between the transformed curves at $f(z_0)$. In mathematical terms, what we want to show is

$$\arg\left(\frac{(f \circ \gamma_2)'(t_2)}{(f \circ \gamma_1)'(t_1)}\right) = \arg\left(\frac{\gamma_2'(t_2)}{\gamma_1'(t_1)}\right).$$

The key is the chain rule for $f(z(t))$, where $z : [a, b] \to \mathbb{C}$ is a smooth function mapping real numbers in the interval $[a, b]$ to the complex plane, and $f : \mathbb{C} \to \mathbb{C}$ is a complex function whose domain contains the range of $z(t)$.

Assuming $f'(z_0)$ is nonzero, we can apply the chain rule in (7.4.1) to get

$$\arg\left(\frac{(f \circ \gamma_2)'(t_2)}{(f \circ \gamma_1)'(t_1)}\right) = \arg\left(\frac{f'(\gamma_2(t_2)) \cdot \gamma_2'(t_2)}{f'(\gamma_1(t_1)) \cdot \gamma_1'(t_1)}\right) = \arg\left(\frac{\gamma_2'(t_2)}{\gamma_1'(t_1)}\right).$$

As a special case, if the two curves are perpendicular to each other in the domain, then it continues to be perpendicular to each other in the co-domain.

The conformal property explains why the conjugate function $f(z) = \bar{z}$ is not complex differentiable anywhere. The conjugate function is a reflection geometrically, and reflection reverses the orientation of angles.

A transformation that is complex differentiable with nonzero derivative throughout the domain is of particular importance in complex analysis.

Definition 7.5.1. We say that two regions R_1 and R_2 in $\mathbb{C}$ are *conformally equivalent* if we can find a holomorphic bijection $f(z)$ from R_1 to R_2 such that $f'(z) \neq 0$ for all z in region R_1.

Example 7.5.1. The derivative of the square function $z \mapsto z^2$ is nonzero when $z \neq 0$. The transformation by the square function is conformal if the range does not contain the origin. For example, if R_1 is the open quarter circle $\{x + iy : x > 0, y > 0, x^2 + y^2 < 1\}$, the mapping from R_1 to the half circle $\{x + iy : y > 0, x^2 + y^2 < 1\}$ is angle-preserving.

More generally, for any positive integer n, the power function $z \mapsto z^n$ is a conformal mapping if the origin is not in the domain.

Example 7.5.2. For nonzero complex constant a, the affine function $az + b$ has constant derivative a, and hence is conformal.

Example 7.5.3. A linear fractional transformation $f(z) = (az+b)/(cz+d)$ is angle-preserving at all points. We consider the non-affine case, i.e., assume $c \neq 0$.

If z is not equal to $-d/c$, the derivative of $f(z)$ equals

$$f'(z) = \frac{a(cz + d) - c(az + b)}{(cz + d)^2} = \frac{ad - cb}{(cz + d)^2}$$

Since it is always assumed that $ad - bc$ is nonzero, the derivative of a linear fractional transformation is nonzero.

When $z = -d/c$, we consider the reciprocal $g(z) = 1/f(z)$. The derivative of $g(z)$

$$g'(z) = \frac{c(az + b) - a(cz + d)}{(az + b)^2} = \frac{cb - ad}{(az + b)^2}$$

is not equal to zero at $z = -d/c$. Finally, we check the derivative at the point at infinity. We make a change of variable $z = 1/w$ and compute the derivative of $g(w) = f(1/w)$ at $w = 0$,

$$g'(w) = \left(\frac{a + bw}{c + dw}\right)' = \frac{b(c + dw) - d(a + bw)}{(c + dw)^2} = \frac{bc - da}{(c + dw)^2}.$$

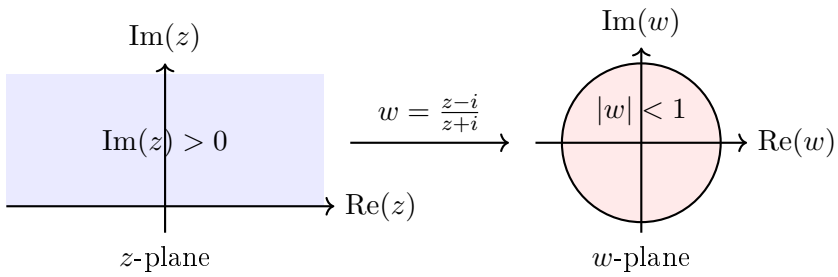


Figure 7.2: Mapping property of linear fractional transformation $f(z) = (z - i)/(z + i)$.

When $w = 0$, the derivative is $(bc - da)/c^2$, which is nonzero. This proves that a linear fractional transformation is a conformal mapping from the Riemann sphere to itself.

Example 7.5.4. The upper half-plane is conformally equivalent to the unit disc. We can apply the linear fractional transformation $f(z) = (z - i)/(z + i)$. Since a linear fractional transformation maps a circle/straight line to circle/straight line, we can check the image of three specific points on the real axis.

$$\begin{aligned} f(1) &= \frac{1-i}{1+i} = \frac{(1-i)^2}{2} = \frac{-2i}{2} = -i \\ f(0) &= \frac{0-i}{0+i} = \frac{-i}{i} = -1 \\ f(-1) &= \frac{-1-i}{-1+i} = \frac{(-1-i)^2}{2} = i \end{aligned}$$

The unique circle that passes through the three image points $-i$, -1 and i is the unit circle. Therefore, the function $f(z)$ maps the real axis to the unit circle. We further check that $f(i) = 0$, i.e., the point i in the upper half plane is mapped to the origin. The above argument shows that $f(z)$ maps the upper half-plane to the interior of the unit disc.

Example 7.5.5. The complex exponential function $f(z) = e^z$ maps the horizontal strip $\{z : -\pi/2 < \mathrm{Im}(z) < \pi/2\}$ conformally and bijectively to the right half plane $\{z : \mathrm{Re}(z) > 0\}$. (See Fig. 7.2).

By the chain rule, composition of two conformal maps is also conformal.

Example 7.5.6. We can map the right half-plane to the unit disk through the linear fractional transformation in Example 7.5.4. We can first map the right half-plane to the upper half-plane by multiplication by i, then apply the function $f(z) = \frac{z-i}{z+i}$. The overall transformation

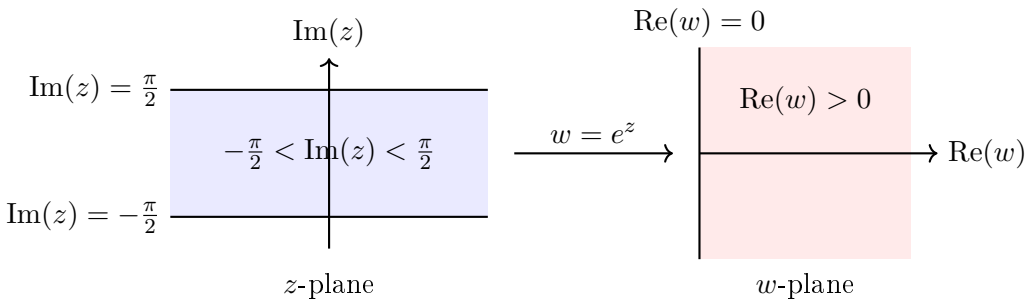


Figure 7.3: Mapping property of the complex exponential function.

is

$$f(iz) = \frac{(iz) - i}{(iz) + i} = \frac{z - 1}{z + 1}.$$

Example 7.5.7. We can find a conformal mapping that transform the horizontal strip $S = \{z : 0 < \mathrm{Im}(z) < \pi\}$ to the unit disk as follows. First, we use the exponential function $z \mapsto \exp(z)$ map S to the upper half-plane. The apply the transformation $f(z)$ in Example 7.5.4 to map it to the unit disk. The composed transformation is

$$z \mapsto \frac{e^z - i}{e^z + i}.$$

7.6 Harmonic functions

Another necessary condition for being holomorphic is that the real and imaginary parts are both harmonic functions. A twice-differentiable real-valued function $F(x, y)$ is said to be harmonic if it satisfies the Laplace equation

$$F_{xx} + F_{yy} = 0,$$

where F_{xx} and F_{yy} are the second partial derivatives of F with respect to x and y, respectively.

Suppose that $u(x, y)$ and $v(x, y)$ are twice differentiable, and the second-order derivatives are continuous functions, so that $u_{xy} = u_{yx}$ and $v_{xy} = v_{yx}$. Under these assumptions, we can deduce that the real and imaginary part of a holomorphic function are both harmonic functions. (We will show in a later lecture that the additional smoothness requirements

always hold.) In general, we say that a function is in C^2 if the second partial derivatives exist and are continuous.

Theorem 7.6.1. *Let $f(z)$ be a holomorphic function on a region R. If the real part $u(x, y)$ and imaginary part $v(x, y)$ are in C^2. Then*

$$u_{xx} + u_{yy} = 0 = v_{xx} + v_{yy}$$

in the region R.

Proof. The complex function $f(z)$ is holomorphic by assumption, hence must satisfy the Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$. Take partial derivatives with respect to x and y, we obtain $u_{xx} = v_{yx}$ and $u_{yy} = -v_{xy}$. Because we can exchange the order of the partial derivatives, we get

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = v_{xy} - v_{xy} = 0.$$

Similarly, we can show that $v(x, y)$ is harmonic by first deriving $u_{xy} = v_{yy}$ and $u_{yx} = -v_{xx}$ from the Cauchy-Riemann equations. Then,-

$$v_{xx} + v_{yy} = -u_{yx} + u_{xy} = -u_{xy} + u_{xy} = 0.$$

□

In the followings, we will assume that $u(x, y)$ and $v(x, y)$ are C^2 functions. We define the notion of harmonic conjugates

Definition 7.6.2. Given a harmonic function $u(x, y)$, we say that $v(x, y)$ is a *harmonic conjugate* of $u(x, y)$ if $u(x, y) + iv(x, y)$ is holomorphic in a domain.

Equivalently, we can define that $v(x, y)$ is a harmonic conjugate to u if u and v satisfy the Cauchy-Riemann equations. It is obvious that harmonic conjugate to a given harmonic functions, if exists, is not unique. It is because, if $v(x, y)$ is a harmonic conjugate to $u(x, y)$, then for any complex constant C, the function $v(x, y) + C$ is also a harmonic conjugate.

The existence of harmonic conjugate depends on the topology of the domain on which the function $u(x, y)$ is defined. It is known that if the domain D is simply connected, then

a harmonic function $u(x, y)$ defined on D has a harmonic conjugate. We can formulate the problem in terms of differential form. We want to find a function $v(x, y)$ whose differential is given by

$$dv = v_x dx + v_y dy = -u_y dx + u_x dy.$$

Since u_y and u_x are given, the problem of finding a harmonic conjugate to $u(x, y)$ reduces to solving the differential equation above. We prove in the appendix of this section that solution always exists if the region of definition D is simply connected (The proof relies an application of the Green's theorem).

Example 7.6.1. We consider the function

$$u(x, y) = \sin(x)\cosh(y).$$

This is harmonic because

$$u_{xx} = -\sin(x)\cosh(y), \quad \text{and} \quad u_{yy} = \sin(x)\cosh(y).$$

From the condition $v_x = -u_y = -\sin(x)\sinh(y)$, we can obtain

$$v(x, y) = \int -\sin(x)\sinh(y)\, dx = \cos(x)\sinh(y) + C(y),$$

where $C(y)$ is a function of y. By applying the other Cauchy-Riemann equation $v_y = u_x$, we get

$$v_y = \cos(x)\cosh(y) + C'(y) = u_x = \cos(x)\cosh(y).$$

Hence the function $C(y)$ should be a constant. The harmonic conjugate of $u(x, y)$ can be written as

$$v(x, y) = \cos(x)\sinh(y) + C$$

for some constant C. The complex holomorphic function is

$$\begin{aligned}
\sin(x)\cosh(y) + i\cos(x)\sinh(y) &= \frac{e^{ix} - e^{-ix}}{2i} \cdot \frac{e^y + e^{-y}}{2} + i\Big(\frac{e^{ix} + e^{-ix}}{2} \cdot \frac{e^y - e^{-y}}{2}\Big) \\
&= \frac{1}{4i}[(e^{ix} - e^{-ix})(e^y + e^{-y}) - (e^{ix} + e^{-ix})(e^y - e^{-y})] \\
&= \frac{e^{i(x+iy)} - e^{-i(x+iy)}}{2i} \\
&= \sin(z).
\end{aligned}$$

Example 7.6.2. Find a harmonic conjugate of the function

$$u(x,y) = -2x^2 + x^3 + 2y^2 - 3xy^2.$$

The function $u(x,y)$ is defined on the whole complex plane. We check that it is harmonic:

$$\begin{aligned} u_x &= -4x + 3x^2 - 3y^2 \\ u_{xx} &= -4 + 6x \\ u_y &= 4y - 6xy \\ u_{yy} &= 4 - 6x. \end{aligned}$$

Hence $u_{xx} + u_{yy}$ is identically equal to zero.

We now proceed to find a harmonic conjugate. We first integrate

$$v_x = -u_y = -4y + 6xy$$

with respect to x and get

$$v(x,y) = \int -4y + 6xy\, dx = -4xy + 3x^2y + C(y),$$

where $C(y)$ is a constant that may involve y. Differentiate the above with respect to y,

$$v_y = -4x + 3x^2 + C'(y).$$

After comparing with u_x, we see that $C'(y) = -3y^2$, and hence $C(y) = -y^3 + C$ for some constant C. The answer is

$$v(x,y) = -4xy + 3x^2y - y^3 + C,$$

where C is any constant.

When $u(x,y)$ is a *polynomial* in x and y, there is a slick method to directly obtain the complex function $f(z)$ whose real part is $u(x,y)$. We can write $z = x + iy$ and exploit the following equalities

$$u\left(\frac{z+\bar{z}}{2}, \frac{z-\bar{z}}{2i}\right) = u(x,y) = \mathrm{Re}(f(z)) = \frac{f(z) + \overline{f(z)}}{2} = \frac{f(z) + f(\bar{z})}{2}.$$

In the last step we can replace $\overline{f(z)}$ by $f(\bar{z})$, because f is a polynomial in z.

We can re-do Example 7.6.2 by using this method.

$$\begin{aligned} u\big(\frac{z+\bar{z}}{2}, \frac{z-\bar{z}}{2i}\big) &= -2\big(\frac{z+\bar{z}}{2}\big)^2 + \big(\frac{z+\bar{z}}{2}\big)^3 + 2\big(\frac{z-\bar{z}}{2i}\big)^2 - 3\big(\frac{z+\bar{z}}{2}\big)\big(\frac{z-\bar{z}}{2i}\big)^2 \\ &= -z^2 - \bar{z}^2 + z^3/2 + \bar{z}^3/2. \end{aligned}$$

By comparing it with $(f(z) + f(\bar{z}))/2$, we obtain $f(z) = -2z^2 + z^3$.

One can further streamline this method by directly computing

$$2u\big(\frac{z}{2}, \frac{z}{2i}\big) - u(0,0). \tag{7.6.1}$$

(We subtract $u(0,0)$ in order to preserve the constant term.) This will be a complex differentiable function f in variable z only and the real part is precisely $u(x,y)$. The complex function in Example 7.6.2 is thus.

$$\begin{aligned} &2u\big(\frac{z}{2}, \frac{z}{2i}\big) - u(0,0) \\ &= 2\big(-2(\frac{z}{2})^2 + (\frac{z}{2})^3 + 2(\frac{z}{2i})^2 - 3(\frac{z}{2})(\frac{z}{2i})^2\big) \\ &= -2z^2 + z^3. \end{aligned}$$

7.7 Application: Smith Chart as a conformal Möbius transformation

Among the simplest examples of a Möbius transformation with interesting geometric behavior is the map

$$\Phi(z) = \frac{z-1}{z+1}.$$

Although this transformation appears in electrical engineering under the name "reflection coefficient," its structure is entirely classical. From the point of view of complex analysis, the so-called *Smith chart* is nothing more than the image of the right half-plane under the Möbius transformation Φ. (See Fig. 7.4)

Mapping the right half–plane to the unit disk. Let

$$H = \{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$$

denote the right half–plane. A direct computation shows that

$$|\Phi(z)| < 1 \quad \Longleftrightarrow \quad |z-1| < |z+1| \quad \Longleftrightarrow \quad \mathrm{Re}(z) > 0.$$

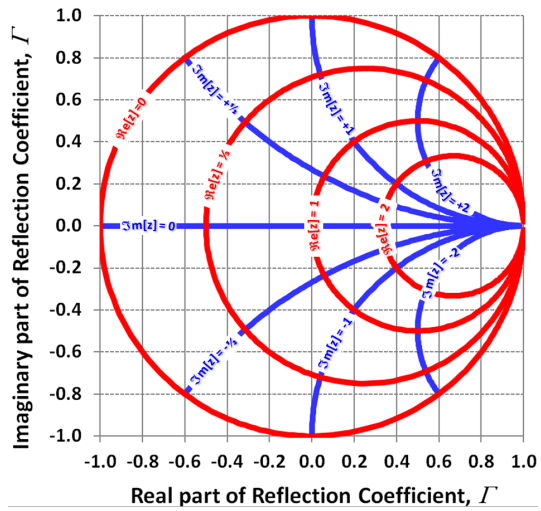


Figure 7.4: Smith chart (from wikipedia)

.

Thus Φ maps H conformally onto the open unit disk

$$D(0,1) = \{w \in \mathbb{C} : |w| < 1\}.$$

The boundary line $\mathrm{Re}(z) = 0$ is mapped to the unit circle $|w| = 1$, and Möbius transformations send generalized circles (circles or lines) to generalized circles. Consequently, the coordinate lines in H become families of circles inside the unit disk.

Geometry of the image. Write $z = x + iy$ with $x > 0$. Then:

- **Vertical lines $x =$ constant** map to circles in the unit disk whose centers lie on the real axis. Each such circle is orthogonal to the unit circle.
- **Horizontal lines $y =$ constant** map to circles tangent to the unit circle at the point $w = 1$. Their centers lie on the real axis to the right of the disk.

Thus the familiar grid of the Smith chart is simply the image of the rectangular coordinate system on H under the Möbius transformation Φ.

7.8 Additional proof for existence of harmonic conjugate

In this optional section we prove that harmonic conjugate of a harmonic function exists if the domain of definition is simply connected. We assume that the function of interest is continuously differentiable. In the followings, the functions u and v must be sufficiently smooth so that the Green's theorem can be applied.

Theorem 7.8.1. *If the domain of definition is simply connected, then harmonic function has a harmonic conjugate in the same domain.*

Proof. Consider the differential form

$$dv = v_x dx + v_y dy = -u_y dx + u_x dy.$$

We claim that the line integral

$$\int_{(x_0,y_0)}^{(x_1,y_1)} -u_y dx + u_x dy \tag{7.8.1}$$

does not depend on path. We can see it from Green's theorem, using the assumption that D is simply connected and $u(x,y)$ satisfies the Laplace equation. If C is a closed path enclosing a region S, we have

$$\oint_C -u_y dx + u_x dy = \iint_S \frac{\partial}{\partial x} u_x - \frac{\partial}{\partial y}(-u_y)\, ds = \iint_S u_{xx} + u_{yy}\, ds = 0.$$

We can thus define a function $v(x,y)$ by first fixing a point (x_0,y_0) in the domain D of $u(x,y)$, and define a function $v(x,y)$ by

$$v(x,y) \triangleq \int_{(x_0,y_0)}^{(x,y)} -u_y dx + u_x dy \tag{7.8.2}$$

for any point (x,y) in D. This is well-defined because the line integral is independent of path, and there is at least one path connecting (x_0,y_0) and (x,y).

We now check that the function $v(x,y)$ so defined is a harmonic conjugate of $u(x,y)$.

Let (a, b) be any point in D. The partial derivative of v with respect to x at this point is

$$\begin{aligned} v_x(a,b) &= \lim_{\Delta x \to 0} \frac{v(a+\Delta x, b) - v(a,b)}{\Delta x} \\ &= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \int_{(a,b)}^{(a+\Delta x, b)} -u_y dx + u_x dy \\ &= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \int_{(a,b)}^{(a+\Delta x, b)} -u_y dx \\ &= -u_y(a,b). \end{aligned}$$

In the last step we have used the assumption that u_y is continuous at (a, b).

Likewise, by consider the partial derivative of v with respect to y, we get

$$\begin{aligned} v_y(a,b) &= \lim_{\Delta y \to 0} \frac{v(a, b+\Delta y) - v(a,b)}{\Delta y} \\ &= \lim_{\Delta y \to 0} \frac{1}{\Delta y} \int_{(a,b)}^{(a, b+\Delta y)} -u_y dx + u_x dy \\ &= \lim_{\Delta y \to 0} \frac{1}{\Delta y} \int_{(a,b)}^{(a, b+\Delta y)} u_x dy \\ &= u_x(a,b). \end{aligned}$$

We have used the continuity of u_x in the last step. This proves that the Cauchy-Riemann equations are satisfied at the point (a, b). By the sufficient condition in Theorem 6.2.5, we conclude that $u(x, y) + iv(x, y)$ is a holomorphic function in the domain D.

We can now conclude that a harmonic conjugate of $u(x, y)$ can be written as in (7.8.2) when D is simply connected. □

This theorem provides the existence proof of harmonic conjugate. If can find a function $v(x, y)$ in closed-form expression such that $dv = -u_y dx + u_x dy$, then this function is a harmonic conjugate of $u(x, y)$. In general, the integration in (7.8.1) is not tractable. The purpose of the proof is to give a sufficient condition under which harmonic conjugate exists.

Summary

- (Holomorphicity) A complex function $f(z)$ is holomorphic at a point z_0 if it is complex differentiable at all points in a neighborhood of z_0. If $f(z)$ is complex differentiable

at all points in a domain D, then $f(z)$ is holomorphic in D. An entire function is a complex function that is holomorphic in the whole complex plane.

- (Differential rules) The differential rules from calculus, such as the product rule, quotient rule, and the chain rule, also apply for holomorphic functions.
- (Holomorphic at ∞) We study the behavior of a complex function $f(z)$ at the point at infinity by making a change of variable $w = 1/z$. By defining $g(w) = f(1/w)$, we say that $f(z)$ is continuous (resp. differentiable, holomorphic) at ∞ if $g(w)$ is continuous (resp. differetiable, holomorphic) at $w = 0$.
- (Smooth curve) A parametric curve in $\mathbb{C}$ is a continuous function $\gamma : [a, b] \to \mathbb{C}$. We say that this is a smooth (resp. piece-wise smooth) curve if γ is a continuously differentiable (resp. piece-wise continuously differentiable).
- (Conformal mapping) The transformation induced by a complex function is conformal if it complex differentiable and the complex derivative is nonzero.
- (Laplace equation) The real and imaginary parts of a holomorphic function satisfy the Laplace equation, if they are twice differentiable.
- (Harmonic conjugate) A harmonic function $v(x, y)$ is called a harmonic conjugate of harmonic conjugate $u(x, y)$ if $u(x, y) + iv(x, y)$ is harmonic.
- (Existence of harmonic conjugate) Given a harmonic function defined on a simple connected domain, harmonic conjugate exists in the same domain.

Exercise

1. Let f denote a holomorphic function in a domain. Prove
   (a) f is a constant function if $\mathrm{Re}(z)$ is a constant function
   (b) f is a constant function if $\mathrm{Im}(z)$ is a constant function.
   (c) f is a constant function if $|z|$ is a constant function.
2. Expand the function in the form of power series $\sum_{k=0}^{\infty} a_k z^k$.

   $(a)\ \dfrac{1+z}{1+iz}$ $\qquad (b)\ \dfrac{1}{(1+z)(i+z)}$ $\qquad (c)\ \dfrac{1}{1+z^3}$

3. By repeated differentiating the function $1/(z+1)^2$, write down the power series expansion of $1/(z+1)^2$.

4. Prove that the radius of convergence R of a series $\sum_{k=0}^{\infty} a_k z^k$ can be expressed as

$$R = \sup\{|z| : \lim_{k\to\infty} a_k z^k = 0\}.$$

5. (Abel limit theorem) Let a_n, for $n \geq 0$ be complex numbers such that $\sum_{n=0}^{\infty} a_n$ converges. Define the power series

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

Show that if $z \to 1$ in such a way that z stays inside the disk $|z - 0.5| \leq 0.5$, then $f(z) \to f(1)$.

(Hint: A complex number z is inside the disk $|z-0.5| \leq 0.5$ if and only if $|1-z|+|z| \leq 1$. By re-defining a_0, we can assume without loss of generality that $\sum_n a_n = 0$. For $n \geq 1$, consider the polynomial $s_n(z) = a_0 + a_1 z + \cdots a_n z^n$, and the partial sum $s_n \triangleq s_n(1) = a_0 + \cdots + a_n$. Prove that

$$s_n(z) = (1-z)(s_0 + s_1 z + \cdots + s_{n-1} z^{n-1}) + s_n z^n,$$

and show that both terms approaches zero.)

6. Consider the power series $\sum_{n=1}^{\infty} \frac{z^n}{n}$.

   (a) Show that when $z = 1$, this series diverges.

   (b) Let $z \neq 1$ be any other point on the unit circle. We first prove the following identity for each integer m,

$$\sum_{n=1}^{m} \frac{z^n}{n} = \frac{z}{1-z}\Big(\sum_{n=1}^{m-1} \frac{1}{n(n+1)} - \sum_{n=1}^{m-1} \frac{z^n}{n(n+1)}\Big) + \frac{z(1-z^m)}{m(1-z)}. \tag{7.8.3}$$

   Using the identity in (7.8.3), we can prove that the power series $\sum_n z^n/n$ converges for $z \neq 1$ on the unit circle by comparing with the convergent power series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Use the fact that

$$\sum_{n=1}^{\infty} \frac{z^n}{n(n+1)}$$

converges absolutely for all z with $|z| = 1$, and the last term $\frac{z(1-z^m)}{m(1-z)}$ in (7.8.3) converges to zero as m tends to infinity.

7. Let $f(z)$ be a polynomial in complex variable z. Prove that $\overline{f(\bar{z})}$ is complex differentiable for all $z \in \mathbb{C}$.

8. (L'Hospital rule for 0/0)

   (a) Let z_0 be a finite complex number Suppose $f(z)$ and $g(z)$ are complex differentiable at z_0, $f(z_0) = g(z_0) = 0$, and $g'(z_0) \neq 0$. Prove that

   $$\lim_{z \to z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

   (b) L'Hospital rule continues to hold at the point at infinity in the following sense. Suppose $f(z)$ (resp. $g(z)$) behaves like a/z (resp. b/z) for some constant a (resp. b) when z tends to infinity. Further, suppose that $b \neq 0$. Then

   $$\lim_{z \to \infty} \frac{f(z)}{g(z)} = \frac{a}{b}.$$

   (The mathematical meaning of $f(z)$ behaves like a/z is $\lim_{z\to\infty} zf(z) = a$.)

9. Suppose $f'(z)$ for $|z| > R$ for some real number R and $\lim_{z\to\infty} f(z)$ exists. Show that if $f(z)$ is differentiable at the point at infinity, then the derivative is given by

   $$f'(\infty) = \lim_{z\to\infty} z^2 f'(z).$$

10. Consider the function

    $$u(x, y) = \sinh(x)\cos(y).$$

    (a) Show that $u(x, y)$ is a harmonic function.

    (b) Find the harmonic conjugate of $u(x, y)$.

    (c) Let $v(x, y)$ be your answer in part (b). Write the holomorphic function $u(x, y) + iv(x, y)$ as a function of z.

11. In general, the Mean Value Theorem does not hold for function in complex variable. Consider the function $f(z) = z^3$, which is complex differentiable everywhere, and the line segment between 1 and i. Represent a point on this segment by $x + (1-x)i$, for $x \in [0,1]$. Show that

$$f'(x + (1-x)i) \neq \frac{f(i) - f(1)}{i - 1}$$

for all $x \in [0,1]$.

12. Solve the system

$$u_x = v_y, \quad u_y = -v_x$$

under the assumption that $u(x,y) = x^2 - y^2 + g(y)$, where $g(y)$ is a function of y. Find all possible holomorphic functions f whose real part is equal to u.

13. Suppose $f(z)$ is holomorphic in some region D. Prove that

$$\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}|f(z)|^2 = |f'(z)|^2$$

for $z \in D$. (Use the property that $\frac{\partial}{\partial \bar{z}}\overline{f(z)} = \overline{\frac{\partial f}{\partial z}(z)}$.)

14. Find a holomorphic function $f(z) = u(x,y) + iv(x,y)$ such that $u(x,y)^2 + iv(x,y)^2$ is also holomorphic.

15. Let $f(x)$ be a real function defined by

$$f(x) \triangleq \begin{cases} 0 & \text{if } x \leq 0 \\ e^{-1/x} & \text{if } x > 0. \end{cases}$$

Show that $f(x)$ is infinitely differentiable at all points, but we cannot represent $f(x)$ using power series at $x = 0$.

16. The Joukowsky map is defined by

$$J(z) \triangleq \frac{1}{2}\left(z + \frac{1}{z}\right)$$

for $z \in \mathbb{C} \setminus \{0\}$. We may extend the domain to the extended complex plane by continuity by setting

$$J(0) \triangleq \infty, \; J(\infty) \triangleq \infty.$$

With this convention, the Joukowsky function has the special property that

$$J(z) = J(1/z)$$

for all $z \in \bar{\mathbb{C}}$.

(a) Show that the derivative of $J(z)$ is equal to

$$J'(z) = \frac{1}{2}\big(1 - \frac{1}{z^2}\big) = \frac{1}{2}\frac{(z-1)(z+1)}{z^2},$$

for $z \neq 0$.

(b) Compute the derivative of $1/J(z)$ at $z = 0$, and show that

$$\frac{d}{dz}\frac{2z}{z^2+1}\Big|_{z=0} = 2.$$

Moreover, show that $\big(1/J(z)\big)'$ approaches 0 as z approaches the point at infinity. Hence, $J(z)$ is conformal in the extended complex plane except ± 1.